66. MIPI DSI

66.1 Overview

This MIPI DSI is a MIPI DSI-2 Host module that has a transmitter function for MIPI Alliance Specification for Display Serial Interface 2 (DSI-2). The DSI-2 Host supports MIPI Alliance Specification for Display Serial Interface 2 (DSI-2) Specification.

Table 66.1 MIPI DSI specifications

Parameter		Specifications
Compliant specification		• MIPI Alliance Specification for Display Serial Interface 2 Version 1.1 (with Errata 01) • MIPI Alliance Specification for D-PHY Version 2.1 (with Errata 01) (80 Mbps to 720 Mbps/lane and up to 2 lanes).
Video mode operation	Available input video format from GLCDC	• Parallel RGB888 (24 bits), little endian • Parallel RGB666 (18 bits), little endian • Parallel RGB565 (16 bits), little endian
	Available output format	• RGB (16 bits, 18 bits, 24 bits)
	Available video mode packet sequence	• Non-Burst mode with sync pulse • Non-Burst mode with sync event • Burst mode
	Others	• Selectable blanking packet or LP-11 during each of blanking interval of HSA, HBP, and HFP
Command mode operation*1	Sequence operation channel 0	LP-only packet generation and LP packet reception from descriptor list
	Sequence operation channel 1	HS or LP packet generation and LP packet reception from descriptor list
DSI Link support functions		• 1 and 2 lane configurations • Unidirectional High-Speed mode transfer (HS-TX) • Bidirectional LP mode transfer/receipt (LP-TX/LP-RX) (Only Lane 0) • ECC/Checksum generation for TX packet • ECC/Checksum verification and ECC error correction for RX packet • Ultra-Low Power mode (ULPS) • Automated power change to LP mode and return to HS mode • Automated clock stop and resume (Non-Continuous Clock mode) • Assignment for virtual channel in Video mode • Assignment for individual virtual channel for each packet in Command mode • Detection for PHY contention error and timeout error • Generation of scrambled packets • Input of TE signal
Module-stop function		Module-stop state can be set to reduce power consumption.
TrustZone Filter		Security and Privilege attribution can be set.

Note 1. Ch0 and Ch1 are exclusive and cannot be used simultaneously.

Figure 62.1 shows the block diagram and Table 62.1 shows the I/O pins.

66.2 Register Description

Access the following registers only in units of 32 bits.

66.2.1 ISR : Interrupt Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	PPI	—	—	—	FERR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RCV	—	—	—	VM	—	—	—	SQ1	—	—	—	SQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SQ0	Sequence Channel-0 Interrupt Flag 0: No interrupt detected 1: Sequence Operation Channel-0 interrupt detected	R
3:1	—	These bits are read as 0.	R
4	SQ1	Sequence Channel-1 Interrupt Flag 0: No interrupt detected 1: Sequence Operation Channel-1 interrupt detected	R
7:5	—	These bits are read as 0.	R
8	VM	Video Mode Interrupt Flag 0: No interrupt detected 1: Video Mode interrupt detected	R
11:9	—	These bits are read as 0.	R
12	RCV	Receive Interrupt Flag 0: No interrupt detected 1: Receive interrupt detected	R
15:13	—	These bits are read as 0.	R
16	FERR	Fatal Error Interrupt Flag 0: No interrupt detected 1: Fatal Error interrupt detected	R
19:17	—	These bits are read as 0.	R
20	PPI	PPI Interrupt Flag 0: No interrupt detected 1: PPI interrupt detected	R
31:21	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

SQ0 bit (Sequence Channel-0 Interrupt Flag)

The SQ0 flag indicates that the Sequence Operation Channel-0 interrupt is detected. The Sequence Operation Channel-0 interrupt is an interrupt shown in the SQCH0SR register.

The SQ0 flag is set to 1 when any one of the flags of the SQCH0SR register elements is set. For details, see section 66.2.54. SQCH0SR : Sequence Channel 0 Status Register.

SQ1 bit (Sequence Channel-1 Interrupt Flag)

The SQ1 flag indicates that the Sequence Operation Channel-1 interrupt is detected. The Sequence Operation Channel-1 interrupt is an interrupt shown in the SQCH1SR register.

The SQ1 flag is set to 1 when any one of the flags of the SQCH1SR register elements is set. For details, see section 66.2.58. SQCH1SR : Sequence Channel 1 Status Register.

VM bit (Video Mode Interrupt Flag)

The VM flag indicates that the Video Mode interrupt is detected. The Video Mode interrupt is an interrupt shown in the VMSR register.

The VM flag is set to 1 when any one of the flags of the VMSR register elements is set. For details, see section 66.2.45. VMSR : Video Mode Status Register.

RCV bit (Receive Interrupt Flag)

The RCV flag indicates that the Receive interrupt is detected. The Receive interrupt is an interrupt shown in the RXSR register.

The RCV flag is set to 1 when any one of the flags of the RXSR register elements is set. For details, see section 66.2.14. RXSR : Receive Status Register.

FERR bit (Fatal Error Interrupt Flag)

The FERR flag indicates that the Fatal Error interrupt is detected. The Fatal Error interrupt is an interrupt shown in the FERRSR register.

The FERR flag is set to 1 when any one of the flags of the FERRSR register elements is set. For details, see section 66.2.35. FERRSR : Fatal Error Status Register.

PPI bit (PPI Interrupt Flag)

The PPI flag indicates that the PPI interrupt is detected. The PPI interrupt is an interrupt shown in the PLSR register.

The PPI flag is set to 1 when any one of the flags of the PLSR register elements is set. For details, see section 66.2.40. PLSR : Physical Lane Status Register.

66.2.2 LINKSR : Link Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	LPBUSY	HSBUSY	—	—	—	VRUN	—	—	—	SQ1RUN	—	—	—	SQ0RUN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SQ0RUN	Sequence Channel-0 Running Flag 0: Sequence Operation Channel-0 stopped 1: Sequence Operation Channel-0 in operation	R
3:1	—	These bits are read as 0.	R
4	SQ1RUN	Sequence Channel-1 Running Flag 0: Sequence Operation Channel-1 stopped 1: Sequence Operation Channel-1 in operation	R
7:5	—	These bits are read as 0.	R
8	VRUN	Video Mode Operation Running Flag 0: Video mode stopped 1: Video mode in operation	R
11:9	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
12	HSBUSY	HS Operation Busy Flag 0: HS mode stopped 1: HS mode in operation	R
13	LPBUSY	LP Operation Busy Flag 0: LP mode stopped 1: LP mode in operation	R
15:14	—	These bits are read as 0.	R
20:16	—	The read values are undefined.	R
23:21	—	These bits are read as 0.	R
28:24	—	The read values are undefined.	R
31:29	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

SQ0RUN bit (Sequence Channel-0 Running Flag)

The SQ0RUN flag is a copy of the SQCH0SR.RUNNING flag. The meaning is also the same. For details, see section 66.2.54. SQCH0SR : Sequence Channel 0 Status Register.

SQ1RUN bit (Sequence Channel-1 Running Flag)

The SQ1RUN flag is a copy of the SQCH1SR.RUNNING flag. The meaning is also the same. For details, see section 66.2.58. SQCH1SR : Sequence Channel 1 Status Register.

VRUN bit (Video Mode Operation Running Flag)

The VRUN flag is a copy of the VMSR.RUNNING bit. The meaning is also the same. For details, see section 66.2.45. VMSR : Video Mode Status Register.

HSBUSY bit (HS Operation Busy Flag)

The HSBUSY flag indicates that an operation related to HS mode is running.

LPBUSY bit (LP Operation Busy Flag)

The LPBUSY flag indicates that an operation related to LP mode is running.

66.2.3 TXSETR : Transmit Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	DLEN	CLEN	—	—	—	—	—	—	NUMLANE[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
1:0	NUMLANE[1:0]	Number of Lane Set the number of lanes for use. 0 0: 1 Lane (Use of Lane-0) 0 1: 2 Lane (Use of Lane-0 and Lane-1) Others Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
8	CLEN	Clock Lane Enable 0: Disable 1: Enable	R/W
9	DLEN	Data Lane Enable 0: Disable 1: Enable	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	—	This bit is read as 1. The write value should be 1.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

NUMLANE[1:0] bits (Number of Lane)

NUMLANE[1:0] bits control the number of lanes to be used. Set the number of lanes for use.

CLEN bit (Clock Lane Enable)

The CLEN bit controls a clock lane of D-PHY. Set the CLEN bit to 1 to enable the clock lane.

DLEN bit (Data Lane Enable)

The DLEN bit controls the data lanes of D-PHY. When the DLEN bit is set to 1, the data lanes specified by the NUMLANE[1:0] bits are enabled.

66.2.4 HSCLKSETR : HS Clock Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HSCLMD	HSCLST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HSCLST	HS Clock Start 0: Stop HS transmission (keep LP state) 1: Start HS transmission	R/W
1	HSCLMD	HS Clock Running Mode 0: Non-continuous clock mode 1: Continuous clock mode	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HSCLST bit (HS Clock Start)

The HSCLST bit controls the HS transmission from the clock lane. When HSCLST = 1, the HS clock is transferred from the clock lane according to the setting of the HSCLMD bit.

It is prohibited to set the HSCLST bit to 1 when TXSETR.CLEN = 0.

HSCLMD bit (HS Clock Running Mode)

The HSCLMD bit controls the HS transmission mode from the clock lane.

When HSCLMD = 0, the HS clock is transferred from the clock lane only when the HS transmit is requested (Non-continuous clock mode).

When HSCLMD = 1, the clock lane keeps the HS transmission (Continuous clock mode).

It is prohibited to change the setting of the HSCLMD bit when HSCLST = 1.

66.2.5 ULPSSETR : ULPS Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x108

Bit position:	31																								7							0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WKUP[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	WKUP[7:0]	ULPS Wakeup Period Set the T_{WAKEUP} period of ULPS.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

WKUP[7:0] bits (ULPS Wakeup Period)

The WKUP[7:0] bits control the T_{WAKEUP} period during which DSI Host drives a Mark-1 state prior to a Stop state in order to initiate an exit from ULPS.

This period is calculated by the following formula:

$$T_{WAKEUP}[ms]^{*1} = WKUP[7:0] \times 128 \times (1 / f_{LPCLK}[MHz]^{*2}) / 1000$$

Note 1. This period should be more than 1 ms.
Note 2. f_{LPCLK} is the frequency of the Low-Power mode clock (LPCLK).

66.2.6 ULPSCR : ULPS Control Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x10C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DLEXIT	DLENT	—	—	CLEXIT	CLENT	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
23:0	—	These bits are read as 0. The write value should be 0.	W

Bit	Symbol	Function	R/W
24	CLENT	CL ULPS Enter 0: No operation 1: Transition Clock Lane to ULPS	W
25	CLEXIT	CL ULPS Exit 0: No operation 1: Clock Lane exits from ULPS	W
27:26	—	These bits are read as 0. The write value should be 0.	W
28	DLENT	DL ULPS Enter 0: No operation 1: Transition Data Lanes to ULPS	W
29	DLEXIT	DL ULPS Exit 0: No operation 1: Data Lanes exit from ULPS	W
31:30	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

CLENT bit (CL ULPS Enter)

The CLENT bit controls the transition of the clock lane to Ultra-low Power State (ULPS).

Set the CLENT bit to 1 during the period when the HSCLKSETR.HSCLST bit is 0.

It is prohibited to set the CLENT bit to 1 when the clock lane is in ULPS.

CLEXIT bit (CL ULPS Exit)

The CLEXIT bit controls the exit of the clock lane from Ultra-low Power State (ULPS).

It is prohibited to set the CLEXIT bit to 1 when the clock lane is not in ULPS.

DLENT bit (DL ULPS Enter)

The DLENT bit controls the transition of the data lanes to Ultra-low Power State (ULPS).

It is prohibited to set the DLENT bit to 1 when the data lanes is in ULPS.

DLEXIT bit (DL ULPS Exit)

The DLEXIT bit controls the exit of the data lanes from Ultra-low Power State (ULPS).

It is prohibited to set the DLEXIT bit to 1 when the data lanes is not in ULPS.

66.2.7 RSTCR : Reset Control Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x110

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FTXSTP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SWRST	Software Reset 0: Complete the software reset 1: Request a software reset	R/W

Bit	Symbol	Function	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
16	FTXSTP	Force TX Stop Mode 0: Finish the Force Stop state 1: Force data lanes into transmit mode and generate Stop state	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SWRST bit (Software Reset)

The SWRST bit is used to request a software reset. See section 66.3.2. Software Reset for the software reset procedure.

FTXSTP bit (Force TX Stop Mode)

When FTXSTP = 1, the data lanes immediately transition into transmit mode and is forced into the Stop state.

It is prohibited to set the FTXSTP bit to 1 when SWRST = 0. The FTXSTP bit is set during the software reset procedure. See section 66.3.2. Software Reset for the software reset procedure.

66.2.8 RSTSR : Reset Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x114

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DL0DIR	—	—	—	—	—	DL1STP	DL0STP	—	—	—	RSTV	RSTAXI	RSTAPB	RSTLP	RSTHS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RSTHS	HS Software Reset Status 0: Not in software reset procedure 1: Running software reset procedure for HS mode	R
1	RSTLP	LP Software Reset Status 0: Not in software reset procedure 1: Running software reset procedure for LP mode	R
2	RSTAPB	APB Software Reset Status 0: Not in software reset procedure 1: Running software reset procedure for APB bus	R
3	RSTAXI	AXI Software Reset Status 0: Not in software reset procedure 1: Running software reset procedure for AXI bus	R
4	RSTV	Video Software Reset Status 0: Not in software reset procedure 1: Running software reset procedure for Video mode	R
7:5	—	These bits are read as 0.	R
8	DL0STP	Data Lane-0 Stop Status 0: Not Stop state 1: Stop state	R
9	DL1STP	Data Lane-1 Stop Status 0: Not Stop state 1: Stop state	R

Bit	Symbol	Function	R/W
14:10	—	These bits are read as 0.	R
15	DL0DIR	Data Lane-0 Direction 0: TX mode 1: RX mode	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

RSTHS bit (HS Software Reset Status)

The RSTHS bit indicates the state of the HS mode initialization process by software reset, which is caused by setting the RSTCR.SWRST bit to 1.

RSTLP bit (LP Software Reset Status)

The RSTLP bit indicates the state of the LP mode initialization process by software reset, which is caused by setting the RSTCR.SWRST bit to 1.

RSTAPB bit (APB Software Reset Status)

The RSTAPB bit indicates the state of the APB initialization process by software reset, which is caused by setting the RSTCR.SWRST bit to 1.

RSTAXI bit (AXI Software Reset Status)

The RSTAXI bit indicates the state of the AXI initialization process by software reset, which is caused by setting the RSTCR.SWRST bit to 1.

RSTV bit (Video Software Reset Status)

The RSTV bit indicates the state of the Video initialization process by software reset, which is caused by setting the RSTCR.SWRST bit to 1.

DL0STP bit (Data Lane-0 Stop Status)

The DL0STP bit indicates the Stop state of the Data Lane-0. The DL0STP bit is a copy of the PLSR.DL0STP bit. The meaning is also the same. See section 66.2.40. PLSR : Physical Lane Status Register for details on the PLSR.DL0STP bit.

DL1STP bit (Data Lane-1 Stop Status)

The DL1STP bit indicates the Stop state of the Data Lane-1. The DL1STP bit is a copy of the PLSR.DL1STP bit. The meaning is also the same. See section 66.2.40. PLSR : Physical Lane Status Register for details on the PLSR.DL1STP bit.

DL0DIR bit (Data Lane-0 Direction)

The DL0DIR bit indicates the direction of the Data Lane-0. The DL0DIR bit is a copy of the PLSR.DL0DIR bit. The meaning is also the same. See section 66.2.40. PLSR : Physical Lane Status Register for details on the PLSR.DL0DIR bit.

66.2.9 DSISETR : DSI Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x120

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EOTPEN	EXTEMD	SCREN	—	—	—	—	—	VC3CRCEN	VC2CRCEN	VC1CRCEN	VC0CRCEN	—	—	—	ECCEN
Value after reset:	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRPSZ[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
15:0	MRPSZ[15:0]	Maximum Return Packet Size Set the maximum return packet size.	R/W
16	ECCEN	ECC Check Enable 0: Disable 1: Enable	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	VC0CRCEN	VC-0 CRC Check Enable 0: Disable 1: Enable	R/W
21	VC1CRCEN	VC-1 CRC Check Enable 0: Disable 1: Enable	R/W
22	VC2CRCEN	VC-2 CRC Check Enable 0: Disable 1: Enable	R/W
23	VC3CRCEN	VC-3 CRC Check Enable 0: Disable 1: Enable	R/W
28:24	—	These bits are read as 0. The write value should be 0.	R/W
29	SCREN	Data Scramble Enable 0: Disable 1: Enable	R/W
30	EXTEMD	External Tearing Effect Detection Sense Select 0: Rising edge 1: Falling edge	R/W
31	EOTPEN	HS Transfer EoTp Enable 0: Disable 1: Enable	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

MRPSZ[15:0] bits (Maximum Return Packet Size)

The MRPSZ[15:0] bits specify the maximum packet size to be received in LP-RX mode.

If the WC of the returned long packet exceeds the value of the MRPSZ[15:0] bits, a Return Packet Size error occurs and the RXSR.RSIZEERR flag is set to 1. In this case, no data is stored in the memory area.

It is prohibited to set the MMRPSZ[15:0] bits to 0.

ECCEN bit (ECC Check Enable)

The ECCEN bit controls the ECC check permission. To enable the ECC check, set the ECCEN bit to 1.

VC0CRCEN bit (VC-0 CRC Check Enable)

The VC0CRCEN bit controls the CRC check permission for Virtual Channel-0 (VC-0). To enable the CRC check for VC-0, set the VC0CRCEN bit to 1.

VC1CRCEN bit (VC-1 CRC Check Enable)

The VC1CRCEN bit controls the CRC check permission for Virtual Channel-1 (VC-1). To enable the CRC check for VC-1, set the VC1CRCEN bit to 1.

VC2CRCEN bit (VC-2 CRC Check Enable)

The VC2CRCEN bit controls the CRC check permission for Virtual Channel-2 (VC-2). To enable the CRC check for VC-2, set the VC2CRCEN bit to 1.

VC3CRCEN bit (VC-3 CRC Check Enable)

The VC3CRCEN bit controls the CRC check permission for Virtual Channel-3 (VC-3). To enable the CRC check for VC-3, set the VC3CRCEN bit to 1.

SCREN bit (Data Scramble Enable)

The SCREN bit controls the data scrambling permission. If the peripheral device does not have a scrambling function, do not set the SCREN bit to 1.

EXTEMD bit (External Tearing Effect Detection Sense Select)

The EXTEMD bit controls the detection edge of the External Tearing Effect (DSI_TE pin) input.

EOTPEN bit (HS Transfer EoTp Enable)

The EOTPEN bit controls the permission of the EoTp transfer in HS-TX mode. To enable EoTp transfer, set the EOTPEN bit to 1. EoTp is always disabled in LP-TX mode.

66.2.10 TXPPD0R : Transmit Packet Payload Data 0 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x160

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA3[7:0]	DATA2[7:0]	DATA1[7:0]	DATA0[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA0[7:0]	Payload Data 0 Set the Long Packet Payload Data 0 to be transferred in Command mode.*1	R/W
15:8	DATA1[7:0]	Payload Data 1 Set the Long Packet Payload Data 1 to be transferred in Command mode.*1	R/W
23:16	DATA2[7:0]	Payload Data 2 Set the Long Packet Payload Data 2 to be transferred in Command mode.*1	R/W
31:24	DATA3[7:0]	Payload Data 3 Set the Long Packet Payload Data 3 to be transferred in Command mode.*1	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when the SQCHnDSCmAR.FMT bit is set to 1 and the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b.

DATA0[7:0] bits (Payload Data 0)

TheDATA0[7:0] bits are used to store the Long Packet payload (Data 0) to be transferred in Command mode using Sequence operation.

DATA1[7:0] bits (Payload Data 1)

TheDATA1[7:0] bits are used to store the Long Packet payload (Data 1) to be transferred in Command mode using Sequence operation.

DATA2[7:0] bits (Payload Data 2)

TheDATA2[7:0] bits are used to store the Long Packet payload (Data 2) to be transferred in Command mode using Sequence operation.

DATA3[7:0] bits (Payload Data 3)

TheDATA3[7:0] bits are used to store the Long Packet payload (Data 3) to be transferred in Command mode using Sequence operation.

66.2.11 TXPPD1R : Transmit Packet Payload Data 1 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x164

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA7[7:0]								DATA6[7:0]								DATA5[7:0]								DATA4[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DATA4[7:0]	Payload Data 4 Set the Long Packet Payload Data 4 to be transferred in Command mode.*1	R/W
15:8	DATA5[7:0]	Payload Data 5 Set the Long Packet Payload Data 5 to be transferred in Command mode.*1	R/W
23:16	DATA6[7:0]	Payload Data 6 Set the Long Packet Payload Data 6 to be transferred in Command mode.*1	R/W
31:24	DATA7[7:0]	Payload Data 7 Set the Long Packet Payload Data 7 to be transferred in Command mode.*1	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when the SQCHnDSCmAR.FMT bit is set to 1 and the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b.

DATA4[7:0] bits (Payload Data 4)

The DATA4[7:0] bits are used to store the Long Packet payload (Data 4) to be transferred in Command mode using Sequence operation.

DATA5[7:0] bits (Payload Data 5)

TheDATA5[7:0] bits are used to store the Long Packet payload (Data 5) to be transferred in Command mode using Sequence operation.

DATA6[7:0] bits (Payload Data 6)

The DATA6[7:0] bits are used to store the Long Packet payload (Data 6) to be transferred in Command mode using Sequence operation.

DATA7[7:0] bits (Payload Data 7)

The DATA7[7:0] bits are used to store the Long Packet payload (Data 7) to be transferred in Command mode using Sequence operation.

66.2.12 TXPPD2R : Transmit Packet Payload Data 2 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x168

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA11[7:0]								DATA10[7:0]								DATA9[7:0]								DATA8[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DATA8[7:0]	Payload Data 8 Set the Long Packet Payload Data 8 to be transferred in Command mode.*1	R/W
15:8	DATA9[7:0]	Payload Data 9 Set the Long Packet Payload Data 9 to be transferred in Command mode.*1	R/W

Bit	Symbol	Function	R/W
23:16	DATA10[7:0]	Payload Data 10 Set the Long Packet Payload Data 10 to be transferred in Command mode.[*1]	R/W
31:24	DATA11[7:0]	Payload Data 11 Set the Long Packet Payload Data 11 to be transferred in Command mode.[*1]	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when the SQCHnDSCmAR.FMT bit is set to 1 and the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b.

DATA8[7:0] bits (Payload Data 8)

The DATA8[7:0] bits are used to store the Long Packet payload (Data 8) to be transferred in Command mode using Sequence operation.

DATA9[7:0] bits (Payload Data 9)

The DATA9[7:0] bits are used to store the Long Packet payload (Data 9) to be transferred in Command mode using Sequence operation.

DATA10[7:0] bits (Payload Data 10)

The DATA10[7:0] bits are used to store the Long Packet payload (Data 10) to be transferred in Command mode using Sequence operation.

DATA11[7:0] bits (Payload Data 11)

The DATA11[7:0] bits are used to store the Long Packet payload (Data 11) to be transferred in Command mode using Sequence operation.

66.2.13 TXPPD3R : Transmit Packet Payload Data 3 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x16C

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA15[7:0]	DATA14[7:0]	DATA13[7:0]	DATA12[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA12[7:0]	Payload Data 12 Set the Long Packet Payload Data 12 to be transferred in Command mode.[*1]	R/W
15:8	DATA13[7:0]	Payload Data 13 Set the Long Packet Payload Data 13 to be transferred in Command mode.[*1]	R/W
23:16	DATA14[7:0]	Payload Data 14 Set the Long Packet Payload Data 14 to be transferred in Command mode.[*1]	R/W
31:24	DATA15[7:0]	Payload Data 15 Set the Long Packet Payload Data 15 to be transferred in Command mode.[*1]	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when the SQCHnDSCmAR.FMT bit is set to 1 and the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b.

DATA12[7:0] bits (Payload Data 12)

The DATA12[7:0] bits are used to store the Long Packet payload (Data 12) to be transferred in Command mode using Sequence operation.

DATA13[7:0] bits (Payload Data 13)

The DATA13[7:0] bits are used to store the Long Packet payload (Data 13) to be transferred in Command mode using Sequence operation.

DATA14[7:0] bits (Payload Data 14)

The DATA14[7:0] bits are used to store the Long Packet payload (Data 14) to be transferred in Command mode using Sequence operation.

DATA15[7:0] bits (Payload Data 15)

The DATA15[7:0] bits are used to store the Long Packet payload (Data 15) to be transferred in Command mode using Sequence operation.

66.2.14 RXSR : Receive Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	—	ECCERRS	—	RSIZEERR	NORESERR	PRTOERR	RXOVFERR	IBERR	CRCERR	WCERR	—	UNEXERR	ECCERRM	MLFERR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EXTEDET	RXACK	—	—	—	RXEOTP	—	RXRESP	—	—	—	—	—	TATO	LRXHTO	BTAREND
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BTAREND	BTA Request End Interrupt Flag 0: Not detected 1: BTA completion detected	R
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Flag 0: Not detected 1: LP-RX Host Processor Timeout (LRX-H_TO) detected	R
2	TATO	Turnaround Acknowledge Timeout Interrupt Flag 0: Not detected 1: Turnaround Acknowledge Timeout (TA_TO) detected	R
7:3	—	These bits are read as 0.	R
8	RXRESP	Response Packet Receive Interrupt Flag 0: No response received 1: Response packet received	R
9	—	This bit is read as 0.	R
10	RXEOTP	EoTp Receive Interrupt Flag 0: No received 1: EoTp received	R
13:11	—	These bits are read as 0.	R
14	RXACK	ACK Trigger Receive Interrupt Flag 0: No trigger received 1: ACK trigger received	R
15	EXTEDET	External Tearing Effect Detect Interrupt Flag 0: Not detected 1: External tearing effect detected	R
16	MLFERR	Malform Error Interrupt Flag 0: No error received 1: A packet of less than 4 bytes received	R
17	ECCERRM	Multi Bit ECC Error Interrupt Flag 0: No error detected 1: A multi-bit ECC error detected	R

Bit	Symbol	Function	R/W
18	UNEXERR	Unexpected Packet Error Interrupt Flag 0: No error 1: Unexpected packet received	R
19	—	This bit is read as 0.	R
20	WCERR	Word Count Error Interrupt Flag 0: No error detected 1: The length of the received packet is shorter than the Word Count (WC) indicated in the packet header	R
21	CRCERR	CRC Error Interrupt Flag 0: No error detected 1: CRC error detected	R
22	IBERR	Internal Bus Error Interrupt Flag 0: No error detected 1: Internal AXI bus write failed	R
23	RXOVFERR	Receive Buffer Overflow Error Interrupt Flag 0: No error detected 1: A buffer overflow error detected on receiving long packets	R
24	PRTOERR	Peripheral Response Timeout Error Interrupt Flag 0: No error occurred 1: Peripheral response timeout occurred	R
25	NORESERR	No Response Error Interrupt Flag 0: No error occurred 1: No triggers or packets returned during BTA period	R
26	RSIZEERR	Return Packet Size Error Interrupt Flag 0: No error detected 1: Oversize error in a received long packet detected	R
27	—	This bit is read as 0.	R
28	ECCERRS	Single Bit ECC Error Interrupt Flag 0: No error detected 1: A single-bit ECC error detected	R
29	—	This bit is read as 0.	R
30	RXAKE	Acknowledge and Error Report Receive Interrupt Flag 0: No received 1: An acknowledge and error report packet received	R
31	—	This bit is read as 0.	R

Note: S-TYPE-3, P-TYPE-3

BTAREND bit (BTA Request End Interrupt Flag)

The BTAREND flag indicates that the BTA request specified by the SQCHnDSCmAR.BTA[1:0] bits is completed.

For example, if the SQCHnDSCmAR.BTA[1:0] bits are set to 10b, the BTAREND flag is set to 1 after reception of the response packet to read request following the BTA is completed.

The BTAREND flag is not cleared automatically, so set the RXSCR.BTAREND bit to 1 to clear the BTAREND flag.

LRXHTO bit (LP-RX Host Processor Timeout Interrupt Flag)

The LRXHTO flag indicates that a LP-RX Host Processor Timeout Interrupt Flag (LRX-H_TO) is detected.

The LRXHTO flag is not cleared automatically, so set the RXSCR.LRXHTO bit to 1 to clear the LRXHTO flag.

TATO bit (Turnaround Acknowledge Timeout Interrupt Flag)

The TATO flag indicates that a Turnaround Acknowledge Timeout (TA_TO) is detected.

The TATO flag is not cleared automatically, so set the RXSCR.TATO bit to 1 to clear the TATO flag.

RXRESP bit (Response Packet Receive Interrupt Flag)

The RXRESP flag indicates that a response packet is received.

The RXRESP flag is not cleared automatically, so set the RXSCR.RXRESP bit to 1 to clear the RXRESP flag.

RXEOTP bit (EoTp Receive Interrupt Flag)

The RXEOTP flag indicates that an EoTp is received.

The RXEOTP flag is not cleared automatically, so set the RXSCR.RXEOTP bit to 1 to clear the RXEOTP flag.

RXACK bit (ACK Trigger Receive Interrupt Flag)

The RXACK flag indicates that an ACK trigger is received.

The RXACK flag is not cleared automatically, so set the RXSCR.RXACK bit to 1 to clear the RXACK flag.

EXTEDET bit (External Tearing Effect Detect Interrupt Flag)

The EXTEDET flag indicates that external tearing effect from DSI_TE pin is detected.

The EXTEDET flag is not cleared automatically, so set the RXSCR.EXTEDET bit to 1 to clear the EXTEDET flag.

MLFERR bit (Malform Error Interrupt Flag)

The MLFERR flag indicates that a packet of less than 4 bytes is received.

The MLFERR flag is not cleared automatically, so set the RXSCR.MLFERR bit to 1 to clear the MLFERR flag.

ECCERRM bit (Multi Bit ECC Error Interrupt Flag)

The ECCERRM flag indicates that a multi-bit ECC error is detected in a received packet.

The ECCERRM flag is not cleared automatically, so set the RXSCR.ECCERRM bit to 1 to clear the ECCERRM flag.

UNEXERR bit (Unexpected Packet Error Interrupt Flag)

The UNEXERR flag indicates that an unexpected Data Type (DT) or unexpected response is received.

The UNEXERR flag is not cleared automatically, so set the RXSCR.UNEXERR bit to 1 to clear the UNEXERR flag.

WCERR bit (Word Count Error Interrupt Flag)

The WCERR flag indicates that the length of the received packet payload is shorter than the Word Count (WC) indicated in the packet header.

The WCERR flag is not cleared automatically, so set the RXSCR.WCERR bit to 1 to clear the WCERR flag.

CRCERR bit (CRC Error Interrupt Flag)

The CRCERR flag indicates that a CRC error is detected in a received packet.

The CRCERR flag is not cleared automatically, so set the RXSCR.CRCERR bit to 1 to clear the CRCERR flag.

IBERR bit (Internal Bus Error Interrupt Flag)

The IBERR flag is set to 1 when the internal AXI bus write failed.

The IBERR flag is not cleared automatically, so set the RXSCR.IBERR bit to 1 to clear the IBERR flag.

RXOVFERR bit (Receive Buffer Overflow Error Interrupt Flag)

The RXOVFERR flag is set to 1 when the receive buffer of long packet overflows.

The RXOVFERR flag is not cleared automatically, so set the RXSCR.RXOVFERR bit to 1 to clear the RXOVFERR flag.

PRTOERR bit (Peripheral Response Timeout Error Interrupt Flag)

The PRTOERR flag is set to 1 when the timeout for peripheral response expires after giving bus possession to the peripheral and entering in LP-RX mode. The timeout period is specified by the following registers depending on the type of Bus Turn-Around (BTA).

- PRESPTOBTASETR or
- PRESPTOLPSETR.LPRTO[15:0] or
- PRESPTOLPSETR.LPWTO[15:0] or
- PRESPTOHSSETR.HSRTO[15:0] or

- PRESPTOHSSETR.HSWTO[15:0]

The PRTOERR flag is not cleared automatically, so set the RXSCR.PRTOERR bit to 1 to clear the PRTOERR flag.

NORESERR bit (No Response Error Interrupt Flag)

The NORESERR flag is set to 1 when no trigger or packet is returned during BTA period.

The NORESERR flag is not cleared automatically, so set the RXSCR.NORESERR bit to 1 to clear the NORESERR flag.

RSIZEERR bit (Return Packet Size Error Interrupt Flag)

The RSIZEERR flag is set to 1 when the Word Count (WC) value of the received long packet is larger than the size set by the DSISETR.MRPSZ[15:0] bits.

The RSIZEERR flag is not cleared automatically, so set the RXSCR.RSIZEERR bit to 1 to clear the RSIZEERR flag.

ECCERRS bit (Single Bit ECC Error Interrupt Flag)

The ECCERRS flag indicates that a single-bit ECC error is detected and corrected in a received packet.

The ECCERRS flag is not cleared automatically, so set the RXSCR.ECCERRS bit to 1 to clear the ECCERRS flag.

RXAKE bit (Acknowledge and Error Report Receive Interrupt Flag)

The RXAKE flag indicates that the acknowledge and error report packet is received from the peripheral.

The RXAKE flag is not cleared automatically, so set the RXSCR.RXAKE bit to 1 to clear the RXAKE flag.

66.2.15 RXSCR : Receive Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	—	ECCERRS	—	RSIZEERR	NORESERR	PRTOERR	RXOVFERR	IBERR	CRCERR	WCERR	—	UNEXERR	ECCERRM	MLFERR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EXTEDET	RXACK	—	—	—	RXEOTP	—	RXRESP	—	—	—	—	—	TATO	LRXHTO	BTAREND
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BTAREND	BTA Request End Interrupt Flag Clear 0: No operation 1: Clear the RXSR.BTAREND flag	W
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Flag Clear 0: No operation 1: Clear the RXSR.LRXHTO flag	W
2	TATO	Turnaround Acknowledge Timeout Interrupt Flag Clear 0: No operation 1: Clear the RXSR.TATO flag	W
7:3	—	These bits are read as 0. The write value should be 0.	W
8	RXRESP	Response Packet Receive Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RXRESP flag	W
9	—	This bit is read as 0. The write value should be 0.	W
10	RXEOTP	EoTp Receive Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RXEOTP flag	W

Bit	Symbol	Function	R/W
13:11	—	These bits are read as 0. The write value should be 0.	W
14	RXACK	ACK Trigger Receive Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RXACK flag	W
15	EXTEDET	External Tearing Effect Detect Interrupt Flag Clear 0: No operation 1: Clear the RXSR.EXTEDET flag	W
16	MLFERR	Malform Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.MLFERR flag	W
17	ECCERRM	Multi Bit ECC Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.ECCERRM flag	W
18	UNEXERR	Unexpected Packet Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.UNEXERR flag	W
19	—	This bit is read as 0. The write value should be 0.	W
20	WCERR	Word Count Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.WCERR flag	W
21	CRCERR	CRC Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.CRCERR flag	W
22	IBERR	Internal Bus Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.IBERR flag	W
23	RXOVFERR	Receive Buffer Overflow Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RXOVFERR flag	W
24	PRTOERR	Peripheral Response Timeout Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.PRTOERR flag	W
25	NORESERR	No Response Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.NORESERR flag	W
26	RSIZEERR	Return Packet Size Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RSIZEERR flag	W
27	—	This bit is read as 0. The write value should be 0.	W
28	ECCERRS	Single Bit ECC Error Interrupt Flag Clear 0: No operation 1: Clear the RXSR.ECCERRS flag	W
29	—	This bit is read as 0. The write value should be 0.	W
30	RXAKE	Acknowledge and Error Report Receive Interrupt Flag Clear 0: No operation 1: Clear the RXSR.RXAKE flag	W
31	—	This bit is read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

BTAREND bit (BTA Request End Interrupt Flag Clear)

Set the BTAREND bit to 1 to clear the RXSR.BTAREND flag.

LRXHTO bit (LP-RX Host Processor Timeout Interrupt Flag Clear)

Set the LRXHTO bit to 1 to clear the RXSR.LRXHTO flag.

TATO bit (Turnaround Acknowledge Timeout Interrupt Flag Clear)

Set the TATO bit to 1 to clear the RXSR.TATO flag.

RXRESP bit (Response Packet Receive Interrupt Flag Clear)

Set the RXRESP bit to 1 to clear the RXSR.RXRESP flag.

RXEOTP bit (EoTp Receive Interrupt Flag Clear)

Set the RXEOTP bit to 1 to clear the RXSR.RXEOTP flag.

RXACK bit (ACK Trigger Receive Interrupt Flag Clear)

Set the RXACK bit to 1 to clear the RXSR.RXACK flag.

EXTEDET bit (External Tearing Effect Detect Interrupt Flag Clear)

Set the EXTEDET bit to 1 to clear the RXSR.EXTEDET flag.

MLFERR bit (Malform Error Interrupt Flag Clear)

Set the MLFERR bit to 1 to clear the RXSR.MLFERR flag.

ECCERRM bit (Multi Bit ECC Error Interrupt Flag Clear)

Set the ECCERRM bit to 1 to clear the RXSR.ECCERRM flag.

UNEXERR bit (Unexpected Packet Error Interrupt Flag Clear)

Set the UNEXERR bit to 1 to clear the RXSR.UNEXERR flag.

WCERR bit (Word Count Error Interrupt Flag Clear)

Set the WCERR bit to 1 to clear the RXSR.WCERR flag.

CRCERR bit (CRC Error Interrupt Flag Clear)

Set the CRCERR bit to 1 to clear the RXSR.CRCERR flag.

IBERR bit (Internal Bus Error Interrupt Flag Clear)

Set the IBERR bit to 1 to clear the RXSR.IBERR flag.

RXOVFERR bit (Receive Buffer Overflow Error Interrupt Flag Clear)

Set the RXOVFERR bit to 1 to clear the RXSR.RXOVFERR flag.

PRTOERR bit (Peripheral Response Timeout Error Interrupt Flag Clear)

Set the PRTOERR bit to 1 to clear the RXSR.PRTOERR flag.

NORESERR bit (No Response Error Interrupt Flag Clear)

Set the NORESERR bit to 1 to clear the RXSR.NORESERR flag.

RSIZEERR bit (Return Packet Size Error Interrupt Flag Clear)

Set the RSIZEERR bit to 1 to clear the RXSR.RSIZEERR flag.

ECCERRS bit (Single Bit ECC Error Interrupt Flag Clear)

Set the ECCERRS bit to 1 to clear the RXSR.ECCERRS flag.

RXAKE bit (Acknowledge and Error Report Receive Interrupt Flag Clear)

Set the RXAKE bit to 1 to clear the RXSR.RXAKE flag.

66.2.16 RXIER : Receive Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x208

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	—	ECCERRS	—	RSIZEERR	NORESERR	PRTOERR	RXOVFERR	IBERR	CRCERR	WCERR	—	UNEXERR	ECCERRM	MLFERR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EXTEDET	RXACK	—	—	—	RXEOTP	—	RXRESP	—	—	—	—	—	TATO	LRXHTO	BTAREND
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BTAREND	BTA Request End Interrupt Enable 0: Disable 1: Enable	R/W
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Enable 0: Disable 1: Enable	R/W
2	TATO	Turnaround Acknowledge Timeout Interrupt Enable 0: Disable 1: Enable	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	RXRESP	Response Packet Receive Interrupt Enable 0: Disable 1: Enable	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	RXEOTP	EoTp Receive Interrupt Enable 0: Disable 1: Enable	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	RXACK	ACK Trigger Receive Interrupt Enable 0: Disable 1: Enable	R/W
15	EXTEDET	External Tearing Effect Detect Interrupt Enable 0: Disable 1: Enable	R/W
16	MLFERR	Malform Error Interrupt Enable 0: Disable 1: Enable	R/W
17	ECCERRM	Multi-bit ECC Error Interrupt Enable 0: Disable 1: Enable	R/W
18	UNEXERR	Unexpected Packet Error Interrupt Enable 0: Disable 1: Enable	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	WCERR	Word Count Error Interrupt Enable 0: Disable 1: Enable	R/W

Bit	Symbol	Function	R/W
21	CRCERR	CRC Error Interrupt Enable 0: Disable 1: Enable	R/W
22	IBERR	Internal Bus Error Interrupt Enable 0: Disable 1: Enable	R/W
23	RXOVFERR	Receive Buffer Overflow Error Interrupt Enable 0: Disable 1: Enable	R/W
24	PRTOERR	Peripheral Response Timeout Error Interrupt Enable 0: Disable 1: Enable	R/W
25	NORESERR	No Response Error Interrupt Enable 0: Disable 1: Enable	R/W
26	RSIZEERR	Return Packet Size Error Interrupt Enable 0: Disable 1: Enable	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
28	ECCERRS	Single Bit ECC Error Interrupt Enable 0: Disable 1: Enable	R/W
29	—	This bit is read as 0. The write value should be 0.	R/W
30	RXAKE	Acknowledge and Error Report Receive Interrupt Enable 0: Disable 1: Enable	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

BTAREND bit (BTA Request End Interrupt Enable)

The BTAREND bit controls BTA Request End Interrupt permission. When a BTA Request End Interrupt occurs, the RXSR.BTAREND flag is set to 1.

To enable BTA Request End Interrupt, set the BTAREND bit to 1.

LRXHTO bit (LP-RX Host Processor Timeout Interrupt Enable)

The LRXHTO bit controls LP-RX Host Processor Timeout Interrupt permission. When an LP-RX Host Processor Timeout Interrupt occurs, the RXSR.LRXHTO flag is set to 1.

To enable LP-RX Host Processor Timeout Interrupt, set the LRXHTO bit to 1.

TATO bit (Turnaround Acknowledge Timeout Interrupt Enable)

The TATO bit controls Turnaround Acknowledge Timeout Interrupt permission. When a Turnaround Acknowledge Timeout Interrupt occurs, the RXSR.TATO flag is set to 1.

To enable Turnaround Acknowledge Timeout Interrupt, set the TATO bit to 1.

RXRESP bit (Response Packet Receive Interrupt Enable)

The RXRESP bit controls Response Packet Receive Interrupt permission. When a Response Packet Receive Interrupt occurs, the RXSR.RXRESP flag is set to 1.

To enable Response Packet Receive Interrupt, set the RXRESP bit to 1.

RXEOTP bit (EoTp Receive Interrupt Enable)

The RXEOTP bit controls EoTp Receive Interrupt permission. When an EoTp Receive Interrupt occurs, the RXSR.RXEOTP flag is set to 1.

To enable EoTp Receive Interrupt, set the RXEOTP bit to 1.

RXACK bit (ACK Trigger Receive Interrupt Enable)

The RXACK bit controls ACK Trigger Receive Interrupt permission. When an ACK Trigger Receive Interrupt occurs, the RXSR.RXACK flag is set to 1.

To enable ACK Trigger Receive Interrupt, set the RXACK bit to 1.

EXTEDET bit (External Tearing Effect Detect Interrupt Enable)

The EXTEDET bit controls External Tearing Effect Detect Interrupt permission. When an External Tearing Effect Detect Interrupt occurs, the RXSR.EXTEDET flag is set to 1.

To enable External Tearing Effect Detect Interrupt, set the EXTEDET bit to 1.

MLFERR bit (Malform Error Interrupt Enable)

The MLFERR bit controls Malform Error Interrupt permission. When a Malform Error Interrupt occurs, the RXSR.MLFERR flag is set to 1.

To enable Malform Error Interrupt, set the MLFERR bit to 1.

ECCERRM bit (Multi-bit ECC Error Interrupt Enable)

The ECCERRM bit controls Multi-bit ECC Error Interrupt permission. When a Multi-bit ECC Error Interrupt occurs, the RXSR.ECCERRM flag is set to 1.

To enable Multi Bit ECC Error Interrupt, set the ECCERRM bit to 1.

UNEXERR bit (Unexpected Packet Error Interrupt Enable)

The UNEXERR bit controls Unexpected Packet Error Interrupt permission. When an Unexpected Packet Error Interrupt occurs, the RXSR.UNEXERR flag is set to 1.

To enable Unexpected Packet Error Interrupt, set the UNEXERR bit to 1.

WCERR bit (Word Count Error Interrupt Enable)

The WCERR bit controls Word Count Error Interrupt permission. When a Word Count Error Interrupt occurs, the RXSR.WCERR flag is set to 1.

To enable Word Count Error Interrupt, set the WCERR bit to 1.

CRCERR bit (CRC Error Interrupt Enable)

The CRCERR bit controls CRC Error Interrupt permission. When a CRC Error Interrupt occurs, the RXSR.CRCERR flag is set to 1.

To enable CRC Error Interrupt, set the CRCERR bit to 1.

IBERR bit (Internal Bus Error Interrupt Enable)

The IBERR bit controls Internal Bus Error Interrupt permission. When an Internal Bus Error Interrupt occurs, the RXSR.IBERR flag is set to 1.

To enable Internal Bus Error Interrupt, set the IBERR bit to 1.

RXOVFERR bit (Receive Buffer Overflow Error Interrupt Enable)

The RXOVFERR bit controls Receive Buffer Overflow Error Interrupt permission. When a Receive Buffer Overflow Error Interrupt occurs, the RXSR.RXOVFERR flag is set to 1.

To enable Receive Buffer Overflow Error Interrupt, set the RXOVFERR bit to 1.

PRTOERR bit (Peripheral Response Timeout Error Interrupt Enable)

The PRTOERR bit controls Peripheral Response Timeout Error Interrupt permission. When a Peripheral Response Timeout Error Interrupt occurs, the RXSR.PRTOERR flag is set to 1.

To enable Peripheral Response Timeout Error Interrupt, set the PRTOERR bit to 1.

NORESERR bit (No Response Error Interrupt Enable)

The NORESERR bit controls No Response Error Interrupt permission. When an No Response Error Interrupt occurs, the RXSR.NORESERR flag is set to 1.

To enable No Response Error Interrupt, set the NORESERR bit to 1.

RSIZEERR bit (Return Packet Size Error Interrupt Enable)

The RSIZEERR bit controls Return Packet Size Error Interrupt permission. When a Return Packet Size Error Interrupt occurs, the RXSR.RSIZEERR flag is set to 1.

To enable Return Packet Size Error Interrupt, set the RSIZEERR bit to 1.

ECCERRS bit (Single Bit ECC Error Interrupt Enable)

The ECCERRS bit controls Single Bit ECC Error Interrupt permission. When a Single Bit ECC Error Interrupt occurs, the RXSR.ECCERRS flag is set to 1.

To enable Single Bit ECC Error Interrupt, set the ECCERRS bit to 1.

RXAKE bit (Acknowledge and Error Report Receive Interrupt Enable)

The RXAKE bit controls Acknowledge and Error Report Receive Interrupt permission. When an Acknowledge and Error Report Receive Interrupt occurs, the RXSR.RXAKE flag is set to 1.

To enable Acknowledge and Error Report Receive Interrupt, set the RXAKE bit to 1.

66.2.17 PRESPTOBTASETR : Peripheral Response Timeout BTA Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x210

Bit position:	31 … 0
Bit field:	PRTBTA[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	PRTBTA[31:0]	Peripheral Response Timeout Count Set the Peripheral Response Timeout value for Bus Turn-Around (BTA). If the setting value is 0, timeout is not detected.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

The PRTBTA register defines the limit to wait for a peripheral response after Bus Turn-Around (BTA) executed by settings SQCHnDSCmAR.BTA[1:0] = 11b.

Set the peripheral response timeout value from the time of entering LP-RX mode until the peripheral responds to a BTA.

The timeout value is calculated by the following formula:

Time [μs] = PRTBTA[31:0] $\times$ (1 / f_{LPCLK} [MHz]^{*1})

When a timeout is detected, the RXSR.PRTOERR bit is set to 1.

Note 1. f_{LPCLK} is the frequency of Low-Power mode clock (LPCLK).

66.2.18 PRESPTOLPSETR : Peripheral Response Timeout LP Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x214

Bit position:	31 … 16	15 … 0
Bit field:	LPRTO[15:0]	LPWTO[15:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	LPWTO[15:0]	LPDT WRITE Request Timeout Set the Peripheral Response Timeout value for LPDT WRITE Request. If the setting value is 0, timeout is not detected.	R/W
31:16	LPRTO[15:0]	LPDT READ Request Timeout Set the Peripheral Response Timeout value for LPDT READ Request. If the setting value is 0, timeout is not detected.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

LPWTO[15:0] bits (LPDT WRITE Request Timeout)

The LPWTO[15:0] bits define the limit to wait for a peripheral response after Bus Turn-Around (BTA) executed by settings SQCHnDSCmAR.BTA[1:0] = 01b and SQCHnDSCmAR.SPD = 1.

Set the peripheral response timeout value from the time of entering LP-RX mode until the peripheral responds to a write request.

The timeout value is calculated by the following formula:

Time [μs] = LPWTO[15:0] × (1 / f_{LPCLK} [MHz]^{*1})

When a timeout is detected, the RXSR.PRTOERR bit is set to 1.

LPRTO[15:0] bits (LPDT READ Request Timeout)

The LPRTO[15:0] bits define the limit to wait for a peripheral response after Bus Turn-Around (BTA) executed by settings SQCHnDSCmAR.BTA[1:0] = 10b and SQCHnDSCmAR.SPD = 1.

Set the peripheral response timeout value from the time of entering LP-RX mode until the peripheral responds to a read request.

The timeout value is calculated by the following formula:

Time [μs] = LPRTO[15:0] × (1 / f_{LPCLK} [MHz]^{*1})

When a timeout is detected, the RXSR.PRTOERR bit is set to 1.

Note 1. f_{LPCLK} is the frequency of Low-Power mode clock (LPCLK).

66.2.19 PRESPTOHSSETR : Peripheral Response Timeout HS Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x218

Bit position:	31 … 16	15 … 0
Bit field:	HSRTO[15:0]	HSWTO[15:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	HSWTO[15:0]	HS WRITE Request Timeout Set the Peripheral Response Timeout value for HS write request. If the setting value is 0, timeout is not detected.	R/W
31:16	HSRTO[15:0]	HS READ Request Timeout Set the Peripheral Response Timeout value for HS read request. If the setting value is 0, timeout is not detected.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

HSWTO[15:0] bits (HS WRITE Request Timeout)

The HSWTO[15:0] bits define the limit to wait for a peripheral response after Bus Turn-Around (BTA) executed by settings SQCHnDSCmAR.BTA[1:0] = 01b and SQCHnDSCmAR.SPD = 0.

Set the peripheral response timeout value from the time of entering LP-RX mode until the peripheral responds to a write request.

The timeout value is calculated by the following formula:

Time [µs] = HSWTO[15:0] × (1 / f_{LPCLK} [MHz]*1)

When a timeout is detected, the RXSR.PRTOERR bit is set to 1.

HSRTO[15:0] bits (HS READ Request Timeout)

The HSRTO[15:0] bits define the limit to wait for a peripheral response after Bus Turn-Around (BTA) executed by settings SQCHnDSCmAR.BTA[1:0] = 10b and SQCHnDSCmAR.SPD = 0.

Set the peripheral response timeout value from the time of entering LP-RX mode until the peripheral responds to a read request.

The timeout value is calculated by the following formula:

Time [µs] = HSRTO[15:0] × (1 / f_{LPCLK} [MHz]*1)

When a timeout is detected, the RXSR.PRTOERR bit is set to 1.

Note 1. f_{LPCLK} is the frequency of Low-Power mode clock (LPCLK).

66.2.20 AKEPLATIR : Acknowledge and Error Report Packet Parameter Latest Info Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x220

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VC[3:0]				EREP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EREP[15:0]	Error Report Received Error Report bits [15:0]	R
19:16	VC[3:0]	Virtual Channel ID An Acknowledge and Error Report was received from the following Virtual Channel ID. 0x0: No Error Report received 0x1: From VC-0 0x2: From VC-1 0x4: From VC-2 0x8: From VC-3 Others: Reserved	R
31:20	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

EREP[15:0] bits (Error Report)

When an Acknowledge and Error Report packet is received from the peripheral device, the Error Report is stored in the EREP[15:0] bits.

VC[3:0] bits (Virtual Channel ID)

The VC[3:0] bits are indicated from which virtual channel the Error Report stored in the EREP[15:0] bits was received.

66.2.21 AKEPACMSR : Acknowledge and Error Report Packet Parameter Accumulate Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x224

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	AVC3	AVC2	AVC1	AVC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AEREP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	AEREP[15:0]	Accumulated Error Report Accumulated Error Report bits [15:0]	R
16	AVC0	Virtual Channel-0 Accumulated Information 0: No Error Report received from VC-0 1: Received an Acknowledge and Error Report from VC-0	R
17	AVC1	Virtual Channel-1 Accumulated Information 0: No Error Report received from VC-1 1: Received an Acknowledge and Error Report from VC-1	R
18	AVC2	Virtual Channel-2 Accumulated Information 0: No Error Report received from VC-2 1: Received an Acknowledge and Error Report from VC-2	R
19	AVC3	Virtual Channel-3 Accumulated Information 0: No Error Report received from VC-3 1: Received an Acknowledge and Error Report from VC-3	R
31:20	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

AEREP[15:0] bits (Accumulated Error Report)

The AKEPLATIR.EREP[15:0] bits contain the most recently received Error Report, while the AEREP[15:0] bits are the accumulated value of past Error Reports.

AVC0 bit (Virtual Channel-0 Accumulated Information)

The AKEPLATIR.VC[3:0] bits contain the virtual channel ID of the most recently received Error Report, while the AVC0 bit is the accumulated value received from Virtual Channel-0.

If the AVC0 bit is set to 1, it means that an Acknowledge and Error Report from Virtual Channel-0 has been received one or more times.

AVC1 bit (Virtual Channel-1 Accumulated Information)

The AKEPLATIR.VC[3:0] bits contain the virtual channel ID of the most recently received Error Report, while the AVC1 bit is the accumulated value received from Virtual Channel-1.

If the AVC1 bit is set to 1, it means that an Acknowledge and Error Report from Virtual Channel-1 has been received one or more times.

AVC2 bit (Virtual Channel-2 Accumulated Information)

The AKEPLATIR.VC[3:0] bits contain the virtual channel ID of the most recently received Error Report, while the AVC2 bit is the accumulated value received from Virtual Channel-2.

If the AVC2 bit is set to 1, it means that an Acknowledge and Error Report from Virtual Channel-2 has been received one or more times.

AVC3 bit (Virtual Channel-3 Accumulated Information)

The AKEPLATIR.VC[3:0] bits contain the virtual channel ID of the most recently received Error Report, while the AVC3 bit is the accumulated value received from Virtual Channel-3.

If the AVC3 bit is set to 1, it means that an Acknowledge and Error Report from Virtual Channel-3 has been received one or more times.

66.2.22 AKEPSCR : Acknowledge and Error Report Packet Parameter Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x228

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	AVC3	AVC2	AVC1	AVC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AEREP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	AEREP[15:0]	Accumulated Error Report Clear 0: No operation 1: Clear the AKEPACMSR.AEREP[15:0] bits	W
16	AVC0	Virtual Channel-0 Accumulated Information Clear 0: No operation 1: Clear the AKEPACMSR.AVC0 bit	W
17	AVC1	Virtual Channel-1 Accumulated Information Clear 0: No operation 1: Clear the AKEPACMSR.AVC1 bit	W
18	AVC2	Virtual Channel-2 Accumulated Information Clear 0: No operation 1: Clear the AKEPACMSR.AVC2 bit	W
19	AVC3	Virtual Channel-3 Accumulated Information Clear 0: No operation 1: Clear the AKEPACMSR.AVC3 bit	W
31:20	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

AEREP[15:0] bits (Accumulated Error Report Clear)

Set the AEREP[15:0] bits to 1 to clear the AKEPACMSR.AEREP[15:0] bits.

Each of the AEREP[15:0] bits correspond to each of the AKEPACMSR.AEREP[15:0] bits to be cleared.

AVC0 bit (Virtual Channel-0 Accumulated Information Clear)

Set the AVC0 bit to 1 to clear the AKEPACMSR.AVC0 bit.

AVC1 bit (Virtual Channel-1 Accumulated Information Clear)

Set the AVC1 bit to 1 to clear the AKEPACMSR.AVC1 bit.

AVC2 bit (Virtual Channel-2 Accumulated Information Clear)

Set the AVC2 bit to 1 to clear the AKEPACMSR.AVC2 bit.

AVC3 bit (Virtual Channel-3 Accumulated Information Clear)

Set the AVC3 bit to 1 to clear the AKEPACMSR.AVC3 bit.

66.2.23 RXRSSR : Receive Result Saved Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x230

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SLT3VLD	SLT2VLD	SLT1VLD	SLT0VLD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SLT0VLD	Slot-0 Valid Flag 0: None received 1: Response packet is received and stored in the RXRSS0R register	R
1	SLT1VLD	Slot-1 Valid Flag 0: None received 1: Response packet is received and stored in the RXRSS1R register	R
2	SLT2VLD	Slot-2 Valid Flag 0: None received 1: Response packet is received and stored in the RXRSS2R register	R
3	SLT3VLD	Slot-3 Valid Flag 0: None received 1: Response packet is received and stored in the RXRSS3R register	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

SLT0VLD bit (Slot-0 Valid Flag)

The SLT0VLD flag indicates that the response packet is received and stored in the RXRSS0R register.

The SLT0VLD flag is set to 1 if a response packet is received when the SQCHnDSCmCR.ACTCODE[7:0] bits are set to 0x00b.

The SLT0VLD flag is not cleared automatically, so set the RXRSSCR.SLT0VLD bit to 1 to clear the SLT0VLD flag.

SLT1VLD bit (Slot-1 Valid Flag)

The SLT1VLD flag indicates that the response packet is received and stored in the RXRSS1R register.

The SLT1VLD flag is set to 1 if a response packet is received when the SQCHnDSCmCR.ACTCODE[7:0] bits are set to 0x01.

The SLT1VLD flag is not cleared automatically, so set the RXRSSCR.SLT1VLD bit to 1 to clear the SLT1VLD flag.

SLT2VLD bit (Slot-2 Valid Flag)

The SLT2VLD flag indicates that the response packet is received and stored in the RXRSS2R register.

The SLT2VLD flag is set to 1 if a response packet is received when the SQCHnDSCmCR.ACTCODE[7:0] bits are set to 0x02.

The SLT2VLD flag is not cleared automatically, so set the RXRSSCR.SLT2VLD bit to 1 to clear the SLT2VLD flag.

SLT3VLD bit (Slot-3 Valid Flag)

The SLT3VLD flag indicates that the response packet is received and stored in the RXRSS3R register.

The SLT3VLD flag is set to 1 if a response packet is received when the SQCHnDSCmCR.ACTCODE[7:0] bits are set to 0x03.

The SLT3VLD flag is not cleared automatically, so set the RXRSSCR.SLT3VLD bit to 1 to clear the SLT3VLD flag.

66.2.24 RXRSSCR : Receive Result Saved Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x234

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SLT3VLD	SLT2VLD	SLT1VLD	SLT0VLD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SLT0VLD	Slot-0 Valid Flag Clear 0: No operation 1: Clear the RXRSSR.SLT0VLD flag	W
1	SLT1VLD	Slot-1 Valid Flag Clear 0: No operation 1: Clear the RXRSSR.SLT1VLD flag	W
2	SLT2VLD	Slot-2 Valid Flag Clear 0: No operation 1: Clear the RXRSSR.SLT2VLD flag	W
3	SLT3VLD	Slot-3 Valid Flag Clear 0: No operation 1: Clear the RXRSSR.SLT3VLD flag	W
31:4	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

SLT0VLD bit (Slot-0 Valid Flag Clear)

Set the SLT0VLD bit to 1 to clear the RXRSSR.SLT0VLD flag.

SLT1VLD bit (Slot-1 Valid Flag Clear)

Set the SLT1VLD bit to 1 to clear the RXRSSR.SLT1VLD flag.

SLT2VLD bit (Slot-2 Valid Flag Clear)

Set the SLT2VLD bit to 1 to clear the RXRSSR.SLT2VLD flag.

SLT3VLD bit (Slot-3 Valid Flag Clear)

Set the SLT3VLD bit to 1 to clear the RXRSSR.SLT3VLD flag.

66.2.25 RXRINFOOWSR : Receive Result Info Overwrite Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x238

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SL3OW	SL2OW	SL1OW	SL0OW
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SL0OW	Slot-0 Information Overwrite Flag 0: No overwritten 1: Slot-0 information overwritten	R
1	SL1OW	Slot-1 Information Overwrite Flag 0: No overwritten 1: Slot-1 information overwritten	R
2	SL2OW	Slot-2 Information Overwrite Flag 0: No overwritten 1: Slot-2 information overwritten	R
3	SL3OW	Slot-3 Information Overwrite Flag 0: No overwritten 1: Slot-3 information overwritten	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

SL0OW bit (Slot-0 Information Overwrite Flag)

The SL0OW flag is copy of the RXRSS0R.INFOOW flag. The meaning is also the same. For details, see section 66.2.27. RXRSSxR : Receive Result Save Slot-x Register (x = 0 to 3).

SL1OW bit (Slot-1 Information Overwrite Flag)

The SL1OW flag is copy of the RXRSS1R.INFOOW flag. The meaning is also the same. For details, see section 66.2.27. RXRSSxR : Receive Result Save Slot-x Register (x = 0 to 3).

SL2OW bit (Slot-2 Information Overwrite Flag)

The SL2OW flag is copy of the RXRSS2R.INFOOW flag. The meaning is also the same. For details, see section 66.2.27. RXRSSxR : Receive Result Save Slot-x Register (x = 0 to 3).

SL3OW bit (Slot-3 Information Overwrite Flag)

The SL3OW flag is copy of the RXRSS3R.INFOOW flag. The meaning is also the same. For details, see section 66.2.27. RXRSSxR : Receive Result Save Slot-x Register (x = 0 to 3).

66.2.26 RXRINFOOWSCR : Receive Result Info Overwrite Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x23C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SL3OW	SL2OW	SL1OW	SL0OW
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SL0OW	Slot-0 Information Overwrite Flag Clear 0: No operation 1: Clear the RXRINFOOWSR.SL0OW flag	W
1	SL1OW	Slot-1 Information Overwrite Flag Clear 0: No operation 1: Clear the RXRINFOOWSR.SL1OW flag	W
2	SL2OW	Slot-2 Information Overwrite Flag Clear 0: No operation 1: Clear the RXRINFOOWSR.SL2OW flag	W
3	SL3OW	Slot-3 Information Overwrite Flag Clear 0: No operation 1: Clear the RXRINFOOWSR.SL3OW flag	W
31:4	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

SL0OW bit (Slot-0 Information Overwrite Flag Clear)

Set the SL0OW bit to 1 to clear the RXRINFOOWSR.SL0OW flag (the RXRSS0R.INFOOW bit).

SL1OW bit (Slot-1 Information Overwrite Flag Clear)

Set the SL1OW bit to 1 to clear the RXRINFOOWSR.SL1OW flag (the RXRSS1R.INFOOW bit).

SL2OW bit (Slot-2 Information Overwrite Flag Clear)

Set the SL2OW bit to 1 to clear the RXRINFOOWSR.SL2OW flag (the RXRSS2R.INFOOW bit).

SL3OW bit (Slot-3 Information Overwrite Flag Clear)

Set the SL3OW bit to 1 to clear the RXRINFOOWSR.SL3OW flag (the RXRSS3R.INFOOW bit).

66.2.27 RXRSSxR : Receive Result Save Slot-x Register (x = 0 to 3)

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x240 + 0x4 × x

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	INFOOW	RXAKE	RXCERR	RXPFAIL	RXFAIL	RXFERR	RXSUC	FMT	VC[1:0]		DT[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA1[7:0]								DATA0[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	DATA0[7:0]	Data 0 Data 0 of received packet header	R
15:8	DATA1[7:0]	Data 1 Data 1 of received packet header	R
21:16	DT[5:0]	Data Type	R
23:22	VC[1:0]	Virtual Channel ID	R
24	FMT	Packet Format 0: Short packet 1: Long packet	R
25	RXSUC	Receive Success 0: No received 1: Response packet or ACK trigger received	R
26	RXFERR	Fatal Error 0: No fatal error 1: Fatal timeout occurred during BTA	R
27	RXFAIL	Receive Fail 0: No error 1: Expected receive did not occur	R
28	RXPFAIL	Receive Packet Data Fail 0: No error 1: Payload data not saved correctly	R
29	RXCERR	Receive Correctable Error 0: No error 1: Correctable error detected	R
30	RXAKE	Receive Acknowledge and Error Report Packet 0: No received 1: An Acknowledge and Error Report packet received	R
31	INFOOW	Information Overwrite 0: No update 1: This register information (RXRSSxR[30:0]) was overwritten	R

Note: S-TYPE-3, P-TYPE-3

DATA0[7:0] bits (Data 0)

Data 0 of the received packet header is stored in the DATA0[7:0] bits.

When the received packet is a long packet, the lower 8 bits of the word count are stored in the DATA0[7:0] bits.

The DATA0[7:0] bits are valid when the RXSUC bit is 1 and the DT[5:0] bits are not 0x00.

DATA1[7:0] bits (Data 1)

Data 1 of the received packet header is stored in the DATA1[7:0] bits.

When the received packet is a long packet, the upper 8 bits of the word count are stored in the DATA1[7:0] bits.

The DATA1[7:0] bits are valid when the RXSUC bit is 1 and the DT[5:0] bits are not 0x00.

DT[5:0] bits (Data Type)

The DT[5:0] bits indicate the data type of the received packet header.

When an ACK trigger is received, the DT[5:0] bits are set to 0x00.

The DT[5:0] bits are valid when the RXSUC bit is 1.

VC[1:0] bits (Virtual Channel ID)

The VC[1:0] bits indicate the Virtual Channel ID of the received packet header.

The VC[1:0] bits are valid when the RXSUC bit is 1 and the DT[5:0] bits are not 0x00.

FMT bit (Packet Format)

The FMT bit indicates the packet format of the received packet header.

The FMT bit is valid when the RXSUC bit is 1 and the DT[5:0] bits are not 0x00. It is also valid when the RXPFAIL bit is set to 1.

RXSUC bit (Receive Success)

The RXSUC bit indicates that a response packet or ACK trigger is received.

The RXSR.RXRESP flag or the RXSR.RXACK flag is also set to 1.

RXFERR bit (Fatal Error)

The RXFERR bit indicates that a fatal timeout occurred during BTA.

The FERRSR.TATO flag or the FERRSR.LRXHTO flag is also set to 1.

RXFAIL bit (Receive Fail)

The RXFAIL bit indicates that an expected receive did not occur.

One or more of the following flags are also set to 1.

- RXSR.PRTOERR
- RXSR.ECCERRM
- RXSR.MLFERR
- RXSR.NORESERR

RXPFAIL bit (Receive Packet Data Fail)

The RXPFAIL bit indicates that the packet header is saved correctly but payload data was not saved correctly.

One or more of the following flags are also set to 1.

- RXSR.CRCERR
- RXSR.WCERR
- RXSR.RSIZEERR
- RXSR.UNEXERR
- RXSR.RXOVFERR
- RXSR.IBERR

When the RXPFAIL bit and the RXSR.RXRESP bit are set to 1, it means that the packet received during BTA is redundant.

RXCERR bit (Receive Correctable Error)

The RXCERR bit indicates that the received packet has a correctable error.

The RXSR.ECCERRS flag is also set to 1.

RXAKE bit (Receive Acknowledge and Error Report Packet)

The RXAKE bit indicates that an Acknowledge and Error Report packet is received.

The RXSR.RXAKE flag is also set to 1.

INFOOW bit (Information Overwrite)

The INFOOW bit indicates that this register information (RXRSSxR[30:0]) is updated when RXRSSR.SLTxVLD = 1.

Set the RXRINFOOWSCR.SLxOW bit to 1 to clear the INFOOW bit.

66.2.28 RXPPD0R : Receive Packet Payload Data 0 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2C0

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA3[7:0]	DATA2[7:0]	DATA1[7:0]	DATA0[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA0[7:0]	Payload Data 0 Long packet payload data 0 received in Command mode[*1]	R
15:8	DATA1[7:0]	Payload Data 1 Long packet payload data 1 received in Command mode[*1]	R
23:16	DATA2[7:0]	Payload Data 2 Long packet payload data 2 received in Command mode[*1]	R
31:24	DATA3[7:0]	Payload Data 3 Long packet payload data 3 received in Command mode[*1]	R

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when SQCHnDSCmBR.DTSEL[1:0] = 00b.

DATA0[7:0] bits (Payload Data 0)

The DATA0[7:0] bits are used to store the long packet payload data 0 to be received in Command mode using Sequence operation.

DATA1[7:0] bits (Payload Data 1)

The DATA1[7:0] bits are used to store the long packet payload data 1 to be received in Command mode using Sequence operation.

DATA2[7:0] bits (Payload Data 2)

The DATA2[7:0] bits are used to store the long packet payload data 2 to be received in Command mode using Sequence operation.

DATA3[7:0] bits (Payload Data 3)

The DATA3[7:0] bits are used to store the long packet payload data 3 to be received in Command mode using Sequence operation.

66.2.29 RXPPD1R : Receive Packet Payload Data 1 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2C4

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA7[7:0]	DATA6[7:0]	DATA5[7:0]	DATA4[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA4[7:0]	Payload Data 4 Long packet payload data 4 received in Command mode*1	R
15:8	DATA5[7:0]	Payload Data 5 Long packet payload data 5 received in Command mode*1	R
23:16	DATA6[7:0]	Payload Data 6 Long packet payload data 6 received in Command mode*1	R
31:24	DATA7[7:0]	Payload Data 7 Long packet payload data 7 received in Command mode*1	R

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when SQCHnDSCmBR.DTSEL[1:0] = 00b.

DATA4[7:0] bits (Payload Data 4)

The DATA4[7:0] bits are used to store the long packet payload data 4 to be received in Command mode using Sequence operation.

DATA5[7:0] bits (Payload Data 5)

The DATA5[7:0] bits are used to store the long packet payload data 5 to be received in Command mode using Sequence operation.

DATA6[7:0] bits (Payload Data 6)

The DATA6[7:0] bits are used to store the long packet payload data 6 to be received in Command mode using Sequence operation.

DATA7[7:0] bits (Payload Data 7)

The DATA7[7:0] bits are used to store the long packet payload data 7 to be received in Command mode using Sequence operation.

66.2.30 RXPPD2R : Receive Packet Payload Data 2 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2C8

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA11[7:0]	DATA10[7:0]	DATA9[7:0]	DATA8[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA8[7:0]	Payload Data 8 Long packet payload data 8 received in Command mode*1	R
15:8	DATA9[7:0]	Payload Data 9 Long packet payload data 9 received in Command mode*1	R

Bit	Symbol	Function	R/W
23:16	DATA10[7:0]	Payload Data 10 Long packet payload data 10 received in Command mode*1	R
31:24	DATA11[7:0]	Payload Data 11 Long packet payload data 11 received in Command mode*1	R

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b.

DATA8[7:0] bits (Payload Data 8)

The DATA8[7:0] bits are used to store the long packet payload data 8 to be received in Command mode using Sequence operation.

DATA9[7:0] bits (Payload Data 9)

The DATA9[7:0] bits are used to store the long packet payload data 9 to be received in Command mode using Sequence operation.

DATA10[7:0] bits (Payload Data 10)

The DATA10[7:0] bits are used to store the long packet payload data 10 to be received in Command mode using Sequence operation.

DATA11[7:0] bits (Payload Data 11)

The DATA11[7:0] bits are used to store the long packet payload data 11 to be received in Command mode using Sequence operation.

66.2.31 RXPPD3R : Receive Packet Payload Data 3 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2CC

Bit position:	31–24	23–16	15–8	7–0
Bit field:	DATA15[7:0]	DATA14[7:0]	DATA13[7:0]	DATA12[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DATA12[7:0]	Payload Data 12 Long packet payload data 12 received in Command mode*1	R
15:8	DATA13[7:0]	Payload Data 13 Long packet payload data 13 received in Command mode*1	R
23:16	DATA14[7:0]	Payload Data 14 Long packet payload data 14 received in Command mode*1	R
31:24	DATA15[7:0]	Payload Data 15 Long packet payload data 15 received in Command mode*1	R

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are valid when SQCHnDSCmBR.DTSEL[1:0] = 00b.

DATA12[7:0] bits (Payload Data 12)

The DATA12[7:0] bits are used to store the long packet payload data 12 to be received in Command mode using Sequence operation.

DATA13[7:0] bits (Payload Data 13)

The DATA13[7:0] bits are used to store the long packet payload data 13 to be received in Command mode using Sequence operation.

DATA14[7:0] bits (Payload Data 14)

The DATA14[7:0] bits are used to store the long packet payload data 14 to be received in Command mode using Sequence operation.

DATA15[7:0] bits (Payload Data 15)

The DATA15[7:0] bits are used to store the long packet payload data 15 to be received in Command mode using Sequence operation.

66.2.32 HSTXTOSETR : HS TX Timeout Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2E0

Bit position:	31 … 0
Bit field:	HTXTO[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	HTXTO[31:0]	HS TX Timeout Count Set HS TX Timeout (HTX_TO) value.*1	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. If the setting value is 0, timeout is not detected.

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

The HSTXTO register defines the limit to wait for the length of DSI Host of HS transmission.

The timeout value is calculated by the following formula:

Time [µs] = HSTXTOSETR[31:0] × 32 × (period of High-Speed serial UI [ns]) / 1000

When a timeout is detected, the FERRSR.HTXTO bit is set to 1.

66.2.33 LRXHTOSETR : LRX-H Timeout Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2E4

Bit position:	31 … 0
Bit field:	LRXHTO[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	LRXHTO[31:0]	LP-RX Host Processor Timeout Set LP-RX Timeout (LRX-H_TO) value. If the setting value is 0, timeout is not detected.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

The LRXHTO register defines the limit to wait for transmission from the peripheral. If a LP TX-Peripheral Timeout (LTX-P_TO) occurs on the peripheral side, then this LP-RX timeout occurs on DSI Host.

The timeout value is calculated by the following formula:

Time [µs] = LRXHTOSETR[31:0] × (1 / f_{LPCLK} [MHz]*1)

When a timeout is detected, the FERRSR.LRXHTO bit is set to 1.

Note 1. f_{LPCLK} is the frequency of Low-Power mode clock (LPCLK).

66.2.34 TATOSETR : TA Timeout Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x2E8

Bit position:	31 … 0
Bit field:	TATO[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	TATO[31:0]	Turnaround Acknowledge Timeout Set Turnaround Acknowledge Timeout (TA_TO) value. If the setting value is 0, timeout is not detected.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

The TATO register defines the limit to wait for the data lane transition sequence processing after Bus Turn-Around (BTA) is issued.

The timeout value is calculated by the following formula:

Time [µs] = TATOSETR[31:0] × (1 / f_{LPCLK} [MHz]^{*1})

When a timeout is detected, the FERRSR.TATO bit is set to 1.

Note 1. f_{LPCLK} is the frequency of Low-Power mode clock (LPCLK).

66.2.35 FERRSR : Fatal Error Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x300

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CLP1S	CLP0S	—	—	—	—	—	—	CLP1	CLP0	CTRL	SYNCESC	ESCENT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	TATO	LRXHTO	HTXTO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HTXTO	HS TX Timeout Interrupt Flag 0: Not detected 1: HS TX Timeout (HTX_TO) detected	R
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Flag 0: Not detected 1: LP-RX Host Processor Timeout (LRX-H_TO) detected	R
2	TATO	Turnaround Acknowledge Timeout Interrupt Flag 0: Not detected 1: Turnaround Acknowledge Timeout (TA_TO) detected	R
15:3	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
16	ESCENT	Escape mode Entry Error Interrupt Flag 0: No error detected 1: Escape mode Entry Error (ErrEsc) detected	R
17	SYNCESC	LPDT Sync Error Interrupt Flag 0: No error detected 1: LPDT Synchronization Error (ErrSyncEsc) detected	R
18	CTRL	Control Error Interrupt Flag 0: No error detected 1: Control Error (ErrControl) detected	R
19	CLP0	LP0 Contention Error Interrupt Flag 0: No error detected 1: LP0 Contention Error (ErrContentionLP0) detected	R
20	CLP1	LP1 Contention Error Interrupt Flag 0: No error detected 1: LP1 Contention Error (ErrContentionLP1) detected	R
26:21	—	These bits are read as 0.	R
27	CLP0S	LP0 Contention Error Status 0: Normal state 1: LP0 Contention Error (ErrContentionLP0) state	R
28	CLP1S	LP1 Contention Error Status 0: Normal state 1: LP1 Contention Error (ErrContentionLP1) state	R
31:29	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

HTXTO flag (HS TX Timeout Interrupt Flag)

The HTXTO flag indicates that a HS TX Timeout (HTX_TO) is detected.

The HTXTO flag is set to 1 when the HS TX Timeout Timer expires. This timer monitors the length of DSI Host from HS transmission.

The HTXTO flag is not cleared automatically, so set the FERRSCR.HTXTO bit to 1 to clear the HTXTO flag.

LRXHTO flag (LP-RX Host Processor Timeout Interrupt Flag)

The LRXHTO flag indicates that a LP-RX Host Processor Timeout (LRX-H_TO) is detected.

The LRXHTO flag is set to 1 when the Host Processor LP-RX Timer expires.

The LRXHTO flag is not cleared automatically, so set the FERRSCR.LRXHTO bit to 1 to clear the LRXHTO flag.

TATO flag (Turnaround Acknowledge Timeout Interrupt Flag)

The TATO flag indicates that a Turnaround Acknowledge Timeout (TA_TO) is detected.

The TATO flag is set to 1 when the timer TA_TO expires.

The TATO flag is not cleared automatically, so set the FERRSCR.TATO bit to 1 to clear the TATO flag.

ESCENT flag (Escape mode Entry Error Interrupt Flag)

The ESCENT flag indicates that an Escape mode Entry Error (ErrEsc) is detected.

The ESCENT flag is not cleared automatically, so set the FERRSCR.ESCENT bit to 1 to clear the ESCENT flag.

SYNCESC flag (LPDT Sync Error Interrupt Flag)

The SYNCESC flag indicates that a Low-Power Data Transmission Synchronization Error (ErrSyncEsc) is detected.

The SYNCESC flag is not cleared automatically, so set the FERRSCR.SYNCESC bit to 1 to clear the SYNCESC flag.

CTRL flag (Control Error Interrupt Flag)

The CTRL flag indicates that a Control Error (ErrControl) is detected.

The CTRL flag is not cleared automatically, so set the FERRSCR.CTRL bit to 1 to clear the CTRL flag.

CLP0 flag (LP0 Contention Error Interrupt Flag)

The CLP0 flag indicates that a LP0 Contention Error (ErrContentionLP0) is detected.

Once the CLP0S bit is detected to be 1, the CLP0 flag is set to 1.

The CLP0 flag is not cleared automatically, so set the FERRSCR.CLP0 bit to 1 to clear the CLP0 flag.

CLP1 flag (LP1 Contention Error Interrupt Flag)

The CLP1 flag indicates that a LP1 Contention Error (ErrContentionLP1) is detected.

Once the CLP1S bit is detected to be 1, the CLP1 flag is set to 1.

The CLP1 flag is not cleared automatically, so set the FERRSCR.CLP1 bit to 1 to clear the CLP1 flag.

CLP0S bit (LP0 Contention Error Status)

The CLP0S bit indicates the latest LP0 Contention Error state of the D-PHY.

The CLP0S bit is set to 1 when the D-PHY Lane module detects a contention situation on a line while trying to drive the line low.

CLP1S bit (LP1 Contention Error Status)

The CLP1S bit indicates the latest LP1 Contention Error state of the D-PHY.

The CLP1S bit is set to 1 when the D-PHY Lane module detects a contention situation on a line while trying to drive the line high.

66.2.36 FERRSCR : Fatal Error Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x304

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CLP1	CLP0	CTRL	SYNCESC	ESCENT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	TATO	LRXHTO	HTXTO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HTXTO	HS TX Timeout Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.HTXTO flag	W
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.LRXHTO flag	W
2	TATO	Turnaround Acknowledge Timeout Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.TATO flag	W
15:3	—	These bits are read as 0. The write value should be 0.	W
16	ESCENT	Escape Mode Entry Error Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.ESCENT flag	W
17	SYNCESC	LPDT Sync Error Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.SYNCESC flag	W

Bit	Symbol	Function	R/W
18	CTRL	Control Error Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.CTRL flag	W
19	CLP0	LP0 Contention Error Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.CLP0 flag	W
20	CLP1	LP1 Contention Error Interrupt Flag Clear 0: No operation 1: Clear the FERRSR.CLP1 flag	W
31:21	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

HTXTO bit (HS TX Timeout Interrupt Flag Clear)

Set the HTXTO bit to 1 to clear the FERRSR.HTXTO flag.

LRXHTO bit (LP-RX Host Processor Timeout Interrupt Flag Clear)

Set the LRXHTO bit to 1 to clear the FERRSR.LRXHTO flag.

TATO bit (Turnaround Acknowledge Timeout Interrupt Flag Clear)

Set the TATO bit to 1 to clear the FERRSR.TATO flag.

ESCENT bit (Escape Mode Entry Error Interrupt Flag Clear)

Set the ESCENT bit to 1 to clear the FERRSR.ESCENT flag.

SYNCESC bit (LPDT Sync Error Interrupt Flag Clear)

Set the SYNCESC bit to 1 to clear the FERRSR.SYNCESC flag.

CTRL bit (Control Error Interrupt Flag Clear)

Set the CTRL bit to 1 to clear the FERRSR.CTRL flag.

CLP0 bit (LP0 Contention Error Interrupt Flag Clear)

Set the CLP0 bit to 1 to clear the FERRSR.CLP0 flag.

CLP1 bit (LP1 Contention Error Interrupt Flag Clear)

Set the CLP1 bit to 1 to clear the FERRSR.CLP1 flag.

66.2.37 FERRIER : Fatal Error Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x308

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CLP1	CLP0	CTRL	SYNCESC	ESCENT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	TATO	LRXHTO	HTXTO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	HTXTO	HS TX Timeout Interrupt Enable 0: Disable 1: Enable	R/W
1	LRXHTO	LP-RX Host Processor Timeout Interrupt Enable 0: Disable 1: Enable	R/W
2	TATO	Turnaround Acknowledge Timeout Interrupt Enable 0: Disable 1: Enable	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	ESCENT	Escape mode Entry Error Interrupt Enable 0: Disable 1: Enable	R/W
17	SYNCESC	LPDT Sync Error Interrupt Enable 0: Disable 1: Enable	R/W
18	CTRL	Control Error Interrupt Enable 0: Disable 1: Enable	R/W
19	CLP0	LP0 Contention Error Interrupt Enable 0: Disable 1: Enable	R/W
20	CLP1	LP1 Contention Error Interrupt Enable 0: Disable 1: Enable	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HTXTO bit (HS TX Timeout Interrupt Enable)

The HTXTO bit controls the HS TX Timeout (HTX_TO) Interrupt permission. When a HS TX Timeout Interrupt occurs, the FERRSR.HTXTO flag is set to 1.

To enable the HS TX Timeout Interrupt, set the HTXTO bit to 1.

LRXHTO bit (LP-RX Host Processor Timeout Interrupt Enable)

The LRXHTO bit controls the LP-RX Host Processor Timeout (LRX-H_TO) Interrupt permission. When a LP-RX Host Processor Timeout Interrupt occurs, the FERRSR.LRXHTO flag is set to 1.

To enable the LP-RX Host Processor Timeout Interrupt, set the LRXHTO bit to 1.

TATO bit (Turnaround Acknowledge Timeout Interrupt Enable)

The TATO bit controls the Turnaround Acknowledge Timeout (TA_TO) Interrupt permission. When a Turnaround Acknowledge Timeout Interrupt occurs, the FERRSR.TATO flag is set to 1.

To enable the Turnaround Acknowledge Timeout Interrupt, set the TATO bit to 1.

ESCENT bit (Escape mode Entry Error Interrupt Enable)

The ESCENT bit controls the Escape mode Entry Error (ErrEsc) Interrupt permission. When an Escape mode Entry Error Interrupt occurs, the FERRSR.ESCENT flag is set to 1.

To enable the Escape mode Entry Error Interrupt, set the ESCENT bit to 1.

SYNCESC bit (LPDT Sync Error Interrupt Enable)

The SYNCESC bit controls the Low-Power Data Transmission Synchronization Error (ErrSyncEsc) Interrupt permission. When a Low-Power Data Transmission Synchronization Error Interrupt occurs, the FERRSR.SYNCESC flag is set to 1.

To enable the Low-Power Data Transmission Synchronization Error Interrupt, set the SYNCESC bit to 1.

CTRL bit (Control Error Interrupt Enable)

The CTRL bit controls the Control Error (ErrControl) Interrupt permission. When a Control Error Interrupt occurs, the FERRSR.CTRL flag is set to 1.

To enable the Control Error Interrupt, set the CTRL bit to 1.

CLP0 bit (LP0 Contention Error Interrupt Enable)

The CLP0 bit controls the LP0 Contention Error (ErrContentionLP0) Interrupt permission. When an LP0 Contention Error Interrupt occurs, the FERRSR.CLP0 flag is set to 1.

To enable the LP0 Contention Error Interrupt, set the CLP0 bit to 1.

CLP1 bit (LP1 Contention Error Interrupt Enable)

The CLP1 bit controls the LP1 Contention Error (ErrContentionLP1) Interrupt permission. When an LP1 Contention Error Interrupt occurs, the FERRSR.CLP1 flag is set to 1.

To enable the LP1 Contention Error Interrupt, set the CLP1 bit to 1.

66.2.38 CLSTPTSETR : Clock Lane Stop Time Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x314

Bit position:	31–24	23–16	15	14	13	12	11–2	1	0
Bit field:	CLKKPT[7:0]	CLKBFHT[7:0]	—	—	—	—	CLKSTPT[9:0]	—	—
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0	0	0	0	0 0 0 0 0 0 0 0 0 0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
11:2	CLKSTPT[9:0]	Clock Stop Time Set the CLKSTPT period shown in Figure 66.1.*1	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
23:16	CLKBFHT[7:0]	Clock Beforehand Time Set the CLKBFHT period shown in Figure 66.1.*2*3	R/W
31:24	CLKKPT[7:0]	Clock Keep Time Set the CLKKPT period shown in Figure 66.1.*2*3	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Set this bit to a value greater than or equal to the value of the calculated period rounded up plus 3.
Note 2. Set the appropriate value (nonzero) before setting the HSCLKSETR.HSCLST bit to 1.
Note 3. Set these bits to a value greater than the desired time.

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

CLKSTPT[9:0] bits (Clock Stop Time)

The CLKSTPT[9:0] bits define the minimum period of time between the transition to LP mode and the return to HS mode on the clock lane.

This setting is used to stop the clock lane when no HS transfer is taking place in the Non-continuous clock mode (HSCLKSETR.HSCLMD = 0).

If there is an interval of more than the period specified by the CLKSTPT[9:0] bits between HS transfers on the clock lane, an LP mode period is inserted on the clock lane.

When the HSCLKSETR.HSCLMD bit is set to 1 (Continuous clock mode), the setting of the CLKSTPT[9:0] bits is irrelevant.

This period corresponds to the time specified in the MIPI D-PHY Specification ($T_{HS\text{-}TRAIL} + T_{CLK\text{-}POST} + T_{CLK\text{-}TRAIL} + T_{HS\text{-}EXIT} + T_{LPX} \times 2 + T_{LPX\ (for\ Clock\ Lane\ HS\ Entry)} + T_{CLK\text{-}PREPARE} + T_{CLK\text{-}ZERO} + T_{CLK\text{-}PRE} + T_{LPX\ (for\ Data\ Lane\ HS\ Entry)} + T_{HS\text{-}PREPARE} + T_{HS\text{-}ZERO} + T_{HS\text{-}SYNC}$). See Figure 66.1.

The period is derived from the following formula:

CLKSTPT period [µs] = CLKSTPT[9:0] × (period of HS serial UI) [µs] × 32

For example:

When CLKSTPT[9:0] = 0x3E8, HS rate = 720 Mbps

CLKSTPT period [µs] = 1000 × (1 / 720 Mbps) × 32 = 44.4 [µs]

For information about each parameter, see the D-PHY specification and Figure 66.1. Of these parameters, it is necessary to set the time set in CLKKPT for ($T_{HS\text{-}TRAIL} + T_{CLK\text{-}POST}$) and that set in CLKBFHT for ($T_{LPX\ (for\ Clock\ Lane\ HS\ Entry)} + T_{CLK\text{-}PREPARE} + T_{CLK\text{-}ZERO} + T_{CLK\text{-}PRE}$). Set the rounded-up value plus at least 3. Setting this field to 0 is prohibited.

$T_{LPX\ (for\ Clock\ Lane\ HS\ Entry)}$ means the period of LP-01 in the HS entry sequence of Clock Lane. $T_{LPX\ (for\ Data\ Lane\ HS\ Entry)}$ means the period of LP-01 in the HS entry sequence of Data Lane. If these periods are not the same as the standard T_{LPX} value of the host device (in the range of 67% to 150% of the T_{LPX} value of the peripheral device), use the actual periods for this calculation.

CLKBFHT[7:0] bits (Clock Beforehand Time)

The CLKBFHT[7:0] bits specify the period during which the clock lane returns to HS mode before the start of HS packet transfer.

If HS transfers are scheduled in LP mode, the clock lane starts HS transfer in advance according to this setting. See Figure 66.1.

The period is derived from the following formula:

CLKBFHT period [µs] = CLKBFHT[7:0] × (period of HS serial UI) [µs] × 32

For example:

When CLKBFHT[7:0] = 0x64, HS rate = 720 Mbps

CLKBFHT period [µs] = 100 × (1 / 720 Mbps) × 32 = 4.4 [µs]

This time corresponds to the D-PHY specification ($T_{LPX\ (for\ Clock\ Lane\ HS\ Entry)} + T_{CLK\text{-}PREPARE} + T_{CLK\text{-}ZERO} + T_{CLK\text{-}PRE}$). The value must be greater than the one calculated. Set an appropriate value (other than 0) before HSCLKSETR.HSCLST is set to 1.

$T_{LPX\ (for\ Clock\ Lane\ HS\ Entry)}$ means the period of LP-01 in the HS entry sequence of Clock Lane. If this period is not the same as the normal T_{LPX} value of host device (in the range of 67% to 150% of the T_{LPX} value of peripheral device), use the actual period for this calculation.

CLKKPT[7:0] bits (Clock Keep Time)

The CLKKPT[7:0] bits specify the period that the clock lane keeps the HS mode active after the last HS packet transfer. See Figure 66.1.

The period is derived from the following formula:

CLKKPT period [µs] = CLKKPT[7:0] × (period of HS serial UI) [µs] × 32

For example:

When CLKKPT[7:0] = 0x64, HS rate = 720 Mbps

CLKKPT period [µs] = 100 × (1 / 720 Mbps) × 32 = 4.4 [µs]

This time corresponds to the D-PHY specifications ($T_{HS\text{-}TRAIL} + T_{CLK\text{-}POST}$).

The value must be greater than the one calculated. Set an appropriate value (other than 0) before HSCLKSETR.HSCLST is set to 1.

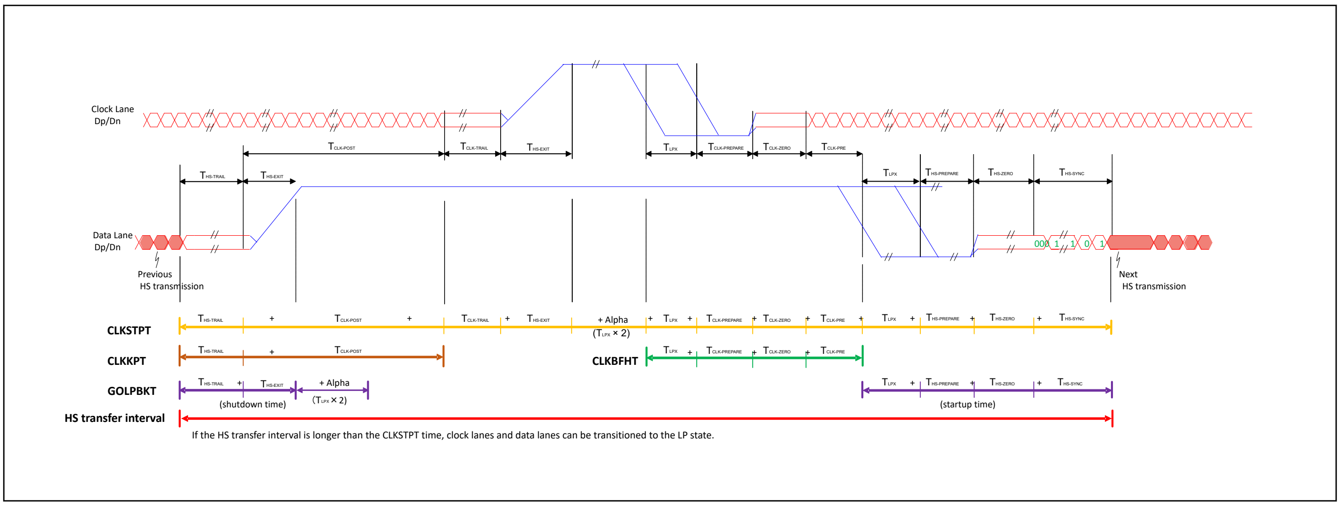


Figure 66.1 Minimum duration of LP mode on the clock lane

66.2.39 LPTRNSTSETR : LP Transition Time Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x318

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	GOLPBKT[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	GOLPBKT[9:0]	Go LP and Back Time Set the GOLPBKT period shown in Figure 66.2. Set this bit to a value greater than or equal to the value of the calculated period rounded up plus 3.	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register can only be set to change during the initialization at startup or software reset process (see section 66.3.2. Software Reset).

GOLPBKT[9:0] bits (Go LP and Back Time)

The GOLPBKT[9:0] bits indicate the minimum period of time between the transition to LP mode and the return to HS mode on the data lane.

This setting is used for the ability of the DSI Host to automatically transition the data lane from HS mode to LP mode during the HSA, HBP, and HFP periods. See section 66.3.6.5. Selection of Blanking Packets or LP-11 for HSA, HBP, and HFP.

If there is an interval of more than the period specified by the GOLPBKT[9:0] bits between HS transfers, an LP mode period is inserted on the data lane.

This period corresponds to the time specified in the MIPI D-PHY Specification ($T_{HS\text{-}TRAIL} + T_{HS\text{-}EXIT} + T_{LPX} \times 2 + T_{LPX}$ (for Data Lane HS Entry) $+ T_{HS\text{-}PREPARE} + T_{HS\text{-}ZERO} + T_{HS\text{-}SYNC}$). See Figure 66.2.

The period is derived from the following formula:

GOLPBKT period [μs] = GOLPBKT[9:0] × (period of HS serial 8 UI) [μs]*1

For example:

When GOLPBKT[9:0] = 0x3E8, HS rate = 720 Mbps

GOLPBKT period [μs] = 1000 × (1 / 720 Mbps) × 8 = 11.1 [μs]

The unit of this setting is 8 UI, so if the result of this formula contains a fraction of less than 8 UI, round it up. Set the rounded-up value plus at least 3. T_{LPX} (for Data Lane HS Entry) means the period of LP-01 in the HS entry sequence of data lane. If this period is not the same as the standard T_{LPX} value of the host device (in the range of 67% to 150% of the TLPX value of the peripheral device), use the actual period for this calculation.

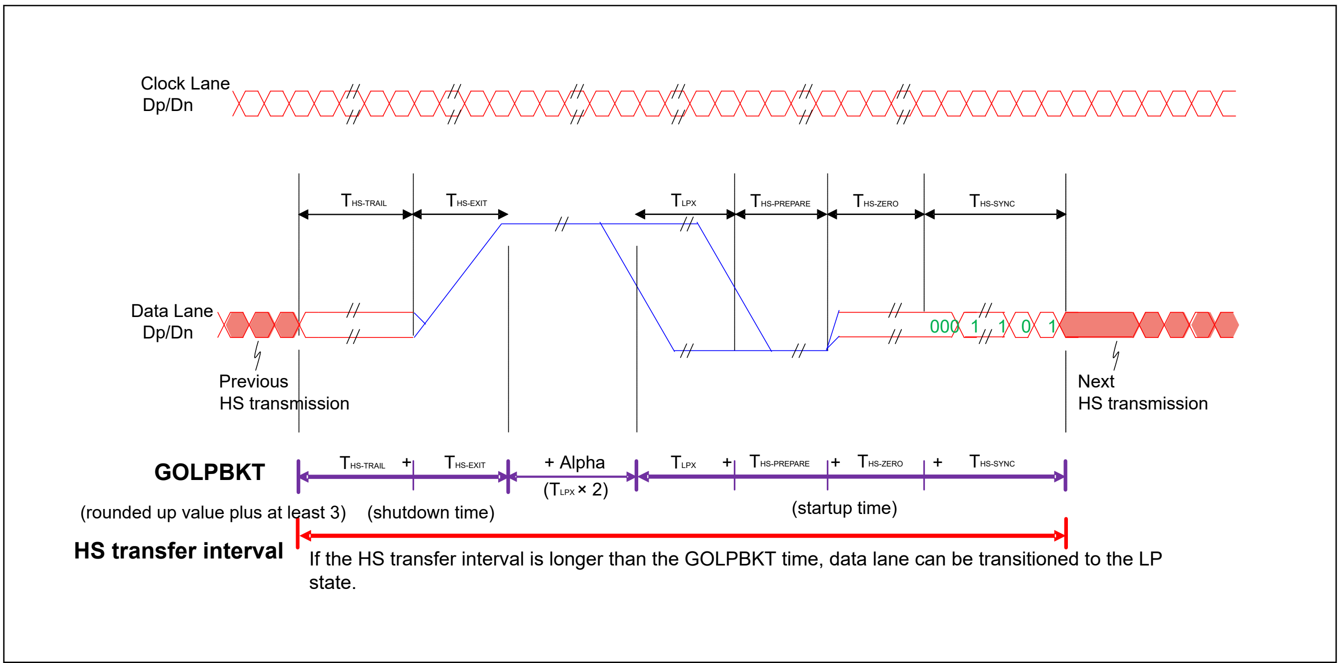


Figure 66.2 Minimum LP mode period of data lane

Note 1. The unit for this setting is 8 UI, so if the result of this equation contains more than half of 8 UI, round it up.

66.2.40 PLSR : Physical Lane Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x320

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DLULPEXT	DLULPENT	CLHS2LP	CLLP2HS	CLULPEXT	CLULPENT	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DL0DIR	—	DL0TX2RX	DL0RX2TX	—	—	DL1STP	DL0STP	—	—	DL1UAN	DL0UAN	DL0RUE	DL0RLE	CLSTP	CLUAN
Value after reset:	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	1

Bit	Symbol	Function	R/W
0	CLUAN	Clock Lane UlpsActiveNot Status Clock Lane state 0: In ULP state 1: Not in ULP state	R
1	CLSTP	Clock Lane Stop Status Clock Lane state 0: Not in Stop state 1: In Stop state	R
2	DL0RLE	Data Lane-0 RxLpdtEsc Status Data Lane-0 state 0: Not in Escape Low-Power Data Receive mode 1: In Escape Low-Power Data Receive mode	R

Bit	Symbol	Function	R/W
3	DL0RUE	Data Lane-0 RxUlpsEsc Status Data Lane-0 state 0: Not in Escape Ultra-Low Power (Receive) mode 1: In Escape Ultra-Low Power (Receive) mode	R
4	DL0UAN	Data Lane-0 UlpsActiveNot Status Data Lane-0 state 0: In ULP state 1: Not in ULP state	R
5	DL1UAN	Data Lane-1 UlpsActiveNot Status Data Lane-1 state 0: In ULP state 1: Not in ULP state	R
7:6	—	These bits are read as 1.	R
8	DL0STP	Data Lane-0 Stop Status Data Lane-0 state 0: Not in Stop state 1: In Stop state	R
9	DL1STP	Data Lane-1 Stop Status Data Lane-1 state 0: Not in Stop state 1: In Stop state	R
11:10	—	These bits are read as 0.	R
12	DL0RX2TX	Data Lane-0 RX to TX Transition Interrupt Flag Data Lane-0 direction transition from receiver to transmitter 0: Not detected 1: Detected	R
13	DL0TX2RX	Data Lane-0 TX to RX Transition Interrupt Flag Data Lane-0 direction transition from transmitter to receiver 0: Not detected 1: Detected	R
14	—	This bit is read as 0.	R
15	DL0DIR	Data Lane-0 Direction Data Lane-0 direction 0: Transmitter 1: Receiver	R
23:16	—	These bits are read as 0.	R
24	CLULPENT	Clock Lane ULPS Enter Interrupt Flag Clock Lane transition to ULPS 0: Not detected 1: Detected	R
25	CLULPEXT	Clock Lane ULPS Exit Interrupt Flag Clock Lane transition from ULPS 0: Not detected 1: Detected	R
26	CLLP2HS	Clock Lane LP to HS Transition Interrupt Flag Clock Lane mode transition from LP to HS 0: Not detected 1: Detected	R
27	CLHS2LP	Clock Lane HS to LP Transition Interrupt Flag Clock Lane mode transition from HS to LP 0: Not detected 1: Detected	R
28	DLULPENT	Data Lane ULPS Enter Interrupt Flag Data Lane transition to ULPS 0: Not detected 1: Detected	R

Bit	Symbol	Function	R/W
29	DLULPEXT	Data Lane ULPS Exit Interrupt Flag Data Lane transition from ULPS 0: Not detected 1: Detected	R
31:30	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

CLUAN bit (Clock Lane UlpsActiveNot Status)

The CLUAN bit indicates that the D-PHY Clock Lane is not in ULP State.*1

CLSTP bit (Clock Lane Stop Status)

The CLSTP bit indicates that the D-PHY Clock Lane module is currently in Stop state. *1

The CLSTP bit is used to indirectly determine if the PHY line levels are in the LP-11 state.

DL0RLE bit (Data Lane-0 RxLpdtEsc Status)

The DL0RLE bit indicates that the D-PHY Data Lane-0 module is in Low-Power data receive mode.*1

DL0RUE bit (Data Lane-0 RxUlpsEsc Status)

The DL0RUE bit indicates that the D-PHY Data Lane-0 module has entered the Ultra-Low Power State.*1

DL0UAN bit (Data Lane-0 UlpsActiveNot Status)

The DL0UAN bit indicates that the D-PHY Data Lane-0 is not in ULP State.*1

DL1UAN bit (Data Lane-1 UlpsActiveNot Status)

The DL1UAN bit indicates that the D-PHY Data Lane-1 is not in ULP State.*1

DL0STP bit (Data Lane-0 Stop Status)

The DL0STP bit indicates that the D-PHY Data Lane-0 module is currently in Stop state.*1

The DL0STP bit is used to indirectly determine if the PHY line levels are in the LP-11 state.

DL1STP bit (Data Lane-1 Stop Status)

The DL1STP bit indicates that the D-PHY Data Lane-1 module is currently in Stop state.*1

The DL1STP bit is used to indirectly determine if the PHY line levels are in the LP-11 state.

DL0RX2TX flag (Data Lane-0 RX to TX Transition Interrupt Flag)

The DL0RX2TX flag indicates that the D-PHY Data Lane-0 has transitioned from the receiver to the transmitter.

The DL0RX2TX flag is not cleared automatically, so set the PLSCR.DL0RX2TX bit to 1 to clear the DL0RX2TX flag.

DL0TX2RX flag (Data Lane-0 TX to RX Transition Interrupt Flag)

The DL0TX2RX flag indicates that the D-PHY Data Lane-0 has transitioned from the transmitter to the receiver.

The DL0TX2RX flag is not cleared automatically, so set the PLSCR.DL0TX2RX bit to 1 to clear the DL0TX2RX flag.

DL0DIR bit (Data Lane-0 Direction)

The DL0DIR bit indicates the status of the lane direction, whether the D-PHY Data Lane-0 is currently a transmitter or a receiver.*1

CLULPENT flag (Clock Lane ULPS Enter Interrupt Flag)

The CLULPENT flag indicates that the Clock Lane has entered ULPS.

The CLULPENT flag is not cleared automatically, so set the PLSCR.CLULPENT bit to 1 to clear the CLULPENT flag.

CLULPEXT flag (Clock Lane ULPS Exit Interrupt Flag)

The CLULPEXT flag indicates that the Clock Lane has exited from ULPS.

The CLULPEXT flag is not cleared automatically, so set the PLSCR.CLULPEXT bit to 1 to clear the CLULPEXT flag.

CLLP2HS flag (Clock Lane LP to HS Transition Interrupt Flag)

The CLLP2HS flag indicates that the Clock Lane has transitioned from LP mode to HS mode.

Ignore the CLLP2HS flag when the HSCLKSETR.HSCLMD bit is set to 0 (Non-continuous clock mode).

The CLLP2HS flag is not cleared automatically, so set the PLSCR.CLLP2HS bit to 1 to clear the CLLP2HS flag.

CLHS2LP flag (Clock Lane HS to LP Transition Interrupt Flag)

The CLHS2LP flag indicates that the Clock Lane has transitioned from HS mode to LP mode.

Ignore the CLHS2LP flag when the HSCLKSETR.HSCLMD bit is set to 0 (Non-continuous clock mode).

The CLHS2LP flag is not cleared automatically, so set the PLSCR.CLHS2LP bit to 1 to clear the CLHS2LP flag.

DLULPENT flag (Data Lane ULPS Enter Interrupt Flag)

The DLULPENT flag indicates that the Data Lanes specified by the TXSETR.NUMLANE[1:0] bits have transitioned from Stop state to ULPS by ULPS Enter.

The DLULPENT flag is not cleared automatically, so set the PLSCR.DLULPENT bit to 1 to clear the DLULPENT flag.

DLULPEXT flag (Data Lane ULPS Exit Interrupt Flag)

The DLULPEXT flag indicates that the Data Lanes specified by the TXSETR.NUMLANE[1:0] bits have transitioned from ULPS to Stop state by ULPS Exit.

The DLULPEXT flag is not cleared automatically, so set the PLSCR.DLULPEXT bit to 1 to clear the DLULPEXT flag.

Note 1. These bits are set after synchronization with LPCLK. Therefore, it takes time for the status to be refrected.

66.2.41 PLSCR : Physical Lane Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x324

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DLULPEXT	DLULPENT	CLHS2LP	CLLP2HS	CLULPEXT	CLULPENT	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	DL0TX2RX	DL0RX2TX	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	—	These bits are read as 0. The write value should be 0.	W
12	DL0RX2TX	Data Lane-0 RX to TX Transition Interrupt Flag Clear 0: No operation 1: Clear the PLSR.DL0RX2TX flag	W
13	DL0TX2RX	Data Lane-0 TX to RX Transition Interrupt Flag Clear 0: No operation 1: Clear the PLSR.DL0TX2RX flag	W
23:14	—	These bits are read as 0. The write value should be 0.	W
24	CLULPENT	Clock Lane ULPS Enter Interrupt Flag Clear 0: No operation 1: Clear the PLSR.CLULPENT flag	W
25	CLULPEXT	Clock Lane ULPS Exit Interrupt Flag Clear 0: No operation 1: Clear the PLSR.CLULPEXT flag	W

Bit	Symbol	Function	R/W
26	CLLP2HS	Clock Lane LP to HS Transition Interrupt Flag Clear 0: No operation 1: Clear the PLSR.CLLP2HS flag	W
27	CLHS2LP	Clock Lane HS to LP Transition Interrupt Flag Clear 0: No operation 1: Clear the PLSR.CLHS2LP flag	W
28	DLULPENT	Data Lane ULPS Enter Interrupt Flag Clear 0: No operation 1: Clear the PLSR.DLULPENT flag	W
29	DLULPEXT	Data Lane ULPS Exit Interrupt Flag Clear 0: No operation 1: Clear the PLSR.DLULPEXT flag	W
31:30	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

DL0RX2TX bit (Data Lane-0 RX to TX Transition Interrupt Flag Clear)

Set the DL0RX2TX bit to 1 to clear the PLSR.DL0RX2TX flag.

DL0TX2RX bit (Data Lane-0 TX to RX Transition Interrupt Flag Clear)

Set the DL0TX2RX bit to 1 to clear the PLSR.DL0TX2RX flag.

CLULPENT bit (Clock Lane ULPS Enter Interrupt Flag Clear)

Set the CLULPENT bit to 1 to clear the PLSR.CLULPENT flag.

CLULPEXT bit (Clock Lane ULPS Exit Interrupt Flag Clear)

Set the CLULPEXT bit to 1 to clear the PLSR.CLULPEXT flag.

CLLP2HS bit (Clock Lane LP to HS Transition Interrupt Flag Clear)

Set the CLLP2HS bit to 1 to clear the PLSR.CLLP2HS flag.

CLHS2LP bit (Clock Lane HS to LP Transition Interrupt Flag Clear)

Set the CLHS2LP bit to 1 to clear the PLSR.CLHS2LP flag.

DLULPENT bit (Data Lane ULPS Enter Interrupt Flag Clear)

Set the DLULPENT bit to 1 to clear the PLSR.DLULPENT flag.

DLULPEXT bit (Data Lane ULPS Exit Interrupt Flag Clear)

Set the DLULPEXT bit to 1 to clear the PLSR.DLULPEXT flag.

66.2.42 PLIER : Physical Lane Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x328

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DLULPEXT	DLULPENT	CLHS2LP	CLLP2HS	CLULPEXT	CLULPENT	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	DL0TX2RX	DL0RX2TX	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	—	These bits are read as 0. The write value should be 0.	R/W
12	DL0RX2TX	Data Lane-0 RX to TX Transition Interrupt Enable 0: Disable 1: Enable	R/W
13	DL0TX2RX	Data Lane-0 TX to RX Transition Interrupt Enable 0: Disable 1: Enable	R/W
23:14	—	These bits are read as 0. The write value should be 0.	R/W
24	CLULPENT	Clock Lane ULPS Enter Interrupt Enable 0: Disable 1: Enable	R/W
25	CLULPEXT	Clock Lane ULPS Exit Interrupt Enable 0: Disable 1: Enable	R/W
26	CLLP2HS	Clock Lane LP to HS Transition Interrupt Enable 0: Disable 1: Enable	R/W
27	CLHS2LP	Clock Lane HS to LP Transition Interrupt Enable 0: Disable 1: Enable	R/W
28	DLULPENT	Data Lane ULPS Enter Interrupt Enable 0: Disable 1: Enable	R/W
29	DLULPEXT	Data Lane ULPS Exit Interrupt Enable 0: Disable 1: Enable	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DL0RX2TX bit (Data Lane-0 RX to TX Transition Interrupt Enable)

The DL0RX2TX bit controls the Data Lane-0 RX to TX Transition Interrupt permission.

When a Data Lane-0 RX to TX Transition Interrupt occurs, the PLSR.DL0RX2TX flag is set to 1.

To enable the Data Lane-0 RX to TX Transition Interrupt, set the DL0RX2TX bit to 1.

DL0TX2RX bit (Data Lane-0 TX to RX Transition Interrupt Enable)

The DL0TX2RX bit controls the Data Lane-0 TX to RX Transition Interrupt permission.

When a Data Lane-0 TX to RX Transition Interrupt occurs, the PLSR.DL0TX2RX flag is set to 1.

To enable the Data Lane-0 TX to RX Transition Interrupt, set the DL0TX2RX bit to 1.

CLULPENT bit (Clock Lane ULPS Enter Interrupt Enable)

The CLULPENT bit controls the Clock Lane ULPS Enter Interrupt permission.

When a Clock Lane ULPS Enter Interrupt occurs, the PLSR.CLULPENT flag is set to 1.

To enable the Clock Lane ULPS Enter Interrupt, set the CLULPENT bit to 1.

CLULPEXT bit (Clock Lane ULPS Exit Interrupt Enable)

The CLULPEXT bit controls the Clock Lane ULPS Exit Interrupt permission.

When a Clock Lane ULPS Exit Interrupt occurs, the PLSR.CLULPEXT flag is set to 1.

To enable the Clock Lane ULPS Exit Interrupt, set the CLULPEXT bit to 1.

CLLP2HS bit (Clock Lane LP to HS Transition Interrupt Enable)

The CLLP2HS bit controls the Clock Lane LP to HS Transition Interrupt permission.

When a Clock Lane LP to HS Transition Interrupt occurs, the PLSR.CLLP2HS flag is set to 1.

To enable the Clock Lane LP to HS Transition Interrupt, set the CLLP2HS bit to 1.

CLHS2LP bit (Clock Lane HS to LP Transition Interrupt Enable)

The CLHS2LP bit controls the Clock Lane HS to LP Transition Interrupt permission.

When a Clock Lane HS to LP Transition Interrupt occurs, the PLSR.CLHS2LP flag is set to 1.

To enable the Clock Lane HS to LP Transition Interrupt, set the CLHS2LP bit to 1.

DLULPENT bit (Data Lane ULPS Enter Interrupt Enable)

The DLULPENT bit controls the Data Lane ULPS Enter Interrupt permission.

When a Data Lane ULPS Enter Interrupt occurs, the PLSR.DLULPENT flag is set to 1.

To enable the Data Lane ULPS Enter Interrupt, set the DLULPENT bit to 1.

DLULPEXT bit (Data Lane ULPS Exit Interrupt Enable)

The DLULPEXT bit controls the Data Lane ULPS Exit Interrupt permission.

When a Data Lane ULPS Exit Interrupt occurs, the PLSR.DLULPEXT flag is set to 1.

To enable the Data Lane ULPS Exit Interrupt, set the DLULPEXT bit to 1.

66.2.43 VMSET0R : Video Mode Set 0 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	HFPNOLP	HBPNOLP	HSANOLP	—	—	—	—	—	—	VSTOP	VSTART
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	VSTART	Video Mode Operation Start 0: No operation 1: Start Video mode operation	W
1	VSTOP	Video Mode Operation Stop 0: No operation 1: Stop Video mode operation	W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	HSANOLP	HSA Period No LP 0: Not suppressing the transition to LP during the HSA period 1: Suppress the transition to LP during the HSA period and keep HS	R/W
9	HBPNOLP	HBP Period No LP 0: Not suppressing the transition to LP during the HBP period 1: Suppress the transition to LP during the HBP period and keep HS	R/W
10	HFPNOLP	HFP Period No LP 0: Not suppressing the transition to LP during the HFP period 1: Suppress the transition to LP during the HFP period and keep HS	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VSTART bit (Video Mode Operation Start)

The VSTART bit is the start trigger for Video mode operation.

Setting 1 to both the VSTART bit and the VSTOP bit is prohibited.

When the VMSR.RUNNING bit is set to 1, setting the VSTART bit to 1 is prohibited.

VSTOP bit (Video Mode Operation Stop)

The VSTOP bit is the stop trigger for Video mode operation.

The Video mode operation is stopped at the end of the video frame. After the Video mode operation is stopped, stop the GLCDC.

Setting 1 to both the VSTOP bit and the VSTART bit is prohibited.

When the VMSR.RUNNING bit is cleared to 0, setting the VSTOP bit to 1 is prohibited.

HSANOLP bit (HSA Period No LP)

The HSANOLP bit controls whether to transition to LP during the HSA period.

When the HSANOLP bit is set to 1, the link does not transition to LP during the HSA period and retains HS mode.

HS mode is kept by sending a blanking packet.

HBPNOLP bit (HBP Period No LP)

The HBPNOLP bit controls whether to transition to LP during the HBP period.

When the HBPNOLP bit is set to 1, the link does not transition to LP during the HBP period and retains HS mode.

HS mode is kept by sending a blanking packet.

HFPNOLP bit (HFP Period No LP)

The HFPNOLP bit controls whether to transition to LP during the HFP period.

When the HFPNOLP bit is set to 1, the link does not transition to LP during the HFP period and retains HS mode.

HS mode is kept by sending a blanking packet.

66.2.44 VMSET1R : Video Mode Set 1 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x404

Bit position:	31																		13											2		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DLY[11:0]												—	—
Value after reset:	0	0	0	1	0	0	0	0	0	1	0	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
13:2	DLY[11:0]	Delay Value Set the delay value the DSI Host until the transfer begins.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
17:16	—	These bits are read as 1. The write value should be 1.	R/W
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	—	This bit is read as 1. The write value should be 1.	R/W
21	—	This bit is read as 0. The write value should be 0.	R/W
22	—	This bit is read as 1. The write value should be 1.	R/W
23	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
29:24	—	The read values are undefined. The write value should be 0.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DLY[11:0] bits (Delay Value)

The DLY[11:0] bits are for the control of the delay time inside the DSI Host to avoid the underflow or the overflow of the video stream from GLCDC.

The delay time is derived from the following formula:

Delay time [µs] = DLY[11:0] × (period of HS serial UI) [µs] × 32

For example:

When DLY[11:0] = 0x3E8, HS line rate = 720 Mbps

Delay time [µs] = 1000 × (1 / 720 Mbps) × 32 = 44.4 [µs]

For more details, see section 66.3.6.7. Delay Adjustment for Video Mode Operation.

It is prohibited to set the VMSET1R.DLY[11:0] bits to 0x000.

If the VMSET0R.HFPNOLP bit is set to 1, set the DLY[11:0] bits to more than HFP period.

If the VMSET0R.HBPNOLP bit is set to 1, set the DLY[11:0] bits to more than HBP period.

If the VMSET0R.HSANOLP bit is set to 1, set the DLY[11:0] bits to more than HSA period.

66.2.45 VMSR : Video Mode Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x410

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	VBUFOVF	VBUFUDF	—	TIMERR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VIRDY	RUNNING	STOP	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	START	Video Mode Operation Start Interrupt Flag 0: Video mode operation has not started 1: Video mode operation has started	R
1	STOP	Video Mode Operation Stop Interrupt Flag 0: Video mode operation has not stopped 1: Video mode operation has stopped	R
2	RUNNING	Video Mode Operation Running Status 0: Video mode operation is stopped 1: Video mode operation is running	R
3	VIRDY	Video Mode Operation Ready Interrupt Flag 0: Video mode operation is not ready 1: Video mode operation is ready	R
19:4	—	These bits are read as 0.	R
20	TIMERR	Timing Error Interrupt Flag 0: Timing error has not occurred 1: Timing error has occurred during the Video mode operation	R

Bit	Symbol	Function	R/W
21	—	This bit is read as 0.	R
22	VBUFUDF	Video Buffer Underflow Error Interrupt Flag 0: Data underflow has not occurred 1: Data underflow has occurred in the video buffer	R
23	VBUFOVF	Video Buffer Overflow Error Interrupt Flag 0: Data overflow has not occurred 1: Data overflow has occurred in the video buffer	R
31:24	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

START flag (Video Mode Operation Start Interrupt Flag)

The START flag indicates that the Video mode operation has started.

The START flag is not cleared automatically, so set the VMSCR.START bit to 1 to clear the START flag.

STOP flag (Video Mode Operation Stop Interrupt Flag)

The STOP flag indicates that the Video mode operation has stopped.

The STOP flag is not cleared automatically, so set the VMSCR.STOP bit to 1 to clear the STOP flag.

RUNNING bit (Video Mode Operation Running Status)

The RUNNING bit indicates the operating state of the Video mode.

Changing the settings of VMSET0R, VMSET1R, VMPPSETR, VMVSSETR, VMVPSETR, VMHSSETR and VMHPSETR is prohibited while the RUNNING bit is 1.

VIRDY flag (Video Mode Operation Ready Interrupt Flag)

The VIRDY flag indicates that the Video mode operation is ready to start working.

The VIRDY flag is not cleared automatically, so set the VMSCR.VIRDY bit to 1 to clear the VIRDY flag.

TIMERR flag (Timing Error Interrupt Flag)

The TIMERR flag indicates that the timing error has occurred during the Video mode operation.

The TIMERR flag is not cleared automatically, so set the VMSCR.TIMERR bit to 1 to clear the TIMERR flag.

VBUFUDF flag (Video Buffer Underflow Error Interrupt Flag)

The VBUFUDF flag indicates that the data underflow has occurred in the video buffer.

The VBUFUDF flag is not cleared automatically, so set the VMSCR.VBUFUDF bit to 1 to clear the VBUFUDF flag.

VBUFOVF flag (Video Buffer Overflow Error Interrupt Flag)

The VBUFOVF flag indicates that the data overflow has occurred in the video buffer.

The VBUFOVF flag is not cleared automatically, so set the VMSCR.VBUFOVF bit to 1 to clear the VBUFOVF flag.

66.2.46 VMSCR : Video Mode Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x414

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	VBUFOVF	VBUFUDF	—	TIMERR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VIRDY	—	STOP	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	START	Video Mode Operation Start Interrupt Flag Clear 0: No operation 1: Clear the VMSR.START flag	W
1	STOP	Video Mode Operation Stop Interrupt Flag Clear 0: No operation 1: Clear the VMSR.STOP flag	W
2	—	This bit is read as 0. The write value should be 0.	W
3	VIRDY	Video Mode Operation Ready Interrupt Flag Clear 0: No operation 1: Clear the VMSR.VIRDY flag	W
19:4	—	These bits are read as 0. The write value should be 0.	W
20	TIMERR	Timing Error Interrupt Flag Clear 0: No operation 1: Clear the VMSR.TIMERR flag	W
21	—	This bit is read as 0. The write value should be 0.	W
22	VBUFUDF	Video Buffer Underflow Error Interrupt Flag Clear 0: No operation 1: Clear the VMSR.VBUFUDF flag	W
23	VBUFOVF	Video Buffer Overflow Error Interrupt Flag Clear 0: No operation 1: Clear the VMSR.VBUFOVF flag	W
31:24	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

START bit (Video Mode Operation Start Interrupt Flag Clear)

Set the START bit to 1 to clear the VMSR.START flag.

STOP bit (Video Mode Operation Stop Interrupt Flag Clear)

Set the STOP bit to 1 to clear the VMSR.STOP flag.

VIRDY bit (Video Mode Operation Ready Interrupt Flag Clear)

Set the VIRDY bit to 1 to clear the VMSR.VIRDY flag.

TIMERR bit (Timing Error Interrupt Flag Clear)

Set the TIMERR bit to 1 to clear the VMSR.TIMERR flag.

VBUFUDF bit (Video Buffer Underflow Error Interrupt Flag Clear)

Set the VBUFUDF bit to 1 to clear the VMSR.VBUFUDF flag.

VBUFOVF bit (Video Buffer Overflow Error Interrupt Flag Clear)

Set the VBUFOVF bit to 1 to clear the VMSR.VBUFOVF flag.

66.2.47 VMIER : Video Mode Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x418

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	VBUF OVF	VBUF UDF	—	TIMER R	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	VIRDY	—	STOP	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	START	Video Mode Operation Start Interrupt Enable 0: Disable 1: Enable	R/W
1	STOP	Video Mode Operation Stop Interrupt Enable 0: Disable 1: Enable	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	VIRDY	Video Mode Operation Ready Interrupt Enable 0: Disable 1: Enable	R/W
19:4	—	These bits are read as 0. The write value should be 0.	R/W
20	TIMERR	Timing Error Interrupt Enable 0: Disable 1: Enable	R/W
21	—	This bit is read as 0. The write value should be 0.	R/W
22	VBUFUDF	Video Buffer Underflow Error Interrupt Enable 0: Disable 1: Enable	R/W
23	VBUFOVF	Video Buffer Overflow Error Interrupt Enable 0: Disable 1: Enable	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

START bit (Video Mode Operation Start Interrupt Enable)

The START bit controls the Video Mode Operation Start Interrupt permission.

When a Video Mode Operation Start Interrupt occurs, the VMSR.START flag is set to 1.

To enable the Video Mode Operation Start Interrupt, set the START bit to 1.

STOP bit (Video Mode Operation Stop Interrupt Enable)

The STOP bit controls the Video Mode Operation Stop Interrupt permission.

When a Video Mode Operation Stop Interrupt occurs, the VMSR.STOP flag is set to 1.

To enable the Video Mode Operation Stop Interrupt, set the STOP bit to 1.

VIRDY bit (Video Mode Operation Ready Interrupt Enable)

The VIRDY bit controls the Video Mode Operation Ready Interrupt permission.

When a Video Mode Operation Ready Interrupt occurs, the VMSR.VIRDY flag is set to 1.

To enable the Video Mode Operation Ready Interrupt, set the VIRDY bit to 1.

TIMERR bit (Timing Error Interrupt Enable)

The TIMERR bit controls the Timing Error Interrupt permission.

When a Timing Error Interrupt occurs, the VMSR.TIMERR flag is set to 1.

To enable the Timing Error Interrupt, set the TIMERR bit to 1.

VBUFUDF bit (Video Buffer Underflow Error Interrupt Enable)

The VBUFUDF bit controls the Video Buffer Underflow Error Interrupt permission.

When a Video Buffer Underflow Error Interrupt occurs, the VMSR.VBUFUDF flag is set to 1.

To enable the Video Buffer Underflow Error Interrupt, set the VBUFUDF bit to 1.

VBUFOVF bit (Video Buffer Overflow Error Interrupt Enable)

The VBUFOVF bit controls the Video Buffer Overflow Error Interrupt permission.

When a Video Buffer Overflow Error Interrupt occurs, the VMSR.VBUFOVF flag is set to 1.

To enable the Video Buffer Overflow Error Interrupt, set the VBUFOVF bit to 1.

66.2.48 VMPPSETR : Video Mode Pixel Packet Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x420

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	VC[1:0]		DT[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TXESYNC	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
14:0	—	These bits are read as 0. The write value should be 0.	R/W
15	TXESYNC	Transmit End of Sync Pulse 0: HSE and VSE are not transmitted 1: HSE and VSE are transmitted	R/W
21:16	DT[5:0]	Video Mode Data Type Set the data type for the pixel stream packet header 0x0E: Packed Pixel Stream, 16-bit RGB 0x1E: Packed Pixel Stream, 18-bit RGB 0x3E: Packed Pixel Stream, 24-bit RGB Others: Setting prohibited	R/W
23:22	VC[1:0]	Video Mode Virtual Channel Set the virtual channel ID of 00b to 11b.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

TXESYNC bit (Transmit End of Sync Pulse)

The setting of 0 is used for Burst mode sequence or Non-burst mode with Sync events sequence.

The setting of 1 is used for Non-burst mode with Sync pulse sequence.

DT[5:0] bits (Video Mode Data Type)

Set these bits to the same format as the input format from GLCDC.

VC[1:0] bits (Video Mode Virtual Channel)

Set these bits to a number between 00b to 11b as the virtual channel ID.

66.2.49 VMVSSETR : Video Mode Vertical Size Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x428

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	VACT[14:0]														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VSPOL	—	—	—	VSA[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	VSA[11:0]	VSA Lines Set the number of VSA (Vertical Sync Active) lines.	R/W
14:12	—	These bits are read as 0. The write value should be 0.	R/W
15	VSPOL	VSYNC Polarity 0: Active-high 1: Active-low	R/W
30:16	VACT[14:0]	Vertical Active Lines Set the number of VACT (Vertical Active) lines.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VSA[11:0] bits (VSA Lines)

The VSA[11:0] bits control the number of VSA (Vertical Sync Active) lines.

VSPOL bit (VSYNC Polarity)

The VSPOL bit controls the polarity of VSYNC the signal.

VACT[14:0] bits (Vertical Active Lines)

The VACT[14:0] bits control the number of VACT (Vertical Active) lines.

66.2.50 VMVPSETR : Video Mode Vertical Porch Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x42C

Bit position:	31			28												16				12												0
Bit field:	—	—	—	VFP[12:0]													—	—	—	VBP[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	VBP[12:0]	VBP Lines Set the number of VBP (Vertical Back Porch) lines.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	VFP[12:0]	VFP Lines Set the number of VFP (Vertical Front Porch) lines.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

VBP[12:0] bits (VBP Lines)

The VBP[12:0] bits control the number of VBP (Vertical Back Porch) lines.

VFP[12:0] bits (VFP Lines)

The VFP[12:0] bits control the number of VFP (Vertical Front Porch) lines.

66.2.51 VMHSSETR : Video Mode Horizontal Size Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x430

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	HACT[14:0]														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	HSPOL	—	—	—	HSA[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	HSA[11:0]	HSA Pixels Set the number of HSA (Horizontal Sync Active) pixels.	R/W
14:12	—	These bits are read as 0. The write value should be 0.	R/W
15	HSPOL	HSYNC Polarity 0: Active-high 1: Active-low	R/W
30:16	HACT[14:0]	HACT Pixels Set the number of HACT (Horizontal Active) pixels.	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HSA[11:0] bits (HSA Pixels)

The HSA[11:0] bits control the number of HSA (Horizontal Sync Active) pixels.

HSPOL bit (HSYNC Polarity)

The HSPOL bit controls the polarity of HSYNC signal.

HACT[14:0] bits (HACT Pixels)

The HACT[14:0] bits control the number of HACT (Horizontal Active) pixels.

66.2.52 VMHPSETR : Video Mode Horizontal Porch Set Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x434

Bit position:	31	30	29	28–16	15	14	13	12–0
Bit field:	—	—	—	HFP[12:0]	—	—	—	HBP[12:0]
Value after reset:	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
12:0	HBP[12:0]	HBP Pixels Set the number of HBP (Horizontal Back Porch) pixels.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	HFP[12:0]	HFP Pixels Set the number of HFP (Horizontal Front Porch) pixels.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HBP[12:0] bits (HBP Pixels)

The HBP[12:0] bits control the number of HBP (Horizontal Back Porch) pixels.

HFP[12:0] bits (HFP Pixels)

The HFP[12:0] bits control the number of HFP (Horizontal Front Porch) pixels.

66.2.53 SQCH0SET0R : Sequence Channel 0 Set 0 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x5C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	START	Sequence Operation Start 0: No operation 1: Start the sequence operation	W
22:1	—	These bits are read as 0. The write value should be 0.	W
23	—	This bit is read as 1. The write value should be 1.	W
31:24	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

START bit (Sequence Operation Start)

The START bit is the trigger to start the sequence operation. The sequence operations are executed in sequence from Descriptor-0.

Setting the START bit to 1 when the SQCH0SR.RUNNING flag or the SQCH1SR.RUNNING flag is set to 1 is prohibited.

66.2.54 SQCH0SR : Sequence Channel 0 Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x5D0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	RUNNING	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0.	R
2	RUNNING	Sequence Operation Running Status 0: Sequence operation is stopped 1: Sequence operation is running	R
3	—	This bit is read as 0.	R
4	AACTFIN	All Actions Finish Interrupt Flag 0: All actions of a descriptor have not finished 1: All actions of a descriptor finished operations	R
7:5	—	These bits are read as 0.	R
8	ADESFIN	All Descriptors Finish Interrupt Flag 0: All descriptors have not finished 1: All descriptors finished operations	R
23:9	—	These bits are read as 0.	R
24	TXIBERR	TX Internal Bus Error Interrupt Flag 0: No error 1: Internal AXI bus read error occurred	R
25	—	This bit is read as 0.	R
26	RXFERR	Receive Fatal Error Interrupt Flag 0: No error 1: Fatal timeout occurred during BTA	R
27	RXFAIL	Receive Fail Interrupt Flag 0: No error 1: Expected receive did not take place	R
28	RXPFAIL	Receive Packet Data Fail Interrupt Flag 0: No error 1: Received packet data is not stored correctly	R
29	RXCORERR	Receive Correctable Error Interrupt Flag 0: No error 1: Received packets have a correctable error	R
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Flag 0: No error 1: Acknowledge and Error Report Packet is received	R
31	—	This bit is read as 0.	R

Note: S-TYPE-3, P-TYPE-3

RUNNING bit (Sequence Operation Running Status)

The RUNNING bit indicates the state of the sequence operation.

Changing the settings of the descriptor is prohibited while the RUNNING bit is 1.

AACTFIN flag (All Actions Finish Interrupt Flag)

The AACTFIN flag is set if all actions of this descriptor are finished with SQCHnDSCmCR.FINACT = 1.

The AACTFIN flag is not cleared automatically, so set the SQCH0SCR.AACTFIN bit to 1 to clear the AACTFIN flag.

ADESFIN flag (All Descriptors Finish Interrupt Flag)

The ADESFIN flag is set when all descriptors finish operations.

When Descriptor-m whose SQCHnDSCmAR.NXACT[1:0] bits are set to 00b finishes the operation, the ADESFIN flag is set to 1.

The ADESFIN flag is not cleared automatically, so set the SQCH0SCR.ADESFIN bit to 1 to clear the ADESFIN flag.

TXIBERR flag (TX Internal Bus Error Interrupt Flag)

The TXIBERR flag is set to 1 when the internal AXI bus read has failed.

The TXIBERR flag is not cleared automatically, so set the SQCH0SCR.TXIBERR bit to 1 to clear the TXIBERR flag.

RXFERR flag (Receive Fatal Error Interrupt Flag)

The RXFERR flag is set to 1 when the fatal timeout occurred during BTA.

The FERRSR.TATO flag or the FERRSR.LRXHTO flag is also set to 1.

The RXFERR flag is not cleared automatically, so set the SQCH0SCR.RXFERR bit to 1 to clear the RXFERR flag.

RXFAIL flag (Receive Fail Interrupt Flag)

The RXFAIL flag is set to 1 when the expected receive did not occur.

The RXSR.PRTOERR flag, the RXSR.ECCERRM flag, the RXSR.MLFERR flag, or the RXSR.NORESERR flag is also set to 1.

The RXFAIL flag is not cleared automatically, so set the SQCH0SCR.RXFAIL bit to 1 to clear the RXFAIL flag.

RXPFAIL flag (Receive Packet Data Fail Interrupt Flag)

The RXPFAIL flag is set to 1 when the received packet header is stored correctly, but the packet data is not stored correctly.

The RXSR.CRCERR flag, the RXSR.WCERR flag, the RXSR.UNEXERR flag, the RXSR.RSIZEERR flag, the RXSR.RXOVFERR flag, or the RXSR.IBERR flag is also set to 1.

The RXPFAIL flag is not cleared automatically, so set the SQCH0SCR.RXPFAIL bit to 1 to clear the RXPFAIL flag.

RXCORERR flag (Receive Correctable Error Interrupt Flag)

The RXCORERR flag is set to 1 when the received packet has correctable errors.

The RXSR.ECCERRS flag is also set to 1.

The RXCORERR flag is not cleared automatically, so set the SQCH0SCR.RXCORERR bit to 1 to clear the RXCORERR flag.

RXAKE flag (Receive Acknowledge and Error Report Packet Interrupt Flag)

The RXAKE flag is set to 1 when the Acknowledge and Error Report Packet is received.

The RXSR.RXAKE flag is also set to 1.

The RXAKE flag is not cleared automatically, so set the SQCH0SCR.RXAKE bit to 1 to clear the RXAKE flag.

66.2.55 SQCH0SCR : Sequence Channel 0 Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x5D4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	W
4	AACTFIN	All Actions Finish Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.AACTFIN flag	W
7:5	—	These bits are read as 0. The write value should be 0.	W
8	ADESFIN	All Descriptors Finish Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.ADESFIN flag	W
23:9	—	These bits are read as 0. The write value should be 0.	W
24	TXIBERR	TX Internal Bus Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.TXIBERR flag	W
25	—	This bit is read as 0. The write value should be 0.	W
26	RXFERR	Receive Fatal Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.RXFERR flag	W
27	RXFAIL	Receive Fail Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.RXFAIL flag	W
28	RXPFAIL	Receive Packet Data Fail Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.RXPFAIL flag	W
29	RXCORERR	Receive Correctable Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.RXCORERR flag	W
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Flag Clear 0: No operation 1: Clear the SQCH0SR.RXAKE flag	W
31	—	This bit is read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

AACTFIN bit (All Actions Finish Interrupt Flag Clear)

Set the AACTFIN bit to 1 to clear the SQCH0SR.AACTFIN flag.

ADESFIN bit (All Descriptors Finish Interrupt Flag Clear)

Set the ADESFIN bit to 1 to clear the SQCH0SR.ADESFIN flag.

TXIBERR bit (TX Internal Bus Error Interrupt Flag Clear)

Set the TXIBERR bit to 1 to clear the SQCH0SR.TXIBERR flag.

RXFERR bit (Receive Fatal Error Interrupt Flag Clear)

Set the RXFERR bit to 1 to clear the SQCH0SR.RXFERR flag.

RXFAIL bit (Receive Fail Interrupt Flag Clear)

Set the RXFAIL bit to 1 to clear the SQCH0SR.RXFAIL flag.

RXPFAIL bit (Receive Packet Data Fail Interrupt Flag Clear)

Set the RXPFAIL bit to 1 to clear the SQCH0SR.RXPFAIL flag.

RXCORERR bit (Receive Correctable Error Interrupt Flag Clear)

Set the RXCORERR bit to 1 to clear the SQCH0SR.RXCORERR flag.

RXAKE bit (Receive Acknowledge and Error Report Packet Interrupt Flag Clear)

Set the RXAKE bit to 1 to clear the SQCH0SR.RXAKE flag.

66.2.56 SQCH0IER : Sequence Channel 0 Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x5D8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	AACTFIN	All Actions Finish Interrupt Enable 0: Disable 1: Enable	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	ADESFIN	All Descriptors Finish Interrupt Enable 0: Disable 1: Enable	R/W
23:9	—	These bits are read as 0. The write value should be 0.	R/W
24	TXIBERR	TX Internal Bus Error Interrupt Enable 0: Disable 1: Enable	R/W
25	—	This bit is read as 0. The write value should be 0.	R/W
26	RXFERR	Receive Fatal Error Interrupt Enable 0: Disable 1: Enable	R/W
27	RXFAIL	Receive Fail Interrupt Enable 0: Disable 1: Enable	R/W
28	RXPFAIL	Receive Packet Data Fail Interrupt Enable 0: Disable 1: Enable	R/W

Bit	Symbol	Function	R/W
29	RXCORERR	Receive Correctable Error Interrupt Enable 0: Disable 1: Enable	R/W
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Enable 0: Disable 1: Enable	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

AACTFIN bit (All Actions Finish Interrupt Enable)

The AACTFIN bit controls the All Actions Finish Interrupt permission.

When an All Actions Finish Interrupt occurs, the SQCH0SR.AACTFIN flag is set to 1.

To enable the All Actions Finish Interrupt, set the AACTFIN bit to 1.

ADESFIN bit (All Descriptors Finish Interrupt Enable)

The ADESFIN bit controls the All Descriptors Finish Interrupt permission.

When an All Descriptors Finish Interrupt occurs, the SQCH0SR.ADESFIN flag is set to 1.

To enable the All Descriptors Finish Interrupt, set the ADESFIN bit to 1.

TXIBERR bit (TX Internal Bus Error Interrupt Enable)

The TXIBERR bit controls the TX Internal Bus Error Interrupt permission.

When a TX Internal Bus Error Interrupt occurs, the SQCH0SR.TXIBERR flag is set to 1.

To enable the TX Internal Bus Error Interrupt, set the TXIBERR bit to 1.

RXFERR bit (Receive Fatal Error Interrupt Enable)

The RXFERR bit controls the Receive Fatal Error Interrupt permission.

When a Receive Fatal Error Interrupt occurs, the SQCH0SR.RXFERR flag is set to 1.

To enable the Receive Fatal Error Interrupt, set the RXFERR bit to 1.

RXFAIL bit (Receive Fail Interrupt Enable)

The RXFAIL bit controls the Receive Fail Interrupt permission.

When a Receive Fail Interrupt occurs, the SQCH0SR.RXFAIL flag is set to 1.

To enable the Receive Fail Interrupt, set the RXFAIL bit to 1.

RXPFAIL bit (Receive Packet Data Fail Interrupt Enable)

The RXPFAIL bit controls the Receive Packet Data Fail Interrupt permission.

When a Receive Packet Data Fail Interrupt occurs, the SQCH0SR.RXPFAIL flag is set to 1.

To enable the Receive Packet Data Fail Interrupt, set the RXPFAIL bit to 1.

RXCORERR bit (Receive Correctable Error Interrupt Enable)

The RXCORERR bit controls the Receive Correctable Error Interrupt permission.

When a Receive Correctable Error Interrupt occurs, the SQCH0SR.RXCORERR flag is set to 1.

To enable the Receive Correctable Error Interrupt, set the RXCORERR bit to 1.

RXAKE bit (Receive Acknowledge and Error Report Packet Interrupt Enable)

The RXAKE bit controls the Receive Acknowledge and Error Report Packet Interrupt permission.

When a Receive Acknowledge and Error Report Packet Interrupt occurs, the SQCH0SR.RXAKE flag is set to 1.

To enable the Receive Acknowledge and Error Report Packet Interrupt, set the RXAKE bit to 1.

66.2.57 SQCH1SET0R : Sequence Channel 1 Set 0 Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x600

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	START	Sequence Operation Start 0: No operation 1: Start the sequence operation	W
22:1	—	These bits are read as 0. The write value should be 0.	W
23	—	This bit is read as 1. The write value should be 1.	W
31:24	—	These bits are read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

START bit (Sequence Operation Start)

The START bit is the trigger to start the sequence operation. The sequence operations are executed in sequence from Descriptor-0.

Setting the START bit to 1 when the SQCH0SR.RUNNING flag or the SQCH1SR.RUNNING flag is set to 1 is prohibited.

66.2.58 SQCH1SR : Sequence Channel 1 Status Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x610

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	SIZEERR	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	RUNNING	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0.	R
2	RUNNING	Sequence Operation Running Status 0: Sequence operation is stopped 1: Sequence operation is running	R
3	—	This bit is read as 0.	R
4	AACTFIN	All Actions Finish Interrupt Flag 0: All actions of a descriptor have not finished 1: All actions of a descriptor finished operations	R

Bit	Symbol	Function	R/W
7:5	—	These bits are read as 0.	R
8	ADESFIN	All Descriptors Finish Interrupt Flag 0: All descriptors have not finished 1: All descriptors finished operations	R
18:9	—	These bits are read as 0.	R
19	SIZEERR	Packet Size Error Interrupt Flag 0: No error 1: Sequence packet size is too large	R
23:20	—	These bits are read as 0.	R
24	TXIBERR	TX Internal Bus Error Interrupt Flag 0: No error 1: Internal AXI bus read error occurred	R
25	—	This bit is read as 0.	R
26	RXFERR	Receive Fatal Error Interrupt Flag 0: No error 1: Fatal timeout occurred during BTA	R
27	RXFAIL	Receive Fail Interrupt Flag 0: No error 1: Expected receive did not take place	R
28	RXPFAIL	Receive Packet Data Fail Interrupt Flag 0: No error 1: Received packet data is not stored correctly	R
29	RXCORERR	Receive Correctable Error Interrupt Flag 0: No error 1: Received packets have correctable error	R
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Flag 0: No error 1: Acknowledge and Error Report Packet is received	R
31	—	This bit is read as 0.	R

Note: S-TYPE-3, P-TYPE-3

RUNNING bit (Sequence Operation Running Status)

The RUNNING bit indicates the state of the sequence operation.

Changing the settings of the descriptor is prohibited while the RUNNING bit is 1.

AACTFIN flag (All Actions Finish Interrupt Flag)

The AACTFIN flag is set if all actions of this descriptor are finished with SQCHnDSCmCR.FINACT =1.

The AACTFIN flag is not cleared automatically, so set the SQCH1SCR.AACTFIN bit to 1 to clear the AACTFIN flag.

ADESFIN flag (All Descriptors Finish Interrupt Flag)

The ADESFIN flag is set when all descriptors finish operations.

When Descriptor-m whose SQCHnDSCmAR.NXACT[1:0] bits are set to 00b finishes the operation, the ADESFIN flag is set to 1.

The ADESFIN flag is not cleared automatically, so set the SQCH1SCR.ADESFIN bit to 1 to clear the ADESFIN flag.

SIZEERR flag (Packet Size Error Interrupt Flag)

The SQCH1SR.SIZEERR flag is set to 1 when the sequence packet is too large.

The SQCH1SR.SIZEERR flag is not cleared automatically, so set the SQCH1SCR.SIZEERR bit to 1 to clear the SQCH1SR.SIZEERR flag.

TXIBERR flag (TX Internal Bus Error Interrupt Flag)

The TXIBERR flag is set to 1 when the internal AXI bus read has failed.

The TXIBERR flag is not cleared automatically, so set the SQCH1SCR.TXIBERR bit to 1 to clear the TXIBERR flag.

RXFERR flag (Receive Fatal Error Interrupt Flag)

The RXFERR flag is set to 1 when the fatal timeout occurred during BTA.

The FERRSR.TATO flag or the FERRSR.LRXHTO flag is also set to 1.

The RXFERR flag is not cleared automatically, so set the SQCH1SCR.RXFERR bit to 1 to clear the RXFERR flag.

RXFAIL flag (Receive Fail Interrupt Flag)

The RXFAIL flag is set to 1 when the expected receive did not occur.

The RXSR.PRTOERR flag, the RXSR.ECCERRM flag, the RXSR.MLFERR flag, or the RXSR.NORESERR flag is also set to 1.

The RXFAIL flag is not cleared automatically, so set the SQCH1SCR.RXFAIL bit to 1 to clear the RXFAIL flag.

RXPFAIL flag (Receive Packet Data Fail Interrupt Flag)

The RXPFAIL flag is set to 1 when the received packet header is stored correctly, but the packet data is not stored correctly.

The RXSR.CRCERR flag, the RXSR.WCERR flag, the RXSR.UNEXERR flag, the RXSR.RSIZEERR flag, the RXSR.RXOVFERR flag, or the RXSR.IBERR flag is also set to 1.

The RXPFAIL flag is not cleared automatically, so set the SQCH1SCR.RXPFAIL bit to 1 to clear the RXPFAIL flag.

RXCORERR flag (Receive Correctable Error Interrupt Flag)

The RXCORERR flag is set to 1 when the received packet has correctable errors.

The RXSR.ECCERRS flag is also set to 1.

The RXCORERR flag is not cleared automatically, so set the SQCH1SCR.RXCORERR bit to 1 to clear the RXCORERR flag.

RXAKE flag (Receive Acknowledge and Error Report Packet Interrupt Flag)

The RXAKE flag is set to 1 when the Acknowledge and Error Report Packet is received.

The RXSR.RXAKE flag is also set to 1.

The RXAKE flag is not cleared automatically, so set the SQCH1SCR.RXAKE bit to 1 to clear the RXAKE flag.

66.2.59 SQCH1SCR : Sequence Channel 1 Status Clear Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x614

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	SIZEERR	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	W
4	AACTFIN	All Actions Finish Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.AACTFIN flag	W
7:5	—	These bits are read as 0. The write value should be 0.	W

Bit	Symbol	Function	R/W
8	ADESFIN	All Descriptors Finish Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.ADESFIN flag	W
18:9	—	These bits are read as 0. The write value should be 0.	W
19	SIZEERR	Packet Size Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.SIZEERR flag	W
23:20	—	These bits are read as 0. The write value should be 0.	W
24	TXIBERR	TX Internal Bus Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.TXIBERR flag	W
25	—	This bit is read as 0. The write value should be 0.	W
26	RXFERR	Receive Fatal Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.RXFERR flag	W
27	RXFAIL	Receive Fail Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.RXFAIL flag	W
28	RXPFAIL	Receive Packet Data Fail Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.RXPFAIL flag	W
29	RXCORERR	Receive Correctable Error Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.RXCORERR flag	W
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Flag Clear 0: No operation 1: Clear the SQCH1SR.RXAKE flag	W
31	—	This bit is read as 0. The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

AACTFIN bit (All Actions Finish Interrupt Flag Clear)

Set the AACTFIN bit to 1 to clear the SQCH1SR.AACTFIN flag.

ADESFIN bit (All Descriptors Finish Interrupt Flag Clear)

Set the ADESFIN bit to 1 to clear the SQCH1SR.ADESFIN flag.

SIZEERR bit (Packet Size Error Interrupt Flag Clear)

Set the SIZEERR bit to 1 to clear the SQCH1SR.SIZEERR flag.

TXIBERR bit (TX Internal Bus Error Interrupt Flag Clear)

Set the TXIBERR bit to 1 to clear the SQCH1SR.TXIBERR flag.

RXFERR bit (Receive Fatal Error Interrupt Flag Clear)

Set the RXFERR bit to 1 to clear the SQCH1SR.RXFERR flag.

RXFAIL bit (Receive Fail Interrupt Flag Clear)

Set the RXFAIL bit to 1 to clear the SQCH1SR.RXFAIL flag.

RXPFAIL bit (Receive Packet Data Fail Interrupt Flag Clear)

Set the RXPFAIL bit to 1 to clear the SQCH1SR.RXPFAIL flag.

RXCORERR bit (Receive Correctable Error Interrupt Flag Clear)

Set the RXCORERR bit to 1 to clear the SQCH1SR.RXCORERR flag.

RXAKE bit (Receive Acknowledge and Error Report Packet Interrupt Flag Clear)

Set the RXAKE bit to 1 to clear the SQCH1SR.RXAKE flag.

66.2.60 SQCH1IER : Sequence Channel 1 Interrupt Enable Register

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x618

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	RXAKE	RXCORERR	RXPFAIL	RXFAIL	RXFERR	—	TXIBERR	—	—	—	—	SIZEERR	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADESFIN	—	—	—	AACTFIN	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	AACTFIN	All Actions Finish Interrupt Enable 0: Disable 1: Enable	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	ADESFIN	All Descriptors Finish Interrupt Enable 0: Disable 1: Enable	R/W
18:9	—	These bits are read as 0. The write value should be 0.	R/W
19	SIZEERR	Packet Size Error Interrupt Enable 0: Disable 1: Enable	R/W
23:20	—	These bits are read as 0. The write value should be 0.	R/W
24	TXIBERR	TX Internal Bus Error Interrupt Enable 0: Disable 1: Enable	R/W
25	—	This bit is read as 0. The write value should be 0.	R/W
26	RXFERR	Receive Fatal Error Interrupt Enable 0: Disable 1: Enable	R/W
27	RXFAIL	Receive Fail Interrupt Enable 0: Disable 1: Enable	R/W
28	RXPFAIL	Receive Packet Data Fail Interrupt Enable 0: Disable 1: Enable	R/W
29	RXCORERR	Receive Correctable Error Interrupt Enable 0: Disable 1: Enable	R/W
30	RXAKE	Receive Acknowledge and Error Report Packet Interrupt Enable 0: Disable 1: Enable	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

AACTFIN bit (All Actions Finish Interrupt Enable)

The AACTFIN bit controls the All Actions Finish Interrupt permission.

When an All Actions Finish Interrupt occurs, the SQCH1SR.AACTFIN flag is set to 1.

To enable the All Actions Finish Interrupt, set the AACTFIN bit to 1.

ADESFIN bit (All Descriptors Finish Interrupt Enable)

The ADESFIN bit controls the All Descriptors Finish Interrupt permission.

When an All Descriptors Finish Interrupt occurs, the SQCH1SR.ADESFIN flag is set to 1.

To enable the All Descriptors Finish Interrupt, set the ADESFIN bit to 1.

SIZEERR bit (Packet Size Error Interrupt Enable)

The SIZEERR bit controls the Packet Size Error Interrupt permission.

When a Packet Size Error Interrupt occurs, the SQCH1SR.SIZEERR flag is set to 1.

To enable the Packet Size Error Interrupt, set the SIZEERR bit to 1.

TXIBERR bit (TX Internal Bus Error Interrupt Enable)

The TXIBERR bit controls the Tx Internal Bus Error Interrupt permission.

When a TX Internal Bus Error Interrupt occurs, the SQCH1SR.TXIBERR flag is set to 1.

To enable the TX Internal Bus Error Interrupt, set the TXIBERR bit to 1.

RXFERR bit (Receive Fatal Error Interrupt Enable)

The RXFERR bit controls the Receive Fatal Error Interrupt permission.

When a Receive Fatal Error Interrupt occurs, the SQCH1SR.RXFERR flag is set to 1.

To enable the Receive Fatal Error Interrupt, set the RXFERR bit to 1.

RXFAIL bit (Receive Fail Interrupt Enable)

The RXFAIL bit controls the Receive Fail Interrupt permission.

When a Receive Fail Interrupt occurs, the SQCH1SR.RXFAIL flag is set to 1.

To enable the Receive Fail Interrupt, set the RXFAIL bit to 1.

RXPFAIL bit (Receive Packet Data Fail Interrupt Enable)

The RXPFAIL bit controls the Receive Packet Data Fail Interrupt permission.

When a Receive Packet Data Fail Interrupt occurs, the SQCH1SR.RXPFAIL flag is set to 1.

To enable the Receive Packet Data Fail Interrupt, set the RXPFAIL bit to 1.

RXCORERR bit (Receive Correctable Error Interrupt Enable)

The RXCORERR bit controls the Receive Correctable Error Interrupt permission.

When a Receive Correctable Error Interrupt occurs, the SQCH1SR.RXCORERR flag is set to 1.

To enable the Receive Correctable Error Interrupt, set the RXCORERR bit to 1.

RXAKE bit (Receive Acknowledge and Error Report Packet Interrupt Enable)

The RXAKE bit controls the Receive Acknowledge and Error Report Packet Interrupt permission.

When a Receive Acknowledge and Error Report Packet Interrupt occurs, the SQCH1SR.RXAKE flag is set to 1.

To enable the Receive Acknowledge and Error Report Packet Interrupt, set the RXAKE bit to 1.

66.2.61 SQCHnDSCmAR : Sequence Channel n Descriptor-m A Register (n = 0 to 1, m = 0 to 7)

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x780 + (0x10 × m) + (0x80 × n)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	NXACT[1:0]		BTA[1:0]		SPD	FMT	VC[1:0]		DT[5:0]					
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA1[7:0]								DATA0[7:0]							
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	DATA0[7:0]	Data 0 Set Data0 of TX Packet Header.*1	R/W
15:8	DATA1[7:0]	Data 1 Set Data1 of TX Packet Header.*1	R/W
21:16	DT[5:0]	Data Type Set the data type of TX Packet Header.	R/W
23:22	VC[1:0]	Virtual Channel Set the ID of the virtual channel of the TX Packet Header.	R/W
24	FMT	Format Set the packet format of TX Packet Header. 0: Short Packet 1: Long Packet	R/W
25	SPD	Speed Set the speed type for TX request. 0: High-speed 1: Low-power	R/W
27:26	BTA[1:0]	Bus Turn Around Set either TX request without Bus Turn-Around (BTA) or no-operation. Or else, set TX request with BTA. 0 0: TX request without BTA or no-operation 0 1: TX non-read request with BTA 1 0: TX read request with BTA 1 1: BTA only	R/W
29:28	NXACT[1:0]	Next Action 0 0: Terminate the sequence operation after this descriptor processing is finished 0 1: Start the next descriptor processing after this descriptor processing is finished Others: Settings prohibited	R/W
31:30	—	The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The maximum size for the Command Transfer mode of Long Packet using Packet Payload Data Register (SQCHnDSCmBR.DTSEL[1:0] = 00b and SQCHnDSCmAR.FMT = 1) is 16 bytes.

This register is not initialized by any reset. Therefore, if sequence operation is used, be sure to set the values according to the specified description for the reserved area as well.

DATA0[7:0] bits (Data 0)

When the TX Packet format is Long Packet (FMT = 1), this value is the lower 8 bits of the Word Count (WC).

DATA1[7:0] bits (Data 1)

When the TX Packet format is Long Packet (FMT = 1), this value is the upper 8 bits of the Word Count (WC).

DT[5:0] bits (Data Type)

The DT[5:0] bits set the data type of the TX Packet Header.

VC[1:0] bits (Virtual Channel)

The VC[1:0] bits set the ID of the virtual channel of the TX Packet Header.

FMT bit (Format)

The FMT bit sets the packet format of the TX Packet Header. This field should be set by the data type.

SPD bit (Speed)

The SPD bit selects the speed type for TX request. It is prohibited to set the SPD bit to 1 during the Video Mode operation.

BTA[1:0] bits (Bus Turn Around)

The BTA[1:0] bits set either TX request without Bus Turn-Around (BTA) or no-operation. Or else, set TX request with BTA.

NXACT[1:0] bits (Next Action)

The NXACT[1:0] bits select the next action after the Descriptor-m processing is finished.

When the SQCHnDSC7AR.NXACT[1:0] bits for Descriptor-7 are set to 01b (start the next descriptor processing after this descriptor processing is finished), the next descriptor after Descriptor-7 is Descriptor-0.

66.2.62 SQCHnDSCmBR : Sequence Channel n Descriptor-m B Register (n = 0 to 1, m = 0 to 7)

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x784 + (0x10 × m) + (0x80 × n)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	DTSEL[1:0]		—	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
23:0	—	The write value should be 0.	R/W
25:24	DTSEL[1:0]	Data Select Select the data buffer for long packet payload 0 0: Use Packet Payload Data Register (TXPPDxR, RXPPDxR) 0 1: Use memory space Others Setting prohibited	R/W
31:26	—	The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register is not initialized by any reset. Therefore, if sequence operation is used, be sure to set the values according to the specified description for the reserved area as well.

DTSEL[1:0] bits (Data Select)

The DTSEL[1:0] bits select the data source for long packet payload. Set the DTSEL[1:0] bits to 00b for write request with short packet.

When the SQCHnDSCmBR.DTSEL[1:0] bits are 00b, DSI Host uses Packet Payload Data Register (TXPPDxR and RXPPDxR, x = 0 to 3) for the long packet payload data buffer. In this setting, the long packet write request data size is the same or lower than 16 bytes. When the read request data to save has over 16-bytes word count, it reports as an error.

When the SQCHnDSCmBR.DTSEL[1:0] bits are 01b, the DSI Host uses memory space for the long packet payload data buffer. DSI Host transfers data stored in memory space at the address specified in the SQCHnDSCmDR register at the write request. DSI Host stores data to memory space at the address specified in the SQCHnDSCmDR register at the read request. The read header is always saved to the RXRSSxR register.

66.2.63 SQCHnDSCmCR : Sequence Channel n Descriptor-m C Register (n = 0 to 1, m = 0 to 7)

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x788 + (0x10 × m) + (0x80 × n)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ACTCODE[7:0]								—	AUXOP	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FINACT
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
0	FINACT	Finish Action Select whether to set the SQCHnSR.AACTFIN bit to 1 when all actions of this Descriptor-m are finished. 0: Disable 1: Enable	R/W
21:1	—	The write value should be 0.	R/W
22	AUXOP	Auxiliary Operation 0: Not using auxiliary operation 1: Executing auxiliary operation*1	R/W
23	—	The write value should be 0.	R/W
31:24	ACTCODE[7:0]	Action Code When not using auxiliary operation (SQCHnDSCmCR.AUXOP = 0): Set the slot number for storing the received result during the BTA action.*2 0 0 0 0 0 0 0 0: Slot-0 (RXRSS0R) 0 0 0 0 0 0 0 1: Slot-1 (RXRSS1R) 0 0 0 0 0 0 1 0: Slot-2 (RXRSS2R) 0 0 0 0 0 0 1 1: Slot-3 (RXRSS3R) Other settings are prohibited. When executing an auxiliary operation (SQCHnDSCmCR.AUXOP = 1): Select the action for the auxiliary operation. 0 0 0 0 0 0 0 0: Send Reset-Trigger 0 0 0 0 1 0 0 0: No-operation Other settings are prohibited.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. It is prohibited to set this bit to 1 while the Video mode operation is running.
Note 2. Setting the same slot number to the SQCHnDSCmCR.ACTCODE[7:0] bits of a different descriptor is prohibited.

This register is not initialized by any reset. Therefore, if sequence operation is used, be sure to set the values according to the specified description for the reserved area as well.

FINACT bit (Finish Action)

The FINACT bit controls whether to set the SQCHnSR.AACTFIN bit to 1 when all actions of this Descriptor-m are finished. Because the operation of each descriptor is not the same, the descriptors may finish out of order. For example, Descriptor-1 may finish before Descriptor-0.

AUXOP bit (Auxiliary Operation)

The AUXOP bit selects whether to execute the auxiliary operation or not. The action of the auxiliary operation is specified by the SQCHnDSCmCR.ACTCODE[7:0] bits. When this AUXOP bit is set to 1, the SQCHnDSCmAR.BTA[1:0] bits should be set to 00b.

ACTCODE[7:0] bits (Action Code)

When not using auxiliary operation (SQCHnDSCmCR.AUXOP = 0), the ACTCODE[7:0] bits select the slot number for storing the received result during the BTA action. Set these bits to 0x00 for non-BTA action. Setting the same slot number to the SQCHnDSCmCR.ACTCODE[7:0] bits of a different descriptor is prohibited.

When executing the auxiliary operation (SQCHnDSCmCR.AUXOP = 1), the ACTCODE[7:0] bits select the action that can be taken as the auxiliary operation.

66.2.64 SQCHnDSCmDR : Sequence Channel n Descriptor-m D Register (n = 0 to 1, m = 0 to 7)

Base address: MIPI_DSI = 0x4034_6000
MIPI_DSI_NS = 0x5034_6000

Offset address: 0x78C + (0x10 × m) + (0x80 × n)

Bit position:	31 … 0
Bit field:	LADDR[31:0]
Value after reset:	x x x x x x x x x x x x x x x x x x x x x x x x x x x x x x x x

Bit	Symbol	Function	R/W
31:0	LADDR[31:0]	Lower Address Set the lower 32-bit address of memory space to store the long packet payload data for Sequence Operation Channel-n Descriptor-m.	R/W

Note: S-TYPE-3, P-TYPE-3

The SQCHnDSCmDR register is not initialized by any reset. Therefore, if sequence operation is used, be sure to set all the bits.

LADDR[31:0] bits (Lower Address)

When memory space is selected for the long packet payload (SQCHnDSCmBR.DTSEL[1:0] = 01b) and the Sequence mode operation format is long packet (SQCHnDSCmAR.FMT = 1), the DSI Host accesses memory space at the address specified in these LADDR[31:0] bits. For write operation, the DSI Host transfers data stored in memory space. For read operation, the DSI Host stores data to memory space.

66.3 Operation

66.3.1 Interrupt

Table 66.2 Interrupt signals (1 of 3)

Interrupt signal	Category	Interrupt condition	Interrupt source register
DSI_SEQ0	Sequence Operation Channel 0 interrupt	Receive Acknowledge and Error Report Packet	SQCH0SR.RXAKE
		Receive Correctable Error	SQCH0SR.RXCORERR
		Receive Packet Data Fail	SQCH0SR.RXPFAIL
		Receive Fail	SQCH0SR.RXFAIL
		Receive Fatal Error	SQCH0SR.RXFERR
		Tx Internal Bus Error	SQCH0SR.TXIBERR
		All-Descriptors Finish	SQCH0SR.ADESFIN
		All Actions Finish	SQCH0SR.AACTFIN

Table 66.2 Interrupt signals (2 of 3)

Interrupt signal	Category	Interrupt condition	Interrupt source register
DSI_SEQ1	Sequence Operation Channel 1 interrupt	Receive Acknowledge and Error Report Packet	SQCH1SR.RXAKE
		Receive Correctable Error	SQCH1SR.RXCORERR
		Receive Packet Data Fail	SQCH1SR.RXPFAIL
		Receive Fail	SQCH1SR.RXFAIL
		Receive Fatal Error	SQCH1SR.RXFERR
		Tx Internal Bus Error	SQCH1SR.TXIBERR
		Packet Size Error	SQCH1SR.SIZEERR
		All-Descriptors Finish	SQCH1SR.ADESFIN
		All Actions Finish	SQCH1SR.AACTFIN
DSI_VIN1	Video Mode Operation Channel 1 interrupt	Video Buffer Overflow Error	VMSR.VBUFOVF
		Video Buffer Underflow	VMSR.VBUFUDF
		Timing Error	VMSR.TIMERR
		Video Mode Operation Ready	VMSR.VIRDY
		Video Mode Operation Stop	VMSR.STOP
		Video Mode Operation Start	VMSR.START
DSI_RCV	DSI Packet Receive interrupt	Acknowledge and Error Report Receive	RXSR.RXAKE
		Single Bit ECC Error	RXSR.ECCERRS
		Return Packet Size Error	RXSR.RSIZEERR
		No Response Error	RXSR.NORESERR
		Peripheral Response Timeout Error	RXSR.PRTOERR
		Receive Buffer Overflow Error	RXSR.RXOVFERR
		Internal Bus Error	RXSR.IBERR
		CRC Error	RXSR.CRCERR
		Word Count Error	RXSR.WCERR
		Unexpected Packet Error	RXSR.UNEXERR
		Multi Bit ECC Error	RXSR.ECCERRM
		Malformed Error	RXSR.MLFERR
		External Tearing Effect Detect	RXSR.EXTEDET
		ACK Trigger Receive	RXSR.RXACK
		EoTp Receive	RXSR.RXEOTP
		Response Packet Receive	RXSR.RXRESP
		Turnaround Acknowledge Timeout	RXSR.TATO
		LP-RX Host Processor Timeout	RXSR.LRXHTO
		BTA Request End	RXSR.BTAREND

Table 66.2 Interrupt signals (3 of 3)

Interrupt signal	Category	Interrupt condition	Interrupt source register
DSI_FERR	DSI Fatal Error interrupt	LP1 Contention Error	FERRSR.CLP1
		LP0 Contention Error	FERRSR.CLP0
		Control Error	FERRSR.CTRL
		LPDT Sync Error	FERRSR.SYNCESC
		Escape mode Entry Error	FERRSR.ESCENT
		Turnaround Acknowledge Timeout	FERRSR.TATO
		LP-RX Host Processor Timeout	FERRSR.LRXHTO
		HS TX Timeout Interrupt	FERRSR.HTXTO
DSI_PPI	DSI D-PHY PPI interrupt	Data Lane transition from ULPS	PLSR.DLULPEXT
		Data Lane transition to ULPS	PLSR.DLULPENT
		Clock Lane mode transition from HS to LP	PLSR.CLHS2LP
		Clock Lane mode transition from LP to HS	PLSR.CLLP2HS
		Clock Lane transition from ULPS	PLSR.CLULPEXT
		Clock Lane transition to ULPS	PLSR.CLULPENT
		Data Lane-0 direction transition from transmitter to receiver	PLSR.DL0TX2RX
		Data Lane-0 direction transition from receiver to transmitter	PLSR.DL0RX2TX

The relationship between interrupt signals and interrupt conditions is shown in Table 66.2

The interrupt signal consists of combinatorial logic shown in Figure 66.3. The interrupt signal from a Status register is enabled by the signal related Interrupt Enable register. These interrupt signals are bound by the OR logic. All interrupt signals are synchronized to pclk and the level signal, and are active-high.

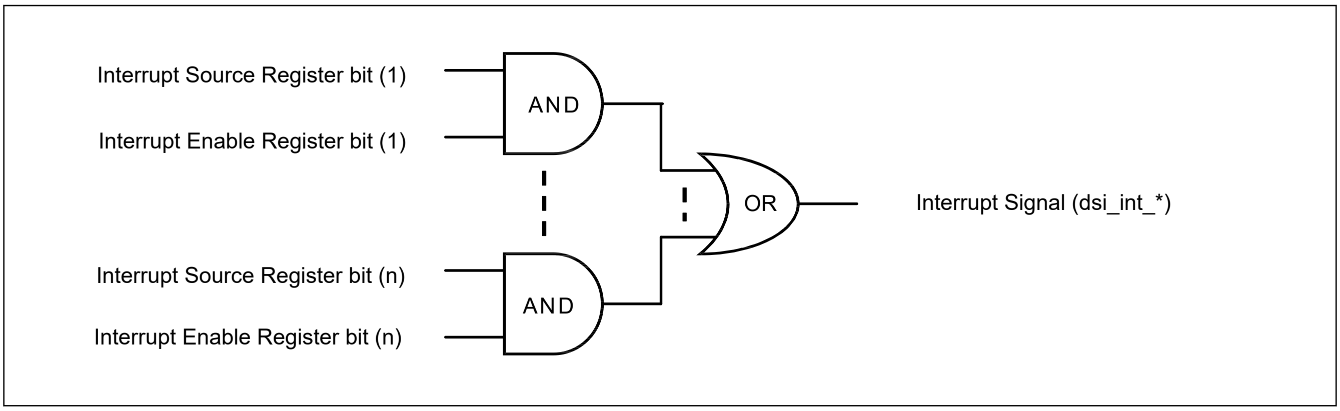


Figure 66.3 Interrupt signal output

66.3.2 Software Reset

The software reset is for the initialization of the DSI Host.*1

Initializing by a software reset is optional. To initialize the DSI Host using a software reset, use the following steps.

1. Set the RSTCR.SWRST bit to 1 for a software reset.
2. Wait for the software reset process to start.
   Wait until the following bits are set to 1.
   (RSTSR.RSTHS, RSTSR.RSTLP, RSTSR.RSTAPB, RSTSR.RSTAXI, RSTSR.RSTV).
3. Set the following registers: (optional)

(TXSETR, ULPSSETR, DSISETR, CLSTPTSETR, LPTRNSTSETR, PRESPTOBTASETR, PRESPTOLPSETR, PRESPTOHSSETR, HSTXTOSETR, LRXHTOSETR, TATOSETR).
These registers are allowed to be changed only during this process or initialization at initial startup.

4. Set the RSTCR.FTXSTP bit to 1 to transition to TX-Stop state (LP-11) for the data lane.
5. Wait for the valid data lane to transition to the stop state.
Wait until the RSTSR.DL0DIR bit is cleared to 0 and the RSTSR.DL1STP and DL0STP bits are set to 1 (RSTSR.DL1STP = 0 and RSTSR.DL0STP = 1 for a single lane).
6. Clear the RSTCR.FTXSTP bit to 0 to finish the operation of the transition to TX-Stop state.
7. Clear the RSTCR.SWRST bit to 0 to finish the software reset process.
8. Wait for the software reset process to end.
Wait until the following bits are cleared to 0.
(RSTSR.RSTHS, RSTSR.RSTLP, RSTSR.RSTAPB, RSTSR.RSTAXI, RSTSR.RSTV).

Note 1. The following registers are not initialized by the software reset.
- RSTCR
- RSTSR
- SQCHnDSCmAR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmBR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmCR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmDR (n = 0 to 1, m = 0 to 7)
- TXSETR
- ULPSSETR
- DSISETR
- CLSTPTSETR
- LPTRNSTSETR
- PRESPTOBTASETR
- PRESPTOLPSETR
- PRESPTOHSSETR
- HSTXTOSETR
- LRXHTOSETR
- TATOSETR

66.3.3 Start of HS Clock

HS clock is controlled by software. Set the registers according to the following flow.

1. Set the TXSETR.CLEN bit to 1 to enable the D-PHY clock lane before starting the Video mode operation or Sequence mode operation.
2. Set the HSCLKSETR.HSCLST bit to 1. At the same time, set the HSCLKSETR.HSCLMD bit to 1 when using the Continuous Clock mode.
In the Continuous Clock mode, the clock lane is always put in the HS state.
In the Non-Continuous Clock mode, the DSI Host automatically controls the transition between LP and HS states of the clock lane.
3. Wait until the PLSR.CLLP2HS bit is set 1 (only in the Continuous Clock mode).

66.3.4 Stop of HS Clock

1. Clear the HSCLKSETR.HSCLST bit to 0 to transition the clock lane to the LP state.
2. Wait until the PLSR.CLHS2LP bit is set 1 (only in the Continuous clock mode).
You can also check the status of the clock lane by reading the PLSR.CLSTP bit.

66.3.5 Command Mode Operation

The DSI Host supports two basic modes of operation.

- Command mode
- Video mode

This section describes how to operate in Command mode.

The Command mode of this DSI Host is intended to be used for communication such as configuration settings for peripheral before sending video signals.

In Command mode, a descriptor list that describes the sending and receiving process in advance is used.

Processing using descriptors is referred to as Sequence operation.

The DSI Host has two channels for Sequence operation (Sequence operation channel n, n = 0 to 1).

Channel 0 supports only LP mode (LP-TX, LP-RX), while channel 1 supports LP mode (LP-TX, LP-RX) and HS mode (HS-TX). There are eight descriptors for each channel.

The Command mode of this DSI Host supports only Sequence operation using these descriptors.

Each descriptor has four control registers for Sequence operation (SQCHnDSCmAR, SQCHnDSCmBR, SQCHnDSCmCR, SQCHnDSCmDR) that specify the transmit and receive operations.

The payload data for Long packet is stored in memory space or Packet Payload Data Register, which is shared among descriptors. Memory space for transfer is dedicated for channel 0 and channel 1, respectively. (Figure 66.4)

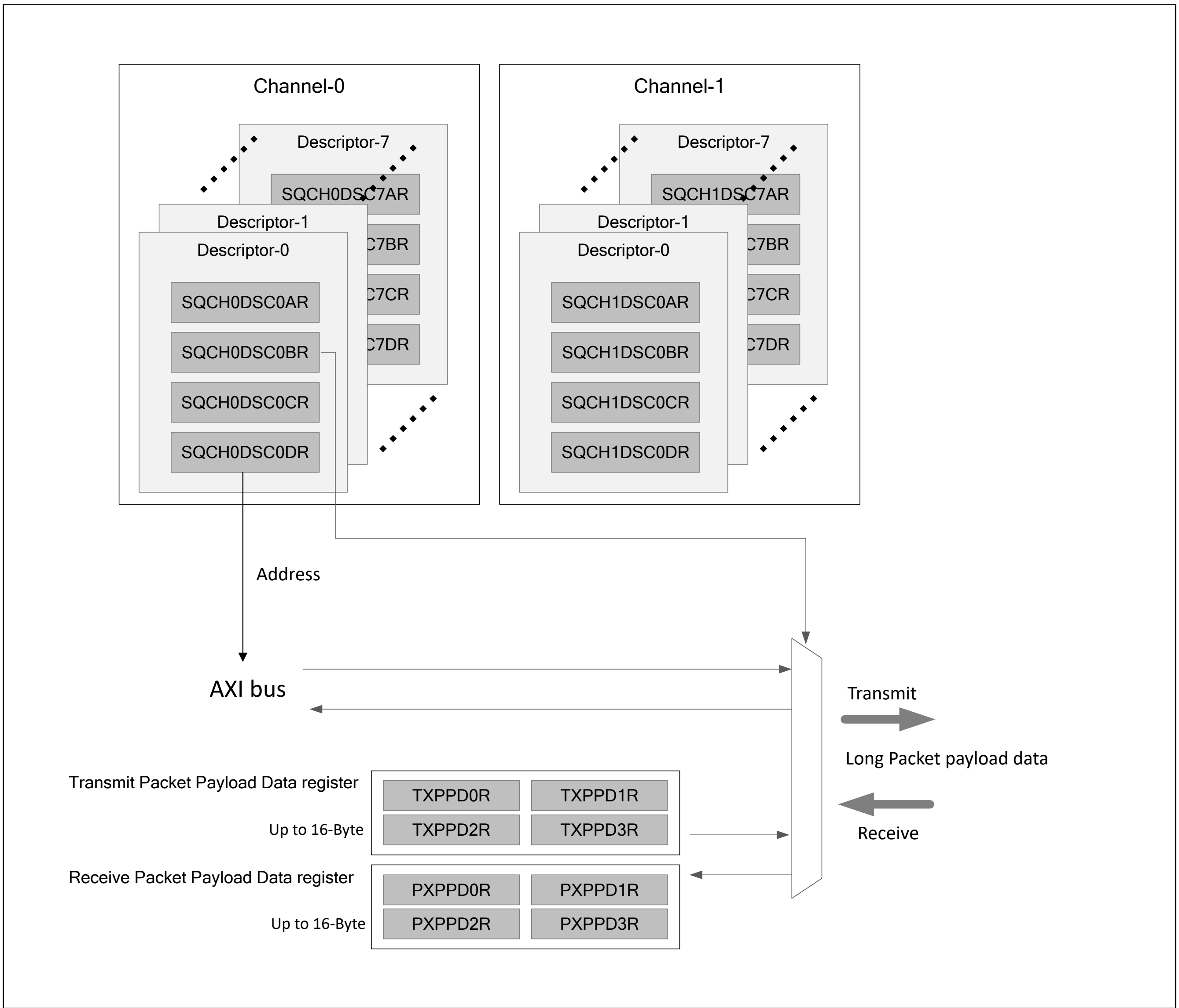


Figure 66.4 Descriptors and packet data buffers for Sequence operation

66.3.5.1 Basic Running Sequence

By setting the SQCHnSET0R.START bit to 1, the Sequence operation channel n starts processing the descriptor. When the process of Descriptor m which the SQCHnDSCmAR.NXACT[1:0] bits are set to 00b is completed, the sequence operations stop.

After setting the SQCHnSET0R.START bit to 1, the SQCHnSR.RUNNING bit is set to 1. Once the SQCHnSR.RUNNING bit is set to 1, it is prohibited to change the settings of the following registers until the Sequence operation is stopped.

- SQCHnDSCmAR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmBR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmCR (n = 0 to 1, m = 0 to 7)
- SQCHnDSCmDR (n = 0 to 1, m = 0 to 7)

66.3.5.2 Single Packet Transmission

This section describes the transmission of the non-read packets in Command mode using Sequence operation.

In the non-read packet transmission, set the following registers before sending the packets.

- SQCHnDSCmAR.NXACT[1:0] : 00b
- SQCHnDSCmAR.FMT : 0 (Short packet) or 1 (Long packet)
- SQCHnDSCmAR.SPD : 0 (High speed) or 1 (Low speed)
- SQCHnDSCmAR.BTA[1:0] : 00b (without BTA) or 01b (with BTA)
- SQCHnDSCmAR.VC[1:0]
- SQCHnDSCmAR.DT[5:0]
- SQCHnDSCmAR.DATA0[7:0]
- SQCHnDSCmAR.DATA1[7:0]
- SQCHnDSCmBR.DTSEL[1:0] : Specify where to store the payload data for the non-read packets

  00b: TXPPDxR (x = 0 to 3). Payloads size is limited up to 16 bytes.

  01b: Memory space. Payloads size is limited up to 4 KB.
- SQCHnDSCmCR.AUXOP : 0

When setting up a Long packet (SQCHnDSCmAR.FMT = 1), the payload data should be in the format of little endian at the location specified by the SQCHnDSCmBR.DTSEL[1:0] bits.

When setting up a Short packet (SQCHnDSCmAR.FMT = 0), no payload data is required.

After the above settings, send a non-read packet by setting the SQCHnSET0R.START bit to 1. When single packet transmission is complete, the SQCHnSR.ADESFIN bit is set to 1.

When the SQCHnDSCmAR.BTA[1:0] bits are set to 01b, BTA takes place after sending a non-read packet. After DSI Host gives bus ownership to the peripheral, an ACK trigger message or an Acknowledge and Error Report packet may be returned from the peripheral.

You can get a summary of the reception result by reading the RXRSSxR register with the number x specified in the SQCHnDSCmCR.ACTCODE[7:0] bits, and you can get the reception result details by reading the RXSR register.

66.3.5.3 Single Packet Reception

This section describes sending the read packets and receiving the response packets in Command mode using Sequence operation.

Since the read packet is a Short packet, set the following registers before sending the packet.

- SQCHnDSCmAR.NXACT[1:0] : 00b
- SQCHnDSCmAR.FMT : 0 (Short packet)
- SQCHnDSCmAR.SPD : 0 (High speed) or 1 (Low speed)

- SQCHnDSCmAR.BTA[1:0] : 10b (Read request with BTA)
- SQCHnDSCmAR.VC[1:0]
- SQCHnDSCmAR.DT[5:0]
- SQCHnDSCmAR.DATA0[7:0]
- SQCHnDSCmAR.DATA1[7:0]
- SQCHnDSCmBR.DTSEL[1:0] : Specify where to store the payload data for response packets

  00b : RXPPDxR (x = 0 to 3). Payloads size is limited up to 16 bytes.

  01b : Memory space. Payloads size is limited up to 128 bytes.
- SQCHnDSCmCR.AUXOP : 0
- SQCHnDSCmCR.ACTCODE[7:0] : Received result save slot number

When the SQCHnDSCmBR.DTSEL[1:0] bits are set to 00b, the payload data reception larger than 16 bytes is not supported. If there is a possibility of receiving the payload data larger than 16 bytes, set the SQCHnDSCmBR.DTSEL[1:0] bits to 01b.

After the above settings, send a read packet by setting the SQCHnSET0R.START bit to 1.

BTA takes place after sending the read packet. After DSI Host gives bus ownership to the peripheral, DSI Host receives a response packet. At this time, DSI Host may also receive an Acknowledge and Error report packet at the same time.

When single packet reception is complete, the SQCHnSR.ADESFIN bit is set to 1.

You can get a summary of the reception result by reading the RXRSSxR register with the number x specified in the SQCHnDSCmCR.ACTCODE[7:0] bits, and you can get the reception result details by reading the RXSR register.

If the response packet received is a Short packet or Long packet with WC = 0, the SQCHnDSCmBR.DTSEL[1:0] bits setting is not meaningful.

66.3.5.4 Reset Trigger Transmission

DSI Host can send the reset trigger in Command mode using Sequence operation.

Set the following registers before starting the Sequence operation.

- SQCHnDSCmAR.NXACT[1:0] : 00b
- SQCHnDSCmCR.AUXOP : 1
- SQCHnDSCmCR.ACTCODE[7:0] : 0x00

After the above settings, send a reset trigger by setting the SQCHnSET0R.START bit to 1.

The SQCHnSR.ADESFIN bit is set to 1 when a reset trigger transmission is complete.

66.3.5.5 Execute Operation in Sequence

This section describes the behavior of the descriptor in Command mode using Sequence operation.

By setting the SQCHnSET0R.START bit to 1, Sequence operation channel n is enabled, and processing is performed from Descriptor 0.

Once Descriptor m process is complete, the next process follows the value of the SQCHnDSCmAR.NXACT[1:0] bits.

When the SQCHnDSCmAR.NXACT[1:0] bits is 00b : After Descriptor m processing is complete, Sequence operation channel n stops.

When the SQCHnDSCmAR.NXACT[1:0] bits is 01b : After Descriptor m processing is complete, Descriptor (m + 1) processing begins.

66.3.5.6 Receiving EoTp

DSI Host can receive the End of Transmission packet (EoTp). When DSI Host receives an EoTp, the RXSR.RXEOTP bit is set to 1.

66.3.5.7 Acknowledge and Error Reporting Mechanism

The reliability of the serial bus can be determined using the Acknowledge and Error reporting mechanism.

If the peripheral detects an error on the serial bus in the peripheral-to-host transmission, the peripheral responds with an Acknowledge and Error Report packet. When DSI Host receives this packet, it sets the RXRSSxR.RXAKE bit, the RXSR.RXAKE bit, and the SQCHnSR.RXAKE bit to 1. The virtual channel identifier and the error report for the latest received Acknowledge and Error Report packet is stored in the AKEPLATIR register and the accumulated information is displayed in the AKEPACMSR register.

66.3.6 Video Mode Operation

66.3.6.1 Start of Video Mode Operation

The procedure to start operating in Video mode is as follows:

1. Set the Video mode parameters (VMPPSETR, VMVSSETR, VMVPSETR, VMHSSETR, VMHPSETR).
2. Set the VMSET1R.DLY[11:0] bits.
3. Set the VMSET0R.VSTART bit to 1 with other VMSET0R setting values.
4. Wait until the VMSR.VIRDY bit is set to 1.
5. Set the BG_EN.VEN bit and GB_EN.EN bit to 1 at the same time for the GLCDC. By this setting, the output of the vertical and horizontal synchronization signals VSYNC, HSYNC, and DE, and the image data (RGB) from the GLCDC starts. See section 66.5.1. GLCDC Register Setting When Using Video Mode Operation.

66.3.6.2 End of Video Mode Operation

The procedure to stop operating in Video mode is as follows:

1. Set the VMSET0R.VSTOP bit to 1 to request that the Video mode operation stops.
   When the DSI Host detects the start of the first frame after setting the VMSET0R.VSTOP bit to 1, it stops sending Video mode packets and sets the VMSR.STOP flag to 1 and VMSR.RUNNING flag to 0.
2. Stop the GLCDC after checking that the VMSR.STOP flag is set to 1.
3. After stopping the GLCDC, wait for the LINKSR.HSBUSY flag is set to 0.

66.3.6.3 Burst Mode

The DSI Host supports Burst mode. Burst mode is a mode for time-compression of pixel data.

Make the bandwidth of D-PHY wider than the bandwidth of the data stream from GLCDC to time-compress the pixel data.

66.3.6.4 Non-Burst Mode

The DSI Host supports the Non-burst mode.

Equalize the bandwidth of D-PHY with the bandwidth of the data stream from GLCDC.

66.3.6.5 Selection of Blanking Packets or LP-11 for HSA, HBP, and HFP

When both of the following two conditions are satisfied, the DSI Host automatically transitions the data lane from HS mode to LP mode during the HSA, HBP, and HFP periods, respectively.

If the DSI Host does not transition from HS mode to LP mode, it sends the blanking packets with the same Virtual Channel (VC) as the VC of the previous packet to maintain HS mode.

1. There is time to transition from HS mode to LP mode and back to HS mode before the next HS transfer starts. The minimum period required to transition to LP mode is specified by the LPTRNSTSETR.GOLPBKT[9:0] bits.
2. The following registers are set for each period of HSA, HBP, and HFP:
   - HSA : VMSET0R.HSANOLP = 0
   - HBP : VMSET0R.HBPNOLP = 0
   - HFP : VMSET0R.HFPNOLP = 0

66.3.6.6 Non-video Packet Action in Video Mode Operation

The DSI Host can send and receive packets using the Sequence operation (see section 66.3.5.2. Single Packet Transmission and section 66.3.5.3. Single Packet Reception) during Video mode operation.

Make sure the operations of all Sequence operation channels are complete before starting the Vide mode operation (before starting the process described in section 66.3.6.1. Start of Video Mode Operation).

During Video mode operation, the Sequence operation is used to send and receive packets after the HSE packet on the first horizontal line if the HSE transfer is permitted (VMPPSETR.TXESYNC = 1) and after the VSS packet if the HSE transfer is not permitted (VMPPSETR.TXESYNC = 0).

Additionally, sending packets using the Sequence operation is only possible with HS mode (SQCHnDSCmAR.SPD = 0). Transmission in LP mode (SQCHnDSCmAR.SPD = 1) is not supported.

Send and receive packets so that they are completed within the BLLP period of the first horizontal line.

Do not use any other functions of the Sequence operation.

If the Sequence operation is executed during Video mode operation, stop the Video mode operation (section 66.3.6.2. End of Video Mode Operation) after confirming that the Sequence operation is completed.

The period of Video mode operation begins the moment the VMSET0R.VSTART bit is set to 1.

Note: The VMSR.RUNNING bit, which indicates that Video mode is operating, is reflected after a delay.

66.3.6.7 Delay Adjustment for Video Mode Operation

The DSI Host has a 4 KB video buffer to input video stream from GLCDC.

It is necessary to use the VMSET1R.DLY[11:0] bits to adjust the delay time for reading from the video buffer to prevent the video buffer from overflowing or underflowing during Video mode operation.

The delay time is derived from the following formula:

Delay time [µs] = VMSET1R.DLY[11:0] × (period of HS serial UI) [µs] × 32

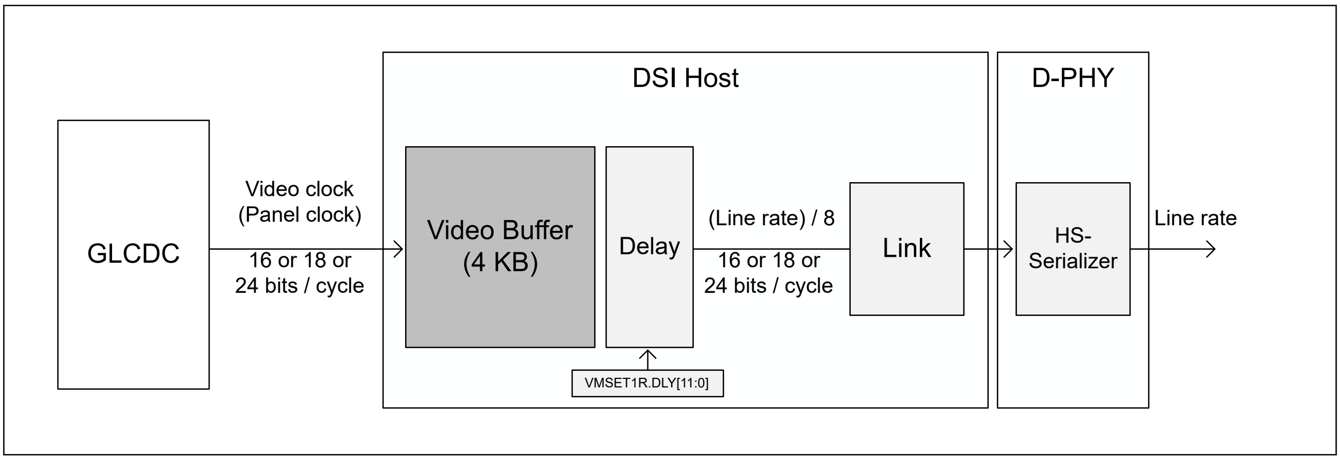


Figure 66.5 Video buffer and delay control

66.3.7 Transition To/From ULPS

This section describes the sequence of transition to or return from ULPS.

The clock lane and data lanes (Lane 0 and Lane 1) can be transitioned to/from ULPS independently.

When the TXSETR.CLEN bit is set to 1, the clock lane can transition to ULPS. It is prohibited to change the TXSETR.CLEN bit to 0 while the clock lane is in ULPS.

When the TXSETR.DLEN bit is set to 1, the data lanes can transition to ULPS. It is prohibited to change the TXSETR.DLEN bit to 0 while the data lanes are in ULPS.

The transition to/from ULPS of the data lanes is performed simultaneously for the data lane specified by the TXSETR.NUMLANE[1:0] bits.

Only a specific data lane cannot be transitioned to ULPS.

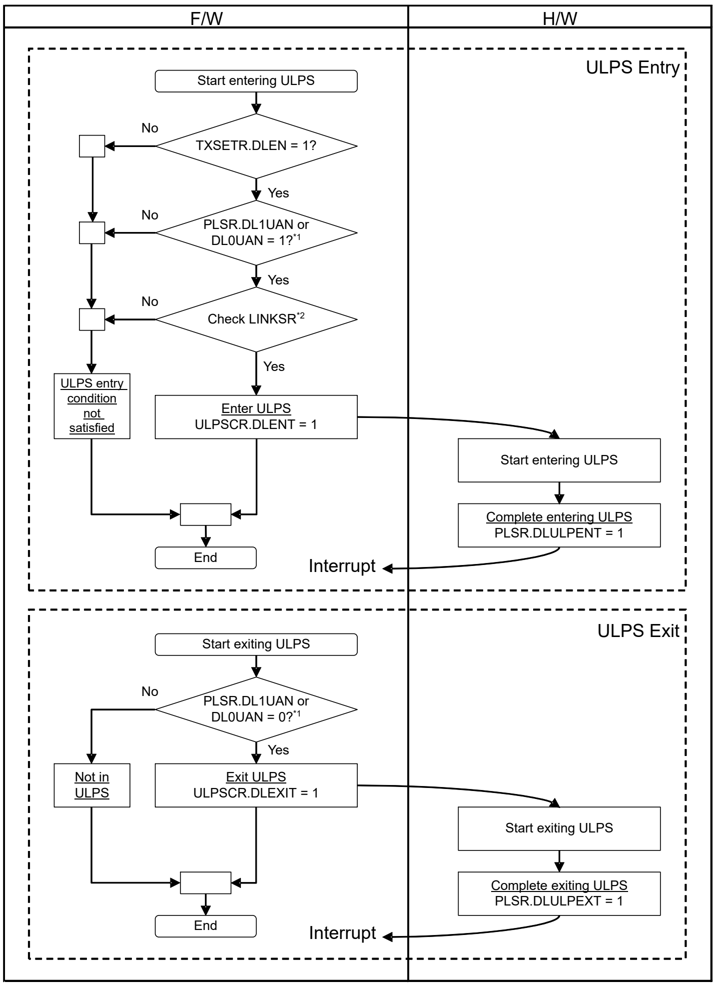
When data lanes enter ULPS, another operation*1 which uses data lanes cannot be performed until exit from ULPS.

When clock lane enters ULPS, another operation*2 which uses clock lane cannot be performed until exit from ULPS.

Note: The High-speed Transmit Word clock (HS mode clock) or Escape Mode Transmit clock (LP mode clock) might be stopped while data lanes and clock lanes are in ULPS depending on the specifications of the system incorporating the DSI-TX module.

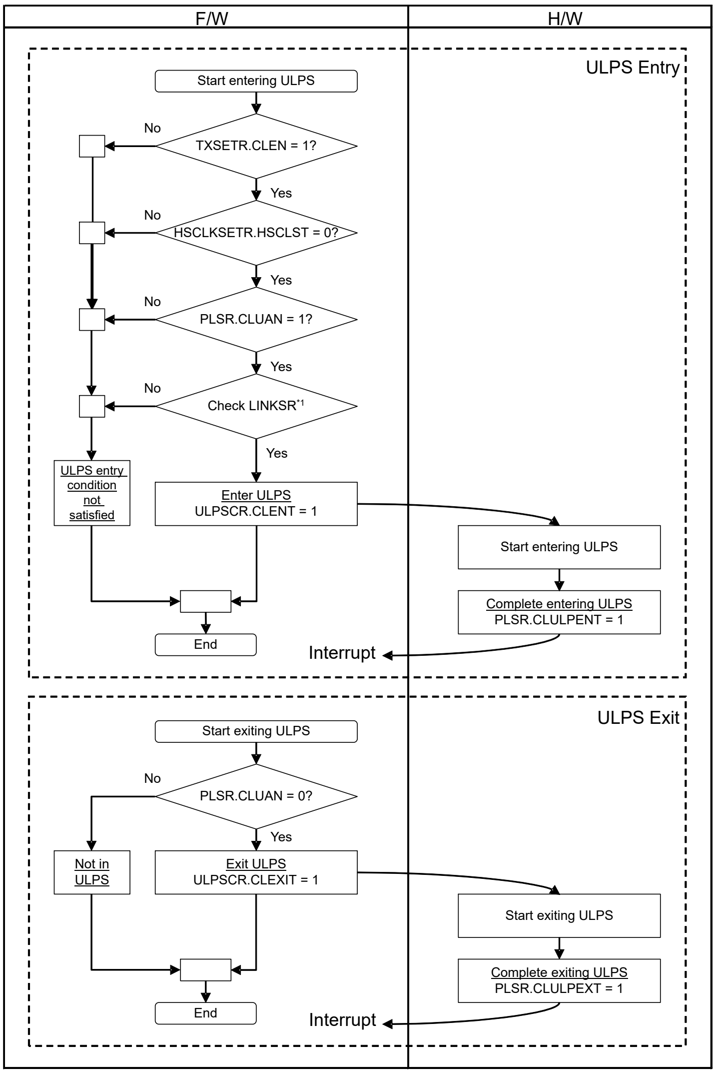
To perform software reset processing (section 66.3.2. Software Reset) in this state, restart the stopped clock before executing the software reset flow.

Note 1. Sequence operation, Video mode operation.
Note 2. HS Sequence operation, Video mode operation.



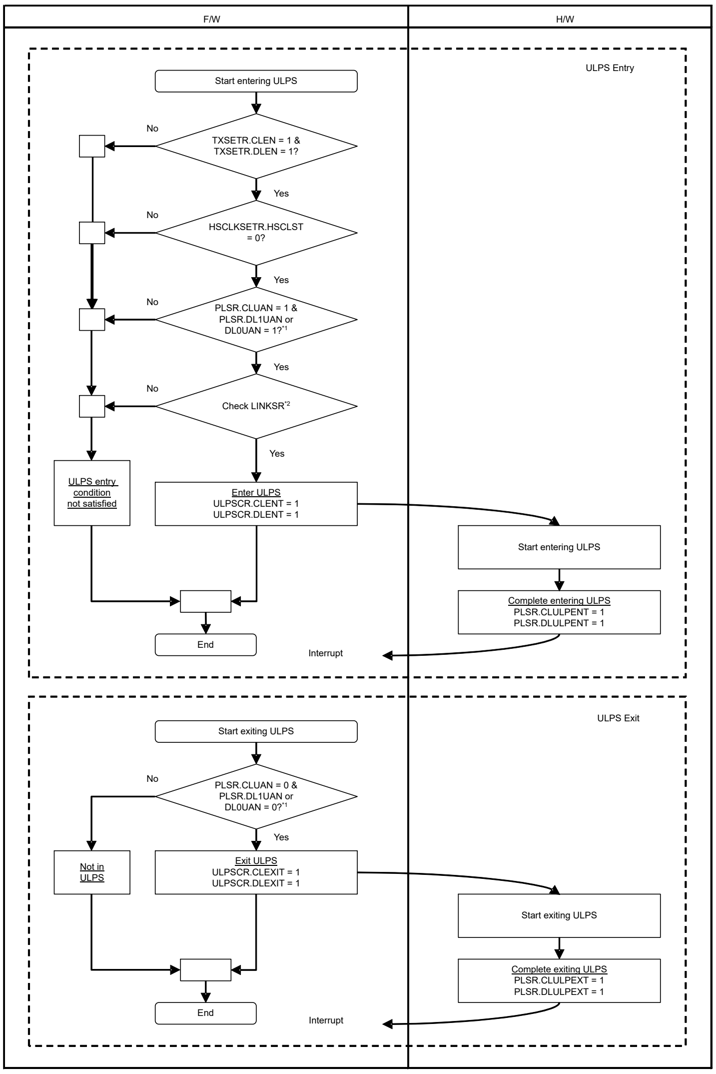
Note 1. Check all enabled data lane which selected by TXSETR.NUMLANE.
Note 2. (SQ0RUN = 0, SQ1RUN = 0, VRUN = 0, HSBUSY = 0 and LPBUSY = 0)?

Figure 66.6 ULPS entry/exit (data lanes)



Note 1. (SQ1RUN = 0, VRUN = 0 and HSBUSY = 0)?

Figure 66.7 ULPS entry/exit (clock lane)



Note 1. Check all enabled data lanes which are selected by TXSETR.NUMLANE.
Note 2. (SQ0RUN = 0, SQ1RUN = 0, VRUN = 0, HSBUSY = 0 and LPBUSY = 0)?

Figure 66.8 ULPS entry/exit (clock and data lanes)

66.3.8 Error Handling

66.3.8.1 Error Detection by Peripheral

The peripheral notifies the DSI Host of errors it detects with the Acknowledge and Error Report packet.

When the DSI Host receives an Acknowledge and Error Report packet, it displays the latest result in the AKEPLATIR register and the accumulated result in the AKEPACMSR register.

If you receive an Acknowledge and Error Report packet, review and change various settings as necessary.

66.3.8.2 Error Detection in D-PHY

When the D-PHY detects an error, the DSI Host sets the corresponding register flag.

The following errors are supported by the DSI Host. See Annex A Logical PHY-Protocol Interface Description of MIPI Alliance Specification for D-PHY Version 2.1, Annex A Logical PHY-Protocol Interface Description.

- ErrEsc
- ErrSyncEsc

- ErrControl
- ErrContentionLP0
- ErrContentionLP1

If this error is detected, perform a software reset (see section 66.3.2. Software Reset) and a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission) as necessary.

When the ErrContentionLP0/ErrContentionLP1 error occurs, be sure to execute a software reset.

In this case, after writing 1 to the RSTCR.FTXSTP bit, wait for a time (equivalent to LRX-H_TO) after which contention is resolved, and then write 0 to the RSTCR.FTXSTP bit. Also, if necessary, execute a reset trigger transmission.

For more information on recovery from contention, see 7.2 Contention Detection and Recovery in the MIPI Alliance Specification for Display Serial Interface 2 Version 1.1.

66.3.8.3 Timeout Error

The DSI Host has timers defined in 7.2.2., Contention Recovery Using Timers in the DSI standard and additional timers.

The following are timers defined in MIPI DSI Specification:

- HS TX Timeout Error (HTX_TO)
- LP-RX Host Processor Timeout Error (LRX-H_TO)
- Turnaround Acknowledge Timeout Error (TA_TO)

Additional timer is the Peripheral Response Timeout Error.

HS TX Timeout (HTX_TO) Error

If this error occurs, execute a software reset (see section 66.3.2. Software Reset). In addition, if necessary, execute a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission).

If this error is detected, the setting value of the HSTXTOSETR register may be too small.

Set the HSTXTOSETR register to a value greater than the HS transmission period.

LP-RX Host Processor Timeout (LRX-H_TO) Error

If this error occurs, execute a software reset (see section 66.3.2. Software Reset). In addition, if necessary, execute a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission).

If this error is detected, the setting value of the LRXHTOSETR register may be too small.

It is possible to receive multiple packets in one reception, such as a response packet and an Acknowledge and Error Report packet.

Set the LRXHTOSETR register to a value greater than the receive period.

Turnaround Acknowledge Timeout (TA_TO) Error

If this error occurs, if necessary, execute a software reset (see section 66.3.2. Software Reset) and a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission).

If this error is detected, the setting value of the TATOSETR register may be too small.

Check the specifications of the peripheral, if the peripheral does not support bidirectional communication, this error also occurs.

Peripheral Response Timeout Error

Refer to the description of Peripheral Response Timeout Error in section 66.3.8.4. Receive Related Error in Sequence Operation.

66.3.8.4 Receive Related Error in Sequence Operation

(1) Receive Packet Related Error in Sequence Operation

The following sections describe the errors in received packets detected by the DSI Host.

A maximum of one error is detected per packet.

Note: A single-bit ECC error may occur at the same time as other errors.

Errors are determined in the following order:

1. Peripheral Response Timeout Error
2. Malformed Packet Error
3. ECC Error
4. Unexpected Packet Error
5. Word Count Error
6. CRC Error
7. No Response Error

Peripheral Response Timeout Error

If no packets or triggers are received within the time set in the register after the bus ownership has been changed from host to peripheral by the BTA, the RXSR.PRTOERR flag is set to 1.

When this error is detected, no packets or triggers are received until the bus ownership shifts from peripheral to host.

Timeout Time Setting Registers

- Peripheral Response Timeout value for BTA is specified by the PRESPTOBTASETR register.
- Peripheral Response Timeout value for LP Read with BTA specified by the PRESPTOLPSETR.LPRTO[15:0] bits.
- Peripheral Response Timeout value for LP Write with BTA specified by the PRESPTOLPSETR.LPWTO[15:0] bits.
- Peripheral Response Timeout value for HS Read with BTA specified by the PRESPTOHSSETR.HSRTO[15:0] bits.
- Peripheral Response Timeout value for HS Write with BTA specified by the PRESPTOHSSETR.HSWTO[15:0] bits.

If this error is detected, the register setting value may be too small, so check the Peripheral specifications.

If this error occurs, perform the necessary processing, such as resending the previous command.

Malformed Packet Error

When a packet of less than 4 bytes is received, the RXSR.MLFERR flag is set to 1. Since there is an error in the received data, perform the necessary processing, such as resending the previous command.

ECC Error

ECC generation is mandatory for peripherals that comply with the MIPI DSI-2 v1.1 standard. However, ECC generation is optional for earlier revision of DSI Peripherals, so the DSI Host can control the enabling of the ECC check by the DSISETR.ECCEN bit.

When connecting to a peripheral that does not support ECC generation, disable the ECC check by setting the DSISETR.ECCEN bit to 0.

Single Bit ECC Error

When a single-bit ECC error is detected, the RXSR.ECCERRS flag is set to 1, and single-bit ECC error correction is performed.

The DSI Host automatically corrects the error and restores the normal packet header, so that the subsequent packet reception process can continue.

Multi Bit ECC Error

When a multi-bit ECC error is detected, the RXSR.ECCERRM flag is set to 1, and the received packet is discarded.

When this error occurs, perform the necessary processing, such as resending the previous command.

Unexpected Packet Error

If an unexpected packet is received, the RXSR.UNEXERR flag is set to 1.

When this error occurs, the packet header is stored in the RXRSSxR register, and subsequent reception processing is stopped until the bus ownership shifts from peripheral to host.

The DSI Host determines the following packets as unexpected packet:

- Reception of a packet whose Data Type field in the packet header is defined as Reserved in Table 23 Data Types for Peripheral-Sourced Packets of the MIPI DSI-2 v1.1 standard.
- Reception of a packet other than Acknowledge and Error Report packet and EoTp after sending a BTA request when the SQCHnDSCmAR.BTA[1:0] bits are set to 01b.
- Reception of a packet other than Acknowledge and Error Report packet and EoTp after sending a BTA request when the SQCHnDSCmAR.BTA[1:0] bits are set to 11b.
- Reception of a response packet after sending a read request following BTA when the SQCHnDSCmAR.BTA[1:0] bits are set to 10b, a response packet is received again.

Word Count Error

When the payload data of a long packet is shorter than the Word Count (WC) value, the RXSR.WCERR flag is set to 1.

When this error occurs, there is an error in the payload data length, but the payload data is not discarded, and the packet header is stored in the RXRSSxR register.

Since there is an error in the received data, perform the necessary processing, such as resending the previous command.

CRC Error

When connecting to a peripheral that does not support CRC generation, the DSI Host can select whether to enable or disable CRC checking for each Virtual Channel using the DSISETR.VCxCRCEN (x = 0 to 3) bit.

When connecting to a peripheral that does not support CRC generation, disable the CRC check by setting the corresponding DSISETR.VCxCRCEN bit to 0.

When this error occurs, there is an error in the payload data, but the payload data is not discarded, and the packet header is stored in the RXRSSxR register.

If this error occurs, take the necessary action, such as resending the previous command.

No Response Error

After bus ownership is transferred from the host to the peripheral by BTA, if it is transferred back to the host without neither packet nor trigger being received, the RXSR.NORESERR flag is set to 1.

If this error occurs, take the necessary action such as resending the command.

If the error recurs, perform a software reset (see section 66.3.2. Software Reset) or a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission) as necessary.

(2) Receive Long Packet Data Payload Related Error in Sequence Operation

Receive Long Packet Data Payload related errors are of the following:

- Maximum Return Packet Size Error
- Internal Bus Error
- Receive Buffer Overflow Error

Maximum Return Packet Size Error

If the WC (Word Count) value of the received long packet is greater than the value set in the DSISETR.MRPSZ[15:0] bits, the RXSR.RSIZEERR flag is set to 1.

When this error occurs, the packet header is stored in the RXRSSxR register and the payload data is discarded.

The payload data is discarded, but errors related to the payload data reception (WC Error, CRC Error) are detected.

If this error occurs, set the DSISETR.MRPSZ[15:0] bits to a value greater than or equal to the value set in the Set Max Return Packet Size command which limits the size of returning packets.

If this error occurs despite the correct settings, consider setting a larger value for the DSISETR.MRPSZ[15:0] bits, as the peripheral may be sending back large packets in violation of the DSI standard.

Internal Bus Error

If an error is detected in Internal AXI Bus Write, the RXSR.IBERR flag is set to 1.

Review the settings of the SQCHnDSCmDR.LADDR[31:0] bits and the DSISETR.MRPSZ[15:0] bits.

Receive Buffer Overflow Error

The RXSR.RXOVFERR flag is set to 1 when the following Receive Buffer Overflow Error is detected.

- When using Packet Payload Data Register (SQCHnDSCmBR.DTSEL = 00b).
  The length of the received payload data is longer than 16 bytes.
- When using memory space (SQCHnDSCmBR.DTSEL = 01b).
  Memory space overflows in LP-RX.

If this error occurs, do the following:

- When using Packet Payload Data Register (SQCHnDSCmBR.DTSEL = 00b)
  - Change to use memory space
  - Change the maximum return packet size of the peripheral device so that it does not exceed 16 bytes.
- When using memory space (SQCHnDSCmBR.DTSEL = 01b)
  - Reduce the maximum return packet size of the peripheral device.

66.3.8.5 Transfer Packet Related Error in Sequence Operation

Internal Bus Error

When an error is detected in Internal AXI Bus Read, the SQCHnSR.TXIBERR flag is set to 1.

Review the settings of the SQCHnDSCmAR.DATA0[7:0] bits, the SQCHnDSCmAR.DATA1[7:0] bits, and the SQCHnDSCmDR.LADDR[31:0] bits.

Packet Size Error

If the packet of Sequence operation is too large to be transmitted within the BLLP period of the first horizontal line of Video mode operation, the SQCH1SR.SIZEERR bit is set to 1.

When this error occurs, the Sequence operation packets are not sent until the Video mode operation is completed.

66.3.8.6 Error in Video Mode Operation

Video Mode Buffer Overflow Error

The VMSR.VBUFOVF flag is set to 1 if the video buffer overflows during the Video mode operation.

If this error occurs, perform a software reset (see section 66.3.2. Software Reset).

Also, take necessary measures such as increasing the line rate of the D-PHY or decreasing the value of the VMSET1R.DLY[11:0] bits.

In addition, perform a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission) if necessary.

Video Mode Buffer Underflow Error

The VMSR.VBUFUDF flag is set to 1 if the video buffer underflows during the Video mode operation.

If this error occurs, perform a software reset (see section 66.3.2. Software Reset).

Also, take necessary measures such as decreasing the line rate of the D-PHY or increasing the value of the VMSET1R.DLY[11:0] bits.

In addition, perform a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission) if necessary.

Timing Error

The VMSR.TIMERR flag is set to 1 if a video packet (including Sync Event packet) is not sent at the intended timing during Video mode operation.

If this error occurs, perform a software reset (see section 66.3.2. Software Reset).

Also, take necessary measures such as reviewing the setting value of the LPTRNSTSETR.GOLPBKT[9:0] bits.

In addition, perform a reset trigger transmission (see section 66.3.5.4. Reset Trigger Transmission) if necessary.

66.4 Interface

66.4.1 Video Mode Operation

66.4.1.1 Video Mode Timing

Video mode timing and Video mode parameters are described in Figure 66.9 and Table 66.3.

In Table 66.3, the actual period of the following parameters depend on the specification of the D-PHY which is connected to the DSI-TX module.

- Period for HS-LP-HS transition, period for HS-LP transition, and period for LP-HS transition
- Period for BTA[host -> peripheral].

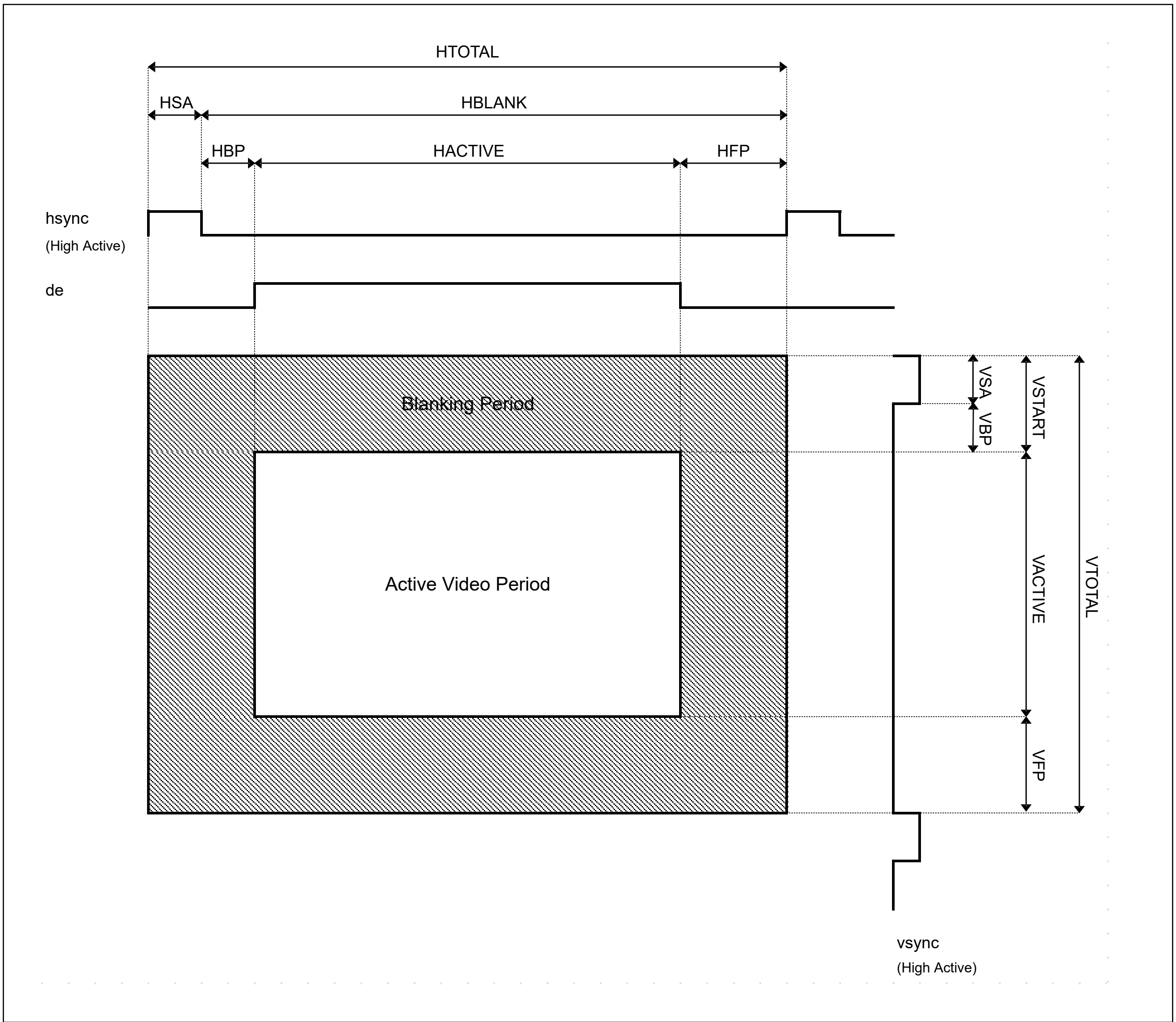


Figure 66.9 Video mode timing

Table 66.3 Configurations for Video mode interface (1 of 3)

Common rules for all modes		
Parameter	**Description**	
All	All parameters related to Input Video should be set to 1 or higher. Examples of the parameters: HSA, HBP, HACTIVE, HFP, VSA, VBP, VACTIVE, VFP	
Parameter	**VMPPSETR.DT**	**VMHSSETR.HACT**
HACTIVE	1Eh: Packed Pixel Stream, 18-bit RGB	This value should be a multiple of 4
	2Ch: Packed Pixel Stream, 16-bit YCbCr 4: 2: 2 0Ch: Loosely Packed Pixel Stream, 20-bit YCbCr 4: 2: 2 1Ch: Packed Pixel Stream, 24-bit YCbCr 4: 2: 2	This value should be a multiple of 2
	0Eh: Packed Pixel Stream, 16-bit RGB 2Eh: Loosely Packed Pixel Stream, 18-bit RGB 3Eh: Packed Pixel Stream, 24-bit RGB	No limitation
HFP	Rounddown (HFP*BPP/ 8, 0) >= 12, (12 = means Pixel Packet Header & footer 6 + Blanking Packet Header & footer 6)	
Parameter	**VMPPSETR.DT**	**VMHSSETR.HSA + VMHPSETR.HBP + VMHSSETR.HACT + VMHPSETR.HFP**
HTOTAL	1Eh: Packed Pixel Stream, 18-bit RGB	This value should be a multiple of 4
	2Eh: Loosely Packed Pixel Stream, 18-bit RGB 0Ch: Loosely Packed Pixel Stream, 20-bit YCbCr 4: 2: 2 1Ch: Packed Pixel Stream, 24-bit YCbCr 4: 2: 2 2Ch: Packed Pixel Stream, 16-bit YCbCr 4: 2: 2 0Eh: Packed Pixel Stream, 16-bit RGB 3Eh: Packed Pixel Stream, 24-bit RGB	No limitation

Table 66.3 Configurations for Video mode interface (2 of 3)

Additional rules for Non-burst Sync Pulse mode	
Parameter	**Description**
Sequence Operation Packet	When a packet is transmitted by Sequence operation during Video mode operation (VMSR.RUNNING = 1), the period of (HBP + HACTIVE + HFP) should be longer than the sum of the following: ● Period for the TX Sync Event packet (4 bytes) ● Period for the packet sent by Sequence operation ● Period for the Header and CRC of a Blanking packet (6 bytes)
HBP + HACTIVE + HFP	HS-LP-HS transition is caused within the period between the current Sync Event packet and the next Sync Event packet, so the period of (HBP + HACTIVE + HFP) should be as follows. (HBP + HACTIVE + HFP) > (Period for the TX Sync Event packet) + (Period for HS-LP-HS transition)
HSA	Blanking packet transmission is caused within the period between the current Sync Event packet and the next Sync Event packet, so the period of HSA should be as follows. $(HAS \times BPP) \div 8 \geq 10$ Note that HSA should not be less than the minimum value of the sum of (Period for the TX Sync Event packet) + (Period for the TX Blanking packet).
HBP	Blanking packet transmission is caused within the period between the current Sync Event packet and the next Sync Event packet, so the period of HBP should be as follows. $(HBP \times BPP) \div 8 \geq 10$ Note that HBP should not be less than the minimum value of the sum of (Period for the TX Sync Event packet) + (Period for the TX Blanking packet).

Table 66.3 Configurations for Video mode interface (3 of 3)

Additional rules for Non-burst Sync Event mode and Burst mode	
Parameter	**Description**
Sequence Operation Packet	When a packet is transmitted by Sequence operation during Video mode operation (VMSR.RUNNING = 1), the period of HTOTAL should be longer than the sum of the following: ● Period for the TX Sync Event packet (4 bytes) ● Period for the packet sent by Sequence operation ● Period for the Header and CRC of a Blanking packet (6 bytes)
HTOTAL	HS-LP-HS transition is caused within the period between the current Sync Event packet and the next Sync Event packet, so the period of HTOTAL should be as follows. HTOTAL > (Period for the TX Sync Event packet) + (Period for HS-LP-HS transition)
HAS + HBP	Blanking packet transmission is caused within the period between the current Sync Event packet and the next Sync Event packet, so the period of (HAS + HBP) should be as follows. ((HAS + HBP) × BPP) ÷ 8 ≥ 10 Note that HAS + HBP should not be less than the minimum value of the sum of (Period for the TX Sync Event packet) + (Period for the TX Blanking packet). The minimum HAS + HBP value currently recognized is 8 + 8 and it is 32 even when 16-bit RGB is the worst condition, which is not subject to this restriction. Therefore, this restriction is not presented up to the limit. Even if this condition is not satisfied, the result of Pixel-Packet is only slightly delayed.

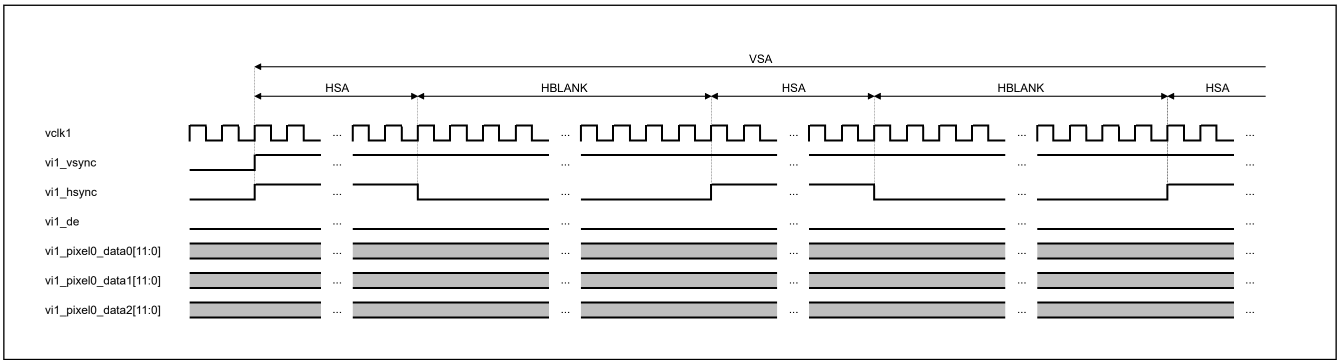


Figure 66.10 VSA start (and VFP end)

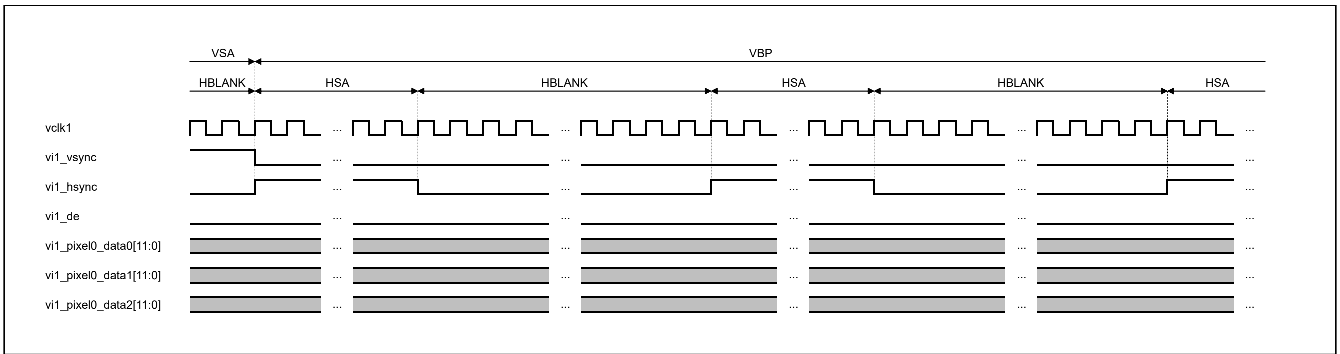


Figure 66.11 VSA end and VBP start

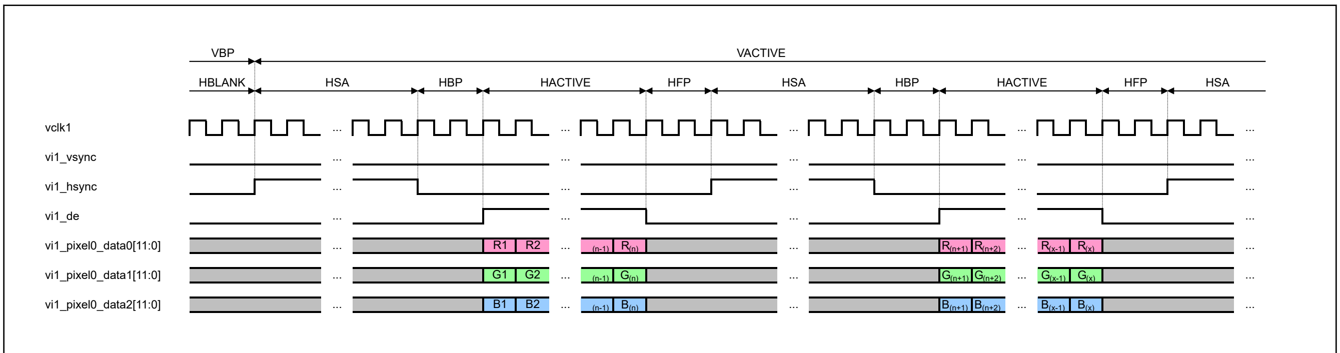


Figure 66.12 VBP end and VACTIVE start (RGB)

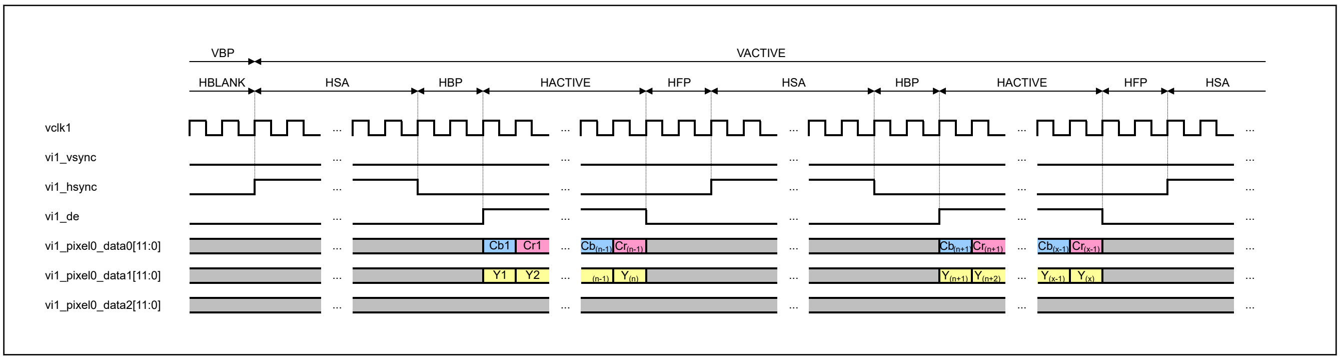


Figure 66.13 VBP end and VACTIVE start (YCbCr 4: 2: 2)

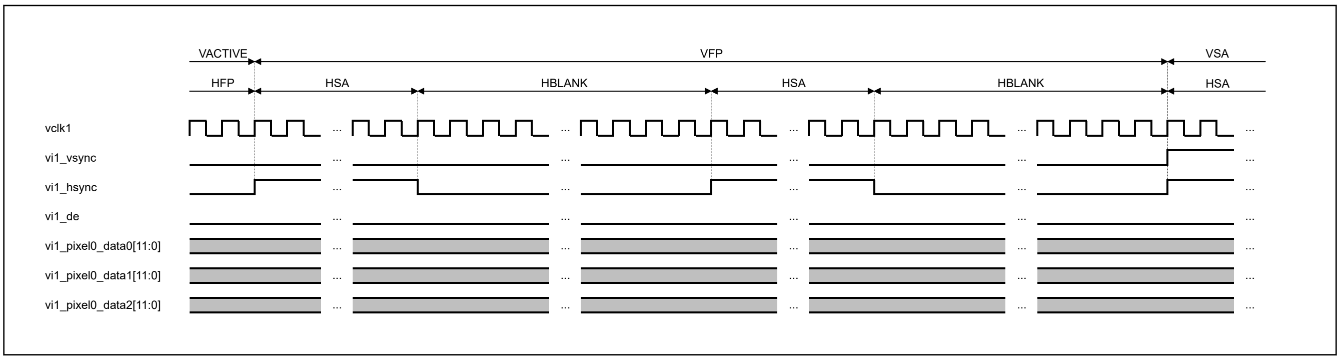


Figure 66.14 VACTIVE end and VFP start

66.5 Usage Notes

66.5.1 GLCDC Register Setting When Using Video Mode Operation

When using Video mode operation, set the GLCDC registers as shown below to specify the video signals (VSYNC, HSYNC, DE, RGB) and supply a video clock.

Common

- SYSCNT_PANEL_CLK.CLKEN = 1 (enable Panel Clock output)
- SYSCNT_PANEL_CLK.PIXSEL = 0 (parallel RGB)
- TCON_STVA1 = Value to specify VSYNC signal
- TCON_STHA1 = Value to specify HSYNC signal
- TCON_STVB1 = Value to specify VE signal
- TCON_STHB1 = Value to specify HE signal
- TCON_STVA2.SEL[2:0] = 000b (assign STVA signal to LCD_TCON0 output)
- TCON_STVB2.SEL[2:0] = 010b (assign STHA signal to LCD_TCON1 output)
- TCON_STHA2.SEL[2:0] = 111b (assign DE signal to LCD_TCON2 output)
- OUT_SET.FRQSEL[1:0] = 00b (parallel RGB)
- OUT_SET.SWAPON = 0 (RGB order)
- OUT_SET.ENDIANON = 0 (little endian)

For RGB888 (24-bit) format

- OUT_SET.FORMAT[1:0] = 00b (RGB888 format)

For RGB666 (18-bit) format

- OUT_SET.FORMAT[1:0] = 01b (RGB666 format)

For RGB565 (16-bit) format

- OUT_SET.FORMAT[1:0] = 10b (RGB565 format)

66.5.2 Power Gating Control or Software Standby Mode

Before transitioning to graphics Power Domain Off or Software Standby mode, stop each operation according to the following sections:

- section 66.3.4. Stop of HS Clock
- section 66.3.6.2. End of Video Mode Operation
- section 66.3.5. Command Mode Operation.

After returning from graphics Power Domain Off or Software Standby mode, resume each operation as necessary, following the descriptions in the following sections:

- section 66.3.3. Start of HS Clock
- section 66.3.6.1. Start of Video Mode Operation
- section 66.3.5. Command Mode Operation.

66.5.3 Module-Stop Function Setting

MIPI DSI operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The MIPI DSI module is initially stopped after reset. Releasing the module-stop state enables access to the registers.